

SEMESTER - I

23U101 CALCULUS AND ITS APPLICATIONS

3 1 0 4

DIFFERENTIAL CALCULUS: Functions of two variables, limits and continuity, partial derivatives, chain rule, extreme values and saddle points, Lagrange multipliers, Taylor's formula for two variables. (9+3)

INTEGRAL CALCULUS: Double and iterated integrals over rectangles, double integrals over general regions, Fubini's theorem, area and volume by double integration, reversing the order of integration, double integrals in polar form. (9+3)

FIRST ORDER ORDINARY DIFFERENTIAL EQUATIONS: Basic concepts, separable differential equations, exact differential equations, integrating factors, linear differential equations, modeling - mixing problems, Newton's law of cooling, decay and growth problems. (9+3)

SECOND ORDER LINEAR DIFFERENTIAL EQUATIONS: Homogeneous linear equations of second order, homogeneous linear ODEs with constant coefficients, Euler-Cauchy equations, solution by variation of parameters, free oscillations mass spring systems, electric circuits. (9+3)

VECTOR CALCULUS: Gradient and directional derivative of a scalar field, divergence and curl of a vector field. Integration in vector field – line integrals, path independence of line integrals, Green's theorem in the plane, divergence theorem of Gauss and Stokes' theorem. (9+3)

Total L: 45 +T: 15 = 60

TEXT BOOKS

1. J. Hass, C. Heil, Maurice D.W "Thomas' Calculus", Pearson Education, New Delhi, 2018.
2. Erwin Kreyszig, "Advanced Engineering Mathematics", Wiley India, New Delhi, 2018.

REFERENCES

1. H. Anton, I. Bivens, S. Davis, "Calculus", John Wiley and Sons, USA, 2016.
2. Wylie C. R and Barrett L. C, "Advanced Engineering Mathematics", Tata McGraw-Hill, New Delhi, 2019.
3. Michael D. G, "Foundations of Applied Mathematics", Dover Publications, INC, New York, 2013.
4. Gilbert Strang, "Calculus", Wellesley Cambridge Press, USA, 2017.

23U102 PHYSICS

3 0 0 3

MECHANICS: Motion in a straight line, average motion, instantaneous velocity, acceleration and constant acceleration, motion in two and three dimensions, force and motion. Application of Newton's laws. Free body diagram. Work, energy and power. Conservation of energy, conservative and non-conservative forces, conservation of mechanical energy. Gravity (9)

RIGID BODY DYNAMICS AND ROTATIONAL MOTION: systems of particles - centre of mass, momentum. Kinetic energy of a system. Elastic and inelastic collisions. Rotational motion, Rotational vectors: velocity, acceleration, torque, angular moment, moment of inertia. – determination of moment of inertia for rigid bodies, rotational energy and rolling motion. (9)

ELECTRIC CHARGE, FORCE & FIELD: Electric charge, Coulomb's law, Electric field intensity, electric dipoles, electric field of a dipole along the axis and equatorial point, Equipotential surfaces. Gauss's law of electrostatics: Application of Gauss's law to line, plane and spherical symmetry. Field at a conductor surface. Applications: Microwave cooking, Liquid Crystals, Shielding and Lightning Safety (9)

ELECTRIC POTENTIAL: Electric potential difference, curved paths and non-uniform fields. Calculating potential difference: potential of a point charge, zero potential, finding potential differences using superposition. Potential difference and Electric field, charged conductors. Electrostatic Energy, Capacitors, Energy in the Electric field. Applications: Corona discharge, pollution control and xerography (9)

ELECTROMAGNETISM: Magnetic force and field, Charged particles in magnetic fields, Magnetic force on a current. Biot-Savart's law, magnetic force between conductors. Magnetic dipoles, Gauss's law of magnetism. Ampere's law, electromagnetic induction. Faraday's law, magnetic circuits, self and mutual induction – magnetic field energy density, Maxwell's equations and propagation of electromagnetic waves in isotropic media. (9)

Total L: 45

TEXT BOOKS:

1. Richard Wolfson "Essential University Physics", Pearson., 2020
2. Physics for Scientists and Engineers with Modern Physics, Raymond A Serway and John W. Jewett, Cengage Publishers, USA, 2019.

REFERENCES:

1. D Halliday, R Resnick and Walker "Fundamentals of Physics", John Wiley and Sons., Twelfth Edition, 2021
2. Gaur R K, Gupta S L, "Engineering Physics", Dhanpat Rai publications., 2014
3. Hugh D Young and Roger A Freedman, "University Physics", Addison Wesley, Fifteenth Edition, 2020

23U103 ELECTRIC CIRCUITS

3 1 0 4

BASIC DEFINITIONS AND LAWS: Overview of electrical circuits: currents, voltages, power and energy –Circuit elements: active and passive elements –Independent and dependent sources –Resistive circuits: Ohm's law and Kirchhoff's laws – Series resistors and voltage division - Parallel resistors and current division –Series voltage source and parallel current source –Y-Δ transformations. (9+3)

DC CIRCUIT ANALYSIS AND CIRCUIT THEOREMS: Node voltage analysis – Mesh current analysis –Circuit theorems: Source transformation –Superposition theorem – Thevenin's theorem –Norton's theorem –Maximum power transfer theorem. (9+3)

COMPLETE RESPONSE ANALYSIS: First order circuits – Source-Free RC Circuits – Source-Free RL Circuits – Step Response of RC Circuits – Step Response of RL Circuits –Second order circuits –Finding Initial and Final Values – Source-Free Series RLCCircuits – Source-Free Parallel RLC Circuits – Step Response of Series RLC Circuits - Step Response of ParallelRLCCircuits (9+3)

AC CIRCUIT ANALYSIS : Single phase circuits – Single phase voltages and currents –Phasor representation –RMS value – Form factor –Energy storage elements: Capacitors and inductors –Phasor relationship for R, L and C –Impedance and admittance – Analysis of Series RLC, Parallel RLC circuits using analytical and phasor methods –Resonance in parallel and series circuits - Bandwidth and Q-factor (9+3)

THREE PHASE CIRCUITS: Three-Phase Voltages and currents –Balanced Y-Y Connection, Y-Δ Connection, Δ-Δ Connection, Δ-Y Connection – Power in Balanced System – Unbalanced Three-Phase Systems –connected source and load - Δ connected source and load – Y to Δ circuit –Power and power factor in three phase (9+3)

Total L: 45 +T: 15 =60

TEXT BOOKS:

1. Charles K. Alexander, Mathew N. O. Sadiku. 'Fundamentals of Electric Circuits', Fifth Edition, Tata McGraw Hill, 2016.
2. John Bird. 'Electrical Circuit Theory and Technology', Fifth Edition, New York: Routledge, 2017

REFERENCES:

1. Richard C. Dorf, James A Svoboda, 'Introduction to Electric Circuits', Ninth Edition John Wiley and Sons Inc, 2017.
2. K.C.A.Smith, R.E. Alley . 'Electrical circuits an introduction', New Delhi: Cambridge University Press, 2017.
3. Allan H. Robbins, Wilhelm C. Miller. 'Circuit Analysis: Theory and Practice', Fifth Edition, Cengage publishers, 2013.
4. Sudhakar A, Shyammoan S. Palli. 'Circuits and Network Analysis and Synthesis', Fifth Edition, Tata McGraw Hill, 2017.

23U104 ELECTRONIC DEVICES AND CIRCUITS

3 0 0 3

DIODES: Semiconductor materials – N type and P type semiconductor – PN Junction – The Ideal diode – Diode operation – Voltage and current characteristics – Diode models - - Diode testing – Zener diode – Zener diode applications –Types of diodes – Diode applications. (9)

BIPOLAR JUNCTION TRANSISTOR: Construction – operation – parameters – characteristics – configuration – biasing – types of biasing – load line - DC Operating point – Transistor as a switch, as an amplifier. (9)

FIELD EFFECT TRANSISTORS: Junction Field Effect Transistor (JFET) Construction – Principle of operation - Parameters – Characteristics – Biasing – application – Metal Oxide Semiconductor Field Effect Semiconductor (MOSFET) – types – Construction – Principle of operation - Parameters –Characteristics – Biasing – application. (9)

AMPLIFIERS: Amplifier concepts – Gain and attenuation –DC and AC analysis - Impedance matching –Small signal amplifiers – BJT amplifier configurations – Introduction to differential amplifier –Power amplifiers –Class A, B, AB and Class C Power amplifiers –Push-Pull amplifiers. (9)

OSCILLATORS: Concept of feedback – Conditions for oscillations –Oscillators with RC feedback circuits - Phase shift oscillator - Wein Bridge oscillator - Oscillators with LC feedback circuits- Colpitts oscillators- UJT relaxation oscillator (9)

Total L: 45

TEXT BOOKS:

1. Albert P. Malvino, David J. Bates and Patrik E. Hoppe, "Electronic Principles", McGraw Hill, Ninth Edition, 2021.
2. Boylestad R. L. and Nashelsky "Electronic Devices and Circuit Theory", Pearson Education India, Eleventh Edition, 2015.

REFERENCES:

1. Thomas L. Floyd, "Electronic Devices", Pearson Education, Tenth Edition, 2021.
2. Bell D. A "Electronic Devices and Circuits", Oxford University Press, Fifth Edition, 2013 .
3. Millman J., Halkias C and Parikh C D "Integrated Electronics", Tata McGraw-Hill., Second Edition, 2017.
4. Donald Neamen, "Electronic Devices: Analysis and Design", McGraw Hill, Third Edition, 2006.

23G105 ENGLISH LANGUAGE PROFICIENCY

3 1 0 4

VOCABULARY: Etymology-Prefixes and suffixes-Synonyms-Antonyms-Guessing meanings from context-Word formation-Single-word substitutes- Different forms of a word-Phrasal verbs-Collocations. (9+3)

LISTENING AND SPEAKING: Understanding listening – Listening techniques - Introducing oneself and others –Seeking and sharing information– Description-Conversation skills– Extempore speaking– Speech practice in varied formal contexts. (9+3)

GRAMMAR: Wh-questions – Yes/no questions– Parts of speech – Articles– Prepositions–Gerunds–Conjunctions–Degrees of comparison– Tenses– Modal verbs – Adverbs - Direct and indirect questions. (9+3)

READING: Reading strategies: Skimming and scanning, Predicting– Reading comprehension: techniques –Practice reading. (9+3)

WRITING: Discourse markers – Dialogue writing - Completing sentences – Jumbled sentences – Paragraph writing –Writing compare & contrast paragraphs – Letter writing. (9+3)

Total: 45 +15 = 60

TEXTBOOKS:

1. Shoba K.N.and Lourdes Joavani Rayen, Communicative English, Cambridge University press, Cambridge, 2021.
2. Raymond Murphy, Intermediate English Grammar, Cambridge University Press, New Delhi, 2020.

REFERENCES:

1. Raymond Murphy, English Grammar in Use, Cambridge University Press, New Delhi, 2020.
2. Sudharshana N P and Savitha C, English for Engineers, Cambridge University Press, New York, 2018.
3. Anne Laws, Writing Skills, Orient Black Swan, Hyderabad, 2011.
4. Helen Naylor with Raymond Murphy, Essential English Grammar, Cambridge University Press, New Delhi, 2019.

23U110 ENGINEERING GRAPHICS

0 0 4 2

INTRODUCTION:

1. Introduction to Engineering Drawing
2. Bureau of Indian Standards
3. Principles of dimensioning
4. Lettering Practice

(12)

ORTHOGRAPHIC PROJECTION:

1. Principles of orthographic projection
2. Projection of points, straight lines, planes and solids.
3. Orthographic projection of simple engineering components – missing view exercises.

(12)

PICTORIAL PROJECTIONS:

1. Principles of pictorial views, isometric view of simple engineering components.
2. Isometric views from given two or three views.

(12)

SECTION OF SOLIDS:

1. Section of regular solids, Need for Sectional views.
2. Sectional views of simple engineering components.

(12)

DEVELOPMENT OF SURFACES:

1. Development of lateral surfaces of regular solids and truncated solids.
2. Preparing parts like tray, funnel, CPU housing using cardboard material.

(12)

Total P: 60

REFERENCES:

1. Venugopal K, Prabhu Raja V “Engineering Graphics”, New Age International Publishers, 2017
2. Natarajan K V “Engineering Drawing and Graphics”, Dhanalakshmi Publishers., 2007

23U111 CIRCUITS AND DEVICES LABORATORY

0 0 4 2

LIST OF EXPERIMENTS:

1. Characteristics of PN Junction Diode
2. Half wave and Full wave diode rectifiers
3. Diode Clipper and Clamper Circuits
4. Design of voltage regulator using Zener diode
5. Bipolar Junction Transistor as a switch
6. Design of single stage amplifier using Bipolar Junction Transistor
7. Junction Field Effect transistor as an amplifier/switch
8. Design of multi-stage amplifier
9. RC phase shift oscillator using BJT
10. Design of passive filters

Total P:60

REFERENCES:

1. David A Bell, ‘Electronic Devices and Circuits Lab Manual’, Oxford University Press, Fourth Edition, 2006.
2. Laboratory manual prepared by the Department of Instrumentation and Control Systems Engineering 2023.

23IP15 INDUCTION PROGRAMME

0 0 0 0

As per AICTE guidelines

SEMESTER – II

23U201 COMPLEX VARIABLES AND TRANSFORMS

3 1 0 4

COMPLEX DIFFERENTIATION: Derivative, analytic function, Cauchy-Riemann equations, Laplace's equation, linear fractional transformations. (9+3)

COMPLEX INTEGRATION: Cauchy's integral theorem, Cauchy's integral formula, derivatives of analytic functions, Laurent series, singularities and zeros, residue integration method (Residue integration of complex integrals only). (9+3)

LAPLACE TRANSFORMS: Laplace transform, linearity, first shifting theorem, transforms of derivatives and integrals, unit step function, second shifting theorem, Dirac's delta function, periodic functions, differentiation and integration of transforms, solving ODEs with constant coefficients and initial value problems. (9+3)

FOURIER ANALYSIS: Fourier series – arbitrary period, even and odd functions, half range expansions. Fourier transforms, Fourier cosine and sine transforms. (9+3)

PARTIAL DIFFERENTIAL EQUATIONS: Basic concepts of PDEs, wave equation, heat equation, steady state two-dimensional heat problems, solution by Fourier series. (9+3)

Total L: 45 + T:15 = 60

TEXT BOOKS

1. Erwin Kreyszig, "Advanced Engineering Mathematics", Wiley India, New Delhi, 2018.
2. Dennis G. Z, "Advanced Engineering Mathematics", Jones and Bartlett Pvt Ltd, New Delhi, 2017.

REFERENCES

1. Dennis G. Z and Patrick D. S, "A first course in Complex Analysis with applications", Jones and Bartlett Pvt Ltd, New Delhi, 2015.
2. Wylie C. R and Barret L. C, "Advanced Engineering Mathematics", Tata McGraw-Hill, New Delhi, 2019.
3. Peter V.O Neil, "Advanced Engineering Mathematics", Cengage, New Delhi, 2018.
4. Dean G. D, "Advanced Engineering Mathematics with MATLAB", CRC Press, USA, 2017.

23U202 APPLIED CHEMISTRY

3 0 0 3

METALS AND ALLOYS: Binary phase diagrams –Cu/Ni, Pb/Ag alloy – interpretation using phase rule, steel – Fe-C phase diagram, heat treatment. Aluminium and titanium alloys. Corrosion – electrode potentials, chemical and electrochemical corrosion – emf series, galvanic series –passivation. Forms of corrosion – galvanic, differential aeration, atmospheric corrosion, stress corrosion cracking, fretting, corrosion fatigue, tribocorrosion. Corrosion rate – factors influencing corrosion. (9)

CORROSION AND WEAR PROTECTION OF METALS: Electropolishing, electroplating – Cu, Ni and Cr, electroless plating – Cu and Ni, anodizing, phosphating, chromating, organic coatings-paints – constituents and functions, powder coating, electrophoretic painting, superhydrophobic and selfhealing coatings. Lubricants-classification, properties, mechanism of lubrication- additives, improvers. (9)

POLYMERS: Polymerisation - chain and condensation polymerization, polymer topologies – linear, branched and crosslinked, degree of polymerisation, polydispersity, average molecular weight. States of aggregation – amorphous and crystalline states – influence on properties of polymers – thermal properties – phase transitions – Glass transition and melting – thermal analysis, classification based on thermal behaviour – thermoplastics and thermosetting plastics, mechanical properties – stress strain behaviour. Modification of properties of polymers – additives- processing, protective and functional additives, composites – fillers – particulate and fibres – carbon, glass, aramid. (9)

INDUSTRIAL AUTOANALYZERS: Performance requirements of analytical instruments. Instrument calibration methods – calibration curve, standard addition, internal standard methods. Automated chemical analysis – segmented flow system and flow injection analysis. Types of analysers – colorimeters and spectrophotometers, ICP-MS, electrochemical analysers-potentiometers, conductivity meters, amperometry. Analysers for water quality monitoring – pH – ion-selective electrode, TDS, conductivity, turbidity determination, Sensors for air quality monitoring –particulate matter, CO, CO₂, VOC., NO_x, SO_x content determination. Sensors for smart farming– soil nutrient content, pH, humidity. (9)

POWER SOURCES FOR IIoT DEVICES: Power requirements for wireless devices. Batteries - characteristics. Primary batteries – construction and working of zinc- carbon cells, lithium cells – Li-MnO₂, Li-FeS₂, Li-SOCl₂. Secondary batteries – lead-acid, lithium ion batteries. Supercapacitors – EDLC and pseudocapacitors. Flexible solar cells – DSSC and polymer photovoltaics – materials – conducting polymers and fullerenes. (9)

Total L: 45

TEXT BOOKS:

1. Shashi Chawla, "A Text Book of Engineering Chemistry", Dhanpat Rai and Co , 2005.
2. Dara SS, UmareSS, "A Textbook of Engineering Chemistry", S Chand and Co , 2010.

REFERENCES

1. Robert W. Messler, Jr, "The Essence of Materials for Engineers", Jones and Barlett India Pvt. Ltd., 2011.
2. Derek Pletcher and Frank C. Walsh. "Industrial Electrochemistry", Chapman and Hall, 1993.
3. Cowie J.M.G, Valeria Arrighi "Polymers: Chemistry and Physics of modern materials", CRC Press, 2007.
4. Khandpur R.S, "Handbook of Analytical Instruments", McGraw Hill Education (India) Private Limited, 2015.

23U203 DIGITAL ELECTRONICS

3 1 04

NUMBER SYSTEMS AND BOOLEAN FUNCTIONS: Review of Number Systems, number complements, Binary arithmetic, Binary codes: weighted and non-weighted codes, alphanumeric codes, Error detection and correction codes, switching functions, Canonical forms, Incompletely specified functions, Simplification of logic functions through Boolean Algebra, K-maps and Quine-McCluskey method, Implementation of logic functions using basic logic gates and universal gates. (9+3)

COMBINATIONAL LOGIC DESIGN: Design of Arithmetic circuits-Adders / subtractors, Carry look-ahead adder, signed number addition and subtraction, BCD adders, IC adders, Magnitude comparator, Decoders, Encoders, Multiplexers and Demultiplexers. Implementation of combinational logic function using multiplexers and demultiplexers - Introduction to HDL (9+3)

SEQUENTIAL LOGIC DESIGN: General model of sequential circuits — Latch, Flip Flops, Design of synchronous sequential circuits—Up-down / Modulus counters, Shift register: Ring counter, Johnson counter, Timing diagram. Mealy / Moore models — Concept of state, State diagram, State table, Design of sequence detector, State reduction procedures using Partitioning and Implication chart. Introduction to Asynchronous Sequential Circuits (9+3)

PROGRAMMABLE LOGIC DEVICES AND HDL: Semicustom design - Introduction to Programmable Logic Devices — Read Only Memory, Programmable Array Logic, Programmable Logic Array, Field Programmable Logic Array, Field Programmable Logic Sequencer, Architecture of Programmable Logic Devices, Implementation of digital functions using HDL (9+3)

DIGITAL LOGIC FAMILIES: Characteristics of digital ICs — Voltage and current ratings, Noise margin, Propagation delay, Power dissipation, Fan-in, Fan-out. TTL logic family—Totem pole, Open collector and tristate outputs. MOS transistor switches — nMOS Inverter / Logic gates, CMOS Inverter / Logic gates, ECL logic families, Comparison of performance of various logic families, Interfacing TTL and CMOS devices. (9+3)

Total L: 45 +T: 15 =60

TEXT BOOKS:

1. M. Morris Mano, Michael D. Cilette, "Digital Design: With an Introduction to the Verilog HDL, VHDL, and System Verilog", Sixth Edition, Prentice Hall, 2022.
2. Donald P Leach, Albert Paul Malvino, Gautam Saha, "Digital Principles and Applications", Eighth Edition, Tata McGraw Hill, 2019

REFERENCES:

1. Thomas L Floyd, "Digital Fundamentals", Eleventh Edition, Prentice Hall, 2021.
2. Donald D Givone. "Digital Principles and Design", Tata McGraw-Hill, 2019.
3. Anand Kumar.A, "Fundamentals of Digital Circuits" Forth Edition, Prentice Hall of India, Pvt Ltd, New Delhi, 2019.
4. Tocci R J, Widmer.N S., "Digital Systems : Principles and applications", Tenth Edition, Pearson Education Pvt. Ltd , 2013.

23U204 LINEAR INTEGRATED CIRCUITS

3 0 0 3

OPERATIONAL AMPLIFIER CHARACTERISTICS: Functional Block Diagram - Symbol - Characteristics of an ideal operational amplifier - Circuit schematic of 741 op-amp - Open loop gain - CMRR - DC characteristics - Frequency response characteristics - stability - frequency compensation — slew rate. (9)

LINEAR APPLICATIONS OF OPERATIONAL AMPLIFIERS: Inverting and Non-inverting amplifiers - Voltage follower - Summing amplifier - Differential amplifier - Instrumentation amplifier: Design and applications - Integrator and Differentiator - Voltage to Current and Current to Voltage converters - Active filters - low pass, high pass, wide band pass, Band stop and notch filter (9)

NON-LINEAR APPLICATIONS OF OPERATIONAL AMPLIFIERS: Comparator - Regenerative comparator - Zero crossing detector - Window detector - Sample and Hold circuit - Precision diode - Half and Full wave rectifiers - Active peak detector - Clipper and Clamper - Logarithmic and Exponential amplifiers - Square and Triangular wave form generators. (9)

VOLTAGE REGULATORS AND SPECIAL FUNCTION ICs: Block diagram of general purpose voltage regulator - Circuit configurations, Fixed and adjustable three terminal regulators - 555 Timer - Functional block diagram and description - Monostable and Astable operation, Applications PLL: Functional Block diagram - VCO- Principle of operation- Applications (9)

SIGNAL CONVERTERS: Digital to Analog Converters - Binary weighted and R-2R Ladder types - Analog to digital converters - Continuous, Counter ramp - Successive approximation, Single slope — Dual slope and Parallel types - Performance characteristics (9)

Total L:45

TEXT BOOKS:

1. Roy Choudhury, ShailJain, "Linear Integrated Circuits", Fifth Edition, New Age International Limited, 2018.
2. Coughlin F R, Driscoll F F, "Operational Amplifiers and Linear Integrated Circuits", Sixth Edition, Pearson Education, Noida, 2016.

REFERENCES:

1. Gayakwad A R, "OP-amps and Linear Integrated circuits", Fourth Edition, Pearson Education, Noida, 2016.
2. Sedra, Smith, "Microelectronic Circuits", Seventh Edition, Oxford University Press, 2017.
3. David A Bell, "Operational Amplifiers and Linear ICs", Third Edition, Prentice Hall of India, 2013.
4. Sergio Franco, "Design with Operational Amplifiers and Analog Integrated Circuits", Fourth Edition, McGraw Hill, 2017

23U210 ANALOG AND DIGITAL ELECTRONICS LABORATORY**0 0 4 2****ANALOG ICS BASED EXPERIMENTS:**

1. Implementation of linear applications of operational amplifier
2. Implementation of Nonlinear applications of operational amplifier
3. Design and implementation of active filters
4. Implementation of voltage regulators
5. Realization of a stable multivibrator using 555 timer

DIGITAL ICS BASED EXPERIMENTS:

1. Realization of Basic Logic Gates using Universal Gates
2. Design and implementation of combinational circuits
3. Design and implementation of sequential circuits
4. Design and implementation of synchronous sequential circuits
5. Design and Implementation of digital functions using FPGA trainer board.

Total P: 60**REFERENCES:**

1. David A Bell, "Laboratory manual for operational amplifiers and Linear ICs", Second Edition, Prentice Hall of India, New Delhi, 2006.
2. Laboratory manual prepared by Department of Instrumentation and Control Systems Engineering, "Laboratory manual", 2023.

23U211 BASIC SCIENCE LABORATORY**0 0 4 2****Physics (Any eight experiments)**

1. Determination of magnetic field along the axis of a coil
2. Determination of Hysteresis loss of a ferromagnetic material
3. Determination of resistivity of metal and alloy using Carey Foster bridge
4. Determination of Temperature Coefficient of Resistance of metallic wire using post office box
5. Study of reverse bias characteristics of Germanium diode and determination of its band gap
6. Study of I-V characteristics of solar cell and determination of its efficiency
7. Thermistor: Measurement of temperature and band gap
8. Hall effect set up - Determination of Hall coefficient
9. Determine e/m (charge to mass ratio) of the electron by Thomson's method.
10. Determine the charge of an electron by Millikan's oil drop method.

Demonstration:

1. Determine the excitation potential of Argon using the Franck Hertz method.
2. Determine the number of turns & radius of the coil and magnetic field of 'Helmholtz' coil using Biot-Savart's Law – apparatus
3. Find the Hall voltage, Hall coefficient, mobility & charge density of the given 'Ge' crystal using Hall apparatus.
4. Determine the Planck's constant using Photo Electric method.
5. Determine the wavelength of the laser source by Michelson Interferometer.
6. Determine the thickness of the film by Mach Zehnder Interferometer.
7. Find the finesse and free spectral region of the Etalon.
8. Analyze the mercury spectrum and find wavelength using polynomial equation.
9. Determine the refractive index of given specimen using Brewster's angle apparatus.

Total P: 30**REFERENCES:**

1. Department of Physics, "Physics laboratory observation", 2022.
2. Jerry D Wilson; Cecilia A Hernandez Hall, "Physics laboratory experiments", Boston, MA : Cengage Learning, 2016.

CHEMISTRY (Any eight experiments)

1. Determination of total hardness of water by EDTA method and determination of water quality - pH, TDS and conductivity.
2. Electroplating of copper and Nickel and determination of cathode efficiency of electroplating.
3. Construction of phase diagram for a simple eutectic system.
4. Demonstration of different forms of metallic corrosion using ferroxyl indicator test and determination of rate of galvanic corrosion using zero-resistance ammeter.
5. Designing a battery and determination of its characteristics.
6. Potentiometric estimation of ferrous iron.
7. Preparation of a chloride ion sensor and ascertaining Nernstian response by construction of calibration curve.
8. Photocolorimetric estimation of Ferric ion in a water sample.

9. Determination of kinematic viscosity of lubricating oil using Redwood viscometer.
10. Determination of flash point and fire point, cloud and pour point of lubricating oil.

Total P:30

REFERENCES:

1. Laboratory manual prepared by the Department of Chemistry, 2023.

23Q213 FOUNDATIONS OF PROBLEM SOLVING

0 0 2 0

PROBLEM SOLVING:

1. Speed mathematics (SAW, Oz, Mirror methods)
2. Speed mathematics (High5, Minion, Butterfly methods)
3. Speed mathematics (Inception, Goldeneye methods)
4. Thinking with numbers
5. Problem solving with visual information
6. Words puzzles
7. Resume writing essentials

Total P: 30

REFERENCE:

1. R S Aggarwal, "Quantitative Aptitude for Competitive Examination", S Chand Publishing, New Delhi, 2017.

SEMESTER - III

23U301 LINEAR ALGEBRA

3 1 0 4

VECTOR SPACES: General vector spaces - real vector spaces - Euclidean n-space - subspaces – linear independence-coordinates and basis - dimension – change of basis - row space, column space and null space – consistency of linear systems - rank and nullity. (9+3)

LINEAR TRANSFORMATIONS: General linear transformation - matrix transformations- geometry of linear operators on R^2 -matrices for general linear transformations - similarity. (9+3)

INNER PRODUCT SPACES: Inner products - angle and orthogonality in inner product spaces, Gram - Schmidt process: QR decomposition, best approximation: least squares - least squares fitting to data. (9+3)

EIGENVALUES AND EIGENVECTORS: Eigenvalues and eigenvectors, diagonalization - orthogonal diagonalization - quadratic forms. (9+3)

MATRIX DECOMPOSITION AND OPTIMIZATION: Spectral decomposition - singular value decomposition - reduced singular value decomposition - optimization using quadratic forms. (9+3)

Total L: 45+T:15 =60

TEXT BOOKS

1. Howard Anton, Chris Rorres, Anton Haul "Elementary Linear Algebra", Wiley India, New Delhi, 2019.
2. David C Lay, Judi J McDonald, Steven R. Lay "Linear Algebra and its Applications", Pearson Education, New Delhi, 2021.

REFERENCES

1. Gareth Williams, "Linear Algebra with Applications", Narosa Publishing House, New Delhi, 2017.
2. Gilbert Strang, "Linear Algebra and Learning from Data", Wellesley-Cambridge Press, USA, 2019.
3. Friedberg, Insel and Spence, "Linear Algebra", Pearson Education, USA, 2015.
4. Steven J Leon, "Linear Algebra with Applications", Pearson Education, USA, 2015

23U302 SENSORS AND TRANSDUCERS

4 0 0 4

SCIENCE OF MEASUREMENT: Units and Standards, General concept and terminology of measurement systems, General input - output configuration , transducer classification , Static and Dynamic characteristics , Calibration techniques (analog and digital) , Statistical analysis of measurement data. (12)

RESISTANCE TRANSDUCERS: Principles of operation, construction details, characteristics of resistance transducers, resistance potentiometers, metal and semiconductor strain gauges. Signal conditioning circuits. Applications of strain gauge for measurement of load and torque (12)

INDUCTIVE TRANSDUCERS: Induction potentiometer, variable reluctance transducers, LVDT, eddy current transducers, synchros and resolvers, magneto - strictive transducers, electromagnetic sensors - associated signal conditioning circuits for above transducers, Proximity sensor. (12)

OTHER SENSORS: Piezo-electric transducer, Piezo resistive sensor, capacitive transducer, Hall effect transducer, tachogenerator, stroboscope, photoelectric transducers, thermocouples, gyroscope, seismic instruments and accelerometers, digital displacement transducers, fibre optic sensor, IC sensor, LIDAR. (12)

INTELLIGENT SENSORS: Smart sensors, Cogent sensors, Virtual sensors, Self adaptive sensors – Adaptation to linearity, Self validating sensors, Temperature compensating sensors – Selection of sensors (12)

Total L: 60

TEXT BOOKS:

1. Doebelin E O and Dhanesh N Manik, "Measurement Systems", Tata McGraw-Hill, Seventh Edition, 2020.
2. John P. Bentley, "Principles of Measurement Systems", Pearson Education, Fourth Edition, 2020.

REFERENCES:

1. Murty D.V.S, "Transducers and Instrumentation", Prentice-Hall of India Private Limited, New Delhi, Second Edition, 2016.
2. James W.Dally, William F.Rileyand, Kenneth G. McConnell, "Instrumentation for Engineering Measurements", Second Edition, Wiley Publishers, 2012.
3. ManabendraBhuyan, "Intelligent Instrumentation: Principles and Applications", CRC Press, 2016.
4. Richards S. Figliola and Donald E Beasley, "Theory and Design for Mechanical Measurements", Seventh Edition, Wiley Publishers, 2019.

23U303 C PROGRAMMING**2 0 0 2**

INTRODUCTION: Analyzing and Defining the Problem - Algorithm - Flow Chart -Types of programming language. C: The C character set - Identifiers and keywords - Data types – Constants - Variables - Declarations -input and output functions- preprocessor directives. (3)

OPERATORS AND EXPRESSIONS: Arithmetic operators - Unary operators - Relational operators - logical operators - Assignment operators - Conditional operators -bitwise operators- comma operator – sizeof operator -precedence and associativity- Library functions. **CONTROL STATEMENTS:**simple if, if..else, nested if .. else ,else..ifladder , switch case - while -do while - for - nested loops - break – continue. (9)

ARRAYS: Defining an array - Processing an array - Multi dimensional arrays -strings. (6)

FUNCTIONS: Function prototype - Defining a function – function call - Passing arguments to a function –nested function – recursive function- Storage classes - auto - static - extern and register variables. (4)

STRUCTURES: Definitions - Processing a structure – Array and structures – Nested structures - Structures and functions.**POINTERS:**Definition - Pointer Arithmetic – types of pointer - const pointer, pointer to a constant, void pointer, null pointer (8)

Total L: 30**TEXT BOOKS:**

1. Paul Deitel and Harvey Deitel, "C How to Program: With an Introduction to C++", Eighth edition, Pearson Education, 2018.
2. Ajay Mittal, "Programming in C - A Practical approach". Pearson, New Delhi, 2010

REFERENCES:

1. Gottfried B., "Programming with C", McGraw Hill Education, New Delhi, 2018.
2. Herbert Schildt, "C: The Complete Reference", McGraw Hill, New Delhi, 2017.
3. Kernighan B. W. and Ritchie D. M., "C Programming Language (ANSI C)", Prentice Hall of India, New Delhi, 2013.
4. ReemaThareja, "PROGRAMMING IN C", Oxford University Press, India,Second edition, 2016

23U304 ELECTRICAL MACHINES**4 0 0 4**

DC MACHINES:Principles of energy conversion - Construction and principle of operation of DC generator - EMF equation - Characteristics of DC generators - Principle of operation of DC motor - Torque equation- Characteristics of DC motors – Starting of DC motors - Speed control of DC motors - Losses and efficiency calculations- Braking-Applications ofDCmachines. (12)

TRANSFORMERS:Construction and principle of operation of single phase transformer- EMF equation- Equivalent circuit - Testing of transformer — Efficiency and voltage regulation - All day efficiency - Operation of auto transformers - Three phase transformer connections - Applications of single phase and three phasetransformer. (12)

SYNCHRONOUS MACHINES:Construction and principle of operation of three phase alternators - EMF equation - Determination of regulation - Theory of operation of synchronous motor - Methods of starting- Operating characteristics: Constant excitation with variable load and constant load with variable excitation –Synchronouscondenser—Applications. (12)

INDUCTION MOTORS : Construction and principle of operation of three phase induction motor - Classification of induction motor –Equivalent circuit - No load and blocked rotor test - Torque production — Torque slip characteristics - Maximum torque - Starting and speed control - Principle of operation of single phase induction motor - Types of single phase induction motors and theirapplications. (12)

SPECIAL MACHINES: Stepper motor — Switched reluctance motor - Universal motor - Brushless DCmotor -Synchros - Servomotor –Linearinductionmotor. (12)

Total L:60**TEXT BOOKS:**

1. Kothari D P and Nagrath I J , "Electric Machines", Fifth Edition, Tata McGraw Hill, 2017.
2. Bimbhra, "Electrical Machinery", Khanna Publishers, 2021.

REFERENCES:

1. Theodore Wildi "Electrical Machines, Drives and Power Systems", Sixth Edition, Pearson Education, 2013.
2. Murugesh Kumar K, "DC Machines and Transformers", Second Edition, Vikas Publishing House, 2016.
3. Bhattacharya S.K, "Electrical Machines", Fourth Edition, Tata McGraw Hill, 2017.
4. Smarajit Ghosh, "Control Systems Theory and Applications", Second Edition, Pearson Education, 2013.

23O305 ENGINEERING ECONOMICS

3 1 0 4

Fundamentals of Economic Analysis: Introduction – Scope of Engineering Economics – Circular Flow in an Economy – Demand and Supply– Types of Efficiency. (7+2)

Cost and Inventory Control: Cost Concepts – Types of Cost – Short-run and Long-run - Cost-Output Relationship - Make or Buy Decision – Criteria – Approaches - Inventory Models – Purchase Model – Manufacturing Model - With or Without Shortages- Depreciation Methods – Problems in Straight Line and Diminishing Balance Method. (9+4)

Pricing, Revenue and Value Engineering: Pricing Practices and Strategies -Revenue Concepts – CVP Analysis- Problems in Break-Even Analysis – Value Analysis - Value Engineering. (9+2)

Project Management: Capital Budgeting – Decisions – Steps Involved in Capital Budgeting – Methods of Project Appraisal – Pay-back Period – Net Present Value and Internal Rate of Return - Project Management - Techniques – PERT – CPM Models – Case Analysis. (10+5)

Economic Growth and Development: Concepts of Macro Economics – National Income – Inflation – Control Measures - Monetary Policy – Fiscal Policy – Technological Innovation in Banking and Economic Development - Sustainable Development Goals – Circular Economy. (10+2)

Total L: 45 + T: 15 =60

Text Books

1. Panneerselvam. R., "Engineering Economics", PHI Learning Private Limited, 2012.
2. Metha P.L, "Managerial Economics – Analysis, Problems and Cases", Sultan Chand & Sons, 2016.

References

1. Zahid A. Khan., Arshad N. Siddiquee, Brajesh Kumar, Mustufa H. Abidi, "Principles of Engineering Economics with Applications", Cambridge University Press, 2018.
2. Varshney, R.L and K.L. Maheshwari, "Managerial Economics", Sultan Chand & Sons, 2014.
3. McEachern and Indira., "Macro ECON", Cengage India Private Limited, 2017.
4. Shalini Goyal Bhalla, "Circular Economy (Re) Emerging Movement", Invincible Publishers, 2020.

23U310 SENSORS AND TRANSDUCERS LABORATORY

0 0 4 2

LIST OF EXPERIMENTS:

1. Determination of weight using strain gauge
2. Measurement of linear displacement using LVDT
3. Characteristics of temperature sensors
4. Measurement of pressure using load cell
5. Measurement of distance using proximity sensor
6. Experimental study of loading effect of potentiometer
7. Implementation of opto-coupler using photoelectric transducers
8. Characteristics of capacitive measurement systems
9. Study of Hall effect sensor and digital transducer
10. Study of vibration measurement using piezoelectric sensor

Total P:60

REFERENCES:

1. John P Bentley, 'Principles of Measurement Systems', Third Edition, Pearson Education, 2015
2. Department of Instrumentation and Control Systems Engineering, "Laboratory Manual", 2023

23U311 C PROGRAMMING LABORATORY

0 0 4 2

Implementation in the following Topics:

1. Working with RAPTOR Tool – Flowchart Interpreter
2. Operators
3. Decision making Statements
4. Loops : while , do..while, for
5. One dimensional array
6. Two dimensional array
7. Strings
8. Functions
9. Recursive functions
10. Structures
11. Structures and arrays.
12. Nested Structures.
13. Pointers

Note: Separate Problem Sheets will be provided during the course

Total P: 60

REFERENCES:

1. Paul Deitel and Harvey Deitel, "C How to Program: With an Introduction to C++", Eighth edition, Pearson Education, 2018.
2. Ajay Mittal, "Programming in C - A Practical approach". Pearson, New Delhi, 2010
3. Gottfried B., "Programming with C", McGraw Hill Education, New Delhi, 2018.
4. Herbert Schildt, "C: The Complete Reference", McGraw Hill, New Delhi, 2017.
5. Kernighan B. W. and Ritchie D. M., "C Programming Language (ANSI C)", Prentice Hall of India, New Delhi, 2013

23Q313 BUILDING COMMUNICATION SKILLS**0 0 2 1****BUILDING COMMUNICATION SKILLS:**

1. Introduction to workplace communication
2. Profile building for internships
3. English in the workplace (grammar and vocabulary)
4. Professional communication (speaking and writing)
5. Workplace communication tools
6. Career exploration
7. Resume update

Total P: 30**REFERENCES:**

1. P C Wren and H Martin, "High School English Grammar and Composition", S Chand Publishing, New Delhi, 2017.
2. Norman Lewis, "Word Power Made Easy", Goyal Publisher, New Delhi, 2011

23K312 ENVIRONMENTAL SCIENCE**2 0 0 0**

INTRODUCTION TO ENVIRONMENT: Environment-Definition, scope and importance. Types and composition of atmosphere — particles, ions and radicals. Ozone layer- significance, formation and depletion. Ecosystems- Structure and functions, components, energy flow, food chains, food web, Biodiversity-levels, values and threats— India as a mega-diversity nation — hotspots of biodiversity — endangered and endemic species of India — conservation of biodiversity. (6)

ENERGY RESOURCES: Introduction — National and International status- exploitation - sustainable strategies- Fossil fuels- classification, composition, physico-chemical characteristics and energy content of coal, petroleum and natural gas; solar energy - introduction, harnessing strategies. Wind energy - availability, wind power plants, wind energy conversion systems, site characteristics, and types of wind turbines. Supporting renewable energy resources - tidal-geothermal-hydroelectric. (6)

ENVIRONMENTAL POLLUTION: Definition—Sources, causes, impacts and control measures of (a) Air pollution (b) Water pollution (c) Soil pollution (d) Marine pollution (e) Noise pollution (f) Thermal pollution (g) Nuclear hazards (h) RF hazards - Role of an individual in prevention of pollution. **DISASTER MANAGEMENT:** Floods, earthquake, cyclone and landslides—Case studies, consequences and rescue measures (6)

WASTE MANAGEMENT: Waste water- Characteristics of domestic and industrial waste water - COD and BOD Various stages of treatment – primary, secondary, tertiary treatment – Biological and advanced oxidation processes. Solid waste management – Characteristics of municipal solid waste (MSW), biomedical, automobile and e-wastes and their management – landfills, incineration, pyrolysis, gasification and composting. (6)

SOCIAL ISSUES AND THE ENVIRONMENT: Environmentally Sustainable work practices – Rainwater harvesting– Role of non-governmental organizations. Human ethics and rights-impact on environment and human health – role of information technology on environment and human kind. Green IT policies, Process of EIA - ISO 14000. Legislation- Environment protection act – Air (Prevention and Control of Pollution) act – Water (Prevention and control of Pollution) act– Wildlife protection act– Forest conservation act. (6)

Total L: 30**TEXT BOOKS:**

1. Gilbert M. Masters, "Introduction to Environmental Engineering and Science", Pearson Education, New Delhi, 2004.
2. DeA K, "Environmental Chemistry", New Age International P Ltd, New Delhi, 2006.

REFERENCES:

1. Benny Joseph, "Environmental Science and Engineering", Tata McGraw-Hill, New Delhi, 2006.
2. Koteswara Rao MVR, "Energy Resources: Conventional & Non-Conventional", BSP Publications, New Delhi, 2006.
3. Deswal S and Deswal A, "A Basic Course in Environmental Studies", Dhanpat Rai and Co, New Delhi, 2004.

23TC01 தமிழ்மரபு / HERITAGE OF TAMILS**1 0 0 1****அலகு - I மொழி மற்றும் இலக்கியம்:****3**

இந்திய மொழிக்கும் பங்கள் – திராவிட மொழிகள் – தமிழ் ஒரு செம்மொழி – தமிழ் செவ்விலக்கியங்கள் – சங்க இலக்கியத்தின் சமயச் சார்பற்ற தன்மை – சங்க இலக்கியத்தில்கிர்தல் அறம் – திருக்குறளில் மேலாண்மைக்கருத்துக்கள் – தமிழ்க் காப்பியங்கள், தமிழகத்தில் சமணப் பௌத்த சமயங்களின் தாக்கம் – பக்தி இலக்கியம், ஆழ்வார்கள் மற்றும் நாயன்மார்கள் – சிற்றிலக்கியங்கள் –

தமிழில்நவீனஇலக்கியத்தின்வளர்ச்சி – தமிழ் இலக்கிய வளர்ச்சியில் பாரதியார் மற்றும் பாரதிதாசன் ஆகியோரின் பங்களிப்பு

அலகு – II மரபு – பாறைஓவியங்கள்முதல்நவீனஓவியங்கள்வரை – சிற்பக்கலை: 3
நடுகல்முதல்நவீனசிற்பங்கள்வரை – ஐம்பொன்சிலைகள் –
பழங்குடியினர்மற்றும்அவர்கள்தயாரிக்கும்கைவினைப்பொருட்கள், பொம்மைகள் –
தேர்செய்யும்கலை – சுடுமண்சிற்பங்கள் – நாட்டுப்புறத்தெய்வங்கள் –
குமரிமுனையில்திருவள்ளுவர்சிலை – இசைக்கருவிகள் – மிருதங்கம், பறை, வீணை, யாழ்,
நாதஸ்வரம் – தமிழர்களின்சமூகபொருளாதாரவாழ்வியல்கோவில்களின்பங்கு.

அலகு – III நாட்டுப்புறக்கலைகள்மற்றும்வீரவிளையாட்டுகள்: 3
தெருக்கூத்து, கரகாட்டம், வில்லுப்பாட்டு, கணியான்கூத்து, ஓயிலாட்டம், தோல்பாவைக்கூத்து,
சிலம்பாட்டம், வளரி, புலியாட்டம், தமிழர்களின்விளையாட்டுகள்.

அலகு - IV தமிழர்களின்திணைக்கோட்பாடுகள்: 3
தமிழகத்தின்தாவரங்களும், விலங்குகளும் –
தொல்காப்பியம்மற்றும்சங்கஇலக்கியத்தில்அகம்மற்றும்புறக்கோட்பாடுகள் –
தமிழர்கள்போற்றியஅறக்கோட்பாடு – சங்ககாலத்தில்தமிழகத்தில்எழுத்தறிவும், கல்வியும் –
சங்ககாலநகரங்களும்துறைமுகங்களும் – சங்ககாலத்தில்ஏற்றுமதிமற்றும்இறக்குமதி –
கடல்கடந்தநாடுகளில்சோழர்களின்வெற்றி.

அலகு -V இந்தியதேசியஇயக்கம்மற்றும்இந்தியபண்பாட்டிற்குத்தமிழர்களின்பங்களிப்பு: 3
இந்திவிடுதலைப்போரில்தமிழர்களின்பங்கு –
இந்தியாவின்பிறப்பகுதிகளில்தமிழ்ப்பண்பாட்டின்தாக்கம் – சுயமரியாதைஇயக்கம் –
இந்தியமருத்துவத்தில், சித்தமருத்துவத்தின்பங்கு – கல்வெட்டுகள், கையெழுத்துப்படிக்கள் –
தமிழ்ப்புத்தகங்களின்அச்சுவரலாறு

TOTAL : 15 PERIODS

TEXT-CUM-REFERENCE BOOKS

1. தமிழகவரலாறு – மக்களும்பண்பாடும் – கே.கே.பிள்ளை (வெளியீடு – தமிழ்நாடுபாடநூல்மற்றும்கல்வியியல்பணிகள்கழகம்).
2. கணினித் தமிழ் – முனைவர் இல.சுந்தரம். (விகடன் பிரசுரம்)
3. கீழடி – வைகை நதிக்கரையில் சங்ககால நகர நாகரிகம் (தொல்லியல் துறை வெளியீடு)
4. பொருளை – ஆற்றங்கரை நாகரிகம். (தொல்லியல் துறை வெளியீடு)
5. Social Life of Tamils (Dr.K.K.Pillay) A joint publication of TNTB & ESC and RMRL – (in print)
6. Social Life of the Tamils – The Classical Period (Dr.S.Singaravelu) (Published by: International Institute of Tamil Studies.)
7. Historical Heritage of the Tamils (Dr.S.V.Subatamanian, Dr.K.D.Thirunavukkarasu) (Published by : International Institute of Tamil Studies).
8. The Contributions of the Tamils to Indian Culture (Dr.M.Valarmathi) (Published by: International Institute of Tamil Studies).
9. Keeladi – ‘Sangam City Civilization on the banks of river Vaigai’ (Jointly Published by : Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
10. Studies in the History of India with Special Reference to Tamil Nadu (Dr.K.K.Pillay) (Published by : The Author)
11. Porunai Civilization (Jointly Published by : Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
12. Journey of Civilization Indus to Vaigal (R.Balakrishnan) (Published by : RMRL) – Reference Book.

SEMESTER - IV

23U401 PROBABILITY, STOCHASTIC PROCESSES AND STATISTICS

3 1 0 4

PROBABILITY AND DISCRETE RANDOM VARIABLES: Probability, axioms, conditional probability, law of total probability, Baye's theorem, discrete random variables, probability mass function,cumulative distribution function, binomial, Poisson and geometric random variables, expected values. (9+3)

CONTINUOUS RANDOM VARIABLES: Cumulative distribution function, probability density function, uniform, exponential and Gaussian random variables, expected values. (9+3)

PAIRS OF RANDOM VARIABLES: Joint cumulative distribution function, joint probability mass function, marginal probability mass function, joint probability density function, marginal probability density function, independent random variables, expected values, covariance, correlation and independence, central limit theorem. (9+3)

STOCHASTIC PROCESSES: Types of stochastic processes, Poisson process, discrete time Markov chains, classification of states, limiting state probabilities. (9+3)

STATISTICAL INFERENCE: Point estimation, interval estimation. Hypothesis testing : one and two tailed tests, tests concerning mean, proportion, and variance, single and two samples, tests for goodness of fit and independence of attributes. (9+3)

Total L: 45 + T: 15 = 60

TEXT BOOKS

1. Roy D Yates and David J Goodman, "Probability and Stochastic Processes", Wiley India, New Delhi, 2021.
2. Ronald E. W, Raymond H. M, Sharon L. M and Keying Ye, "Probability and Statistics for Engineers and Scientists", Pearson Education, New Delhi, 2016.

REFERENCES

1. Saeed Ghahramani, "Fundamentals of Probability with Stochastic Processes", CRC Press, USA, 2018.
2. Douglas C Montgomery and George C Runger, "Applied Statistics and Probability for Engineers", Wiley India, New Delhi, 2018.
3. Athanasios P and Unnikrishna P. S, "Probability, Random Variables and Stochastic Processes", Tata McGraw Hill, New Delhi, 2017.
4. Arnold O. A, "Probability, Statistics and Queueing theory: with computer science applications", Academic press, USA, 2014.

23U402 ELECTRICAL AND ELECTRONIC MEASUREMENTS

3 0 0 3

MEASUREMENT ERROR ANALYSIS: Errors in measurement- types of errors - measurement error combinations- probability of errors: normal distribution of errors- probable error- limiting error- propagation of errors- linear and weighted regression- uncertainty analysis. (9)

ANALOG INDICATING INSTRUMENTS: Types of ammeters and voltmeters- PMMC Instruments -Moving Iron Instruments - Dynamometer type Instruments. Measurement of power and energy- Dynamometer type wattmeter - Single phase Induction type energy meter, Calibration of meters, Instrument Transformers: Ratio and phase angle errors, Phasor diagram. (9)

MEASUREMENT OF R, L and C: Classification of resistances, Measurement of resistances: Ammeter Voltmeter method, Substitution method, Wheatstone bridge, Kelvin's double bridge method, General equation for AC bridge balance – Maxwell's bridge–Wien's bridge–Hay's bridge–Schering bridge – Anderson bridge. (9)

TEST AND MEASURING INSTRUMENTS: Evolution of test and measuring instruments, Digital voltmeter of Ramp and Integrating types, Digital multimeter, Digital Storage Oscilloscope, Mixed Signal Oscilloscope, Waveform analysers– Distortion meter –Power analyzer - Spectrum Analyzer, Signal generators. (9)

ELECTROMAGNETIC COUPLING: Introduction, Interference coupling mechanism, basics of circuit layout and grounding, concepts of interfaces, filtering and shielding - Safety: Introduction, electrical hazards, hazardous areas and classification - Non hazardous areas, enclosures — NEMA types, fuses and circuit breakers -Protection methods: purging, explosion proofing and intrinsic safety. (9)

Total L:45

TEXT BOOKS:

1. Golding E W, Widdis F C, "Electrical Measurements and Measuring Instruments", Third Edition, Reem Publications, New Delhi, 2011.
2. David A Bell, "Electronic Instrumentation and Measurements", Oxford Publisher, 2017.

REFERENCES:

1. Kalsi H S, "Electronic Instrumentation", Third Edition, Tata McGraw-Hill, New Delhi, 2017.
2. Sawhney A K, "A Course in Electrical and Electronic Measurement and Instrumentation", Dhanpat Rai And Sons, New Delhi, 2017.
3. Albert D, Helfrick, William D Cooper, "Modern Electronic Instrumentation and Measurement Techniques", Pearson Education, New Delhi, 2016.
4. Rangan C S, Sharma G R, Mani V S, "Instrumentation Devices and Systems", McGraw Hill Education, Chennai, 2017.

23U403 CONTROL SYSTEMS

3 1 0 4

INTRODUCTION AND MODELING OF CONTROL SYSTEMS: Basic Components of Control Systems – Types of Control Systems – Types of Feedback Control Schemes – Modeling of dynamic systems – Mechanical systems, Electrical systems – Analogies - Linearization of Non-linear systems - Block diagram model – Block diagram and transfer function signal flow graph – Application of Mason's gain formula – Conversion between block diagrams to signal flow graphs (9+3)

ANALYSIS AND DESIGN IN TIME DOMAIN: Introduction - Standard test signals - Time response of prototype first order and second order systems – Specification of step response - Analysis of steady state error: static error constants - Concepts of

stability: absolute and relative stability – Routh-Hurwitz criterion. Introduction – Root locus analysis – Design using root locus Plots – Control system design – Compensators (9+3)

ANALYSIS AND DESIGN IN FREQUENCY DOMAIN: Introduction – Frequency response analysis – Frequency response specifications – Frequency response plots - Bode plot, Gain margin, Phase margin - Nyquist stability criterion - Stability analysis - Relative stability analysis - Control system design by frequency response analysis– Compensators (9+3)

MATHEMATICAL MODELING IN STATE SPACE: State variable representation: State variable concepts, State variable modeling - Modeling of mechanical, Electrical – State diagrams – Transformation of state variables: Similarity Transformation, Invariance property, Conversion of transfer functions to canonical state variable models - Conversion of state variable model to transfer functions (9+3)

ANALYSIS AND DESIGN IN STATE SPACE: Characteristic equation - Eigen values and Eigen vectors - Solution of state equation – State transition matrix - Concepts of controllability and observability - Controllability tests – Observability tests – Controllable and observable companion forms – Duality– Pole placement design – Design of state observers: Full order and reduced order observers. (9+3)

Total L: 45 + T: 15 = 60

TEXT BOOKS:

1. M. Gopal, "Control Systems: Principles and Design", Fourth Edition, Tata McGraw Hill, New Delhi, 2012
2. K Ogata, "Modern Control Engineering", Pearson, Fifth Edition, 2018

REFERENCES:

1. William J Palm III, "System Dynamics", Second Edition, Tata McGraw Hill, 2012
2. Norman S Nise, 'Control Systems Engineering', Sixth Edition, John Wiley & Sons, 2011.
3. Farid Galnarghi & Benjamin C. Kuo, "Automatic Control System", Ninth Edition, John Wiley & Sons, Inc, 2010
4. Anand Natarajan & P Ramesh Babu, "Control Systems Engineering", Second Edition, Scitech, 2010,

23U404 MICROPROCESSORS AND MICROCONTROLLERS

4 0 0 4

MICROPROCESSOR: Architecture of 8-bit microprocessor, bus configurations, Instruction classifications, data format and storage – Overview of 8085 Instruction set – Programming exercise – Instruction execution time. (12)

MEMORY AND I/O INTERFACING: Interface requirements – Address space partitioning – Buffering of Buses Memory control signals – Typical EPROM, RAM Interfacing. I/O interfacing: Memory mapped I/O scheme – I/O mapped I/O scheme – I/O ports – Programmable peripheral interface. (12)

MICROCONTROLLER: Microprocessor versus Microcontroller – 8051 Microcontroller Block diagram – Internal Data RAM – Special Function Registers – Internal register banks and stack – Addressing mode and Instruction set – 8051 Assembler and Intel Hex file format. (12)

ON-CHIP PERIPHERALS AND INTERFACING: Input / Output Ports structure - Counter and Timers, Serial Data Input / Output - Interrupts, LCD, LED and Keyboard Interfacing, ADC, DAC and Sensor interfacing, interfacing to external memory. (12)

APPLICATIONS AND PROGRAMMING IN C : Stepper and DC motor interfacing, 8051 programming in C: Data types, I/O programming, Logic operation, Accessing code ROM space. (12)

Total L: 60

TEXT BOOKS:

1. Ramesh S Goankar, "Microprocessor Architecture, Programming and Applications with the 8085", Sixth Edition, Penram International, 2013.
2. Muhammad Ali Mazidi, Janice Gillispie Mazidi, Rolin D Mckinlay, "8051 Microcontroller and Embedded Systems", Second Edition, Pearson Education, 2022.

REFERENCES:

1. Kenneth Ayala, "The 8051 Microcontroller", Third Edition, Thomson Delmar Learning, 2016.
2. Krishna Kant, "Microprocessors and Microcontrollers: Architecture, Programming and System Design 8085, 8086, 8051, 8096", Second Edition, PHI Learning Pvt. Ltd., 2016
3. Douglas V Hall, "Microprocessor and Interfacing : Programming and Hardware", Third Edition, McGraw Hill Inc, 2013.
4. Subrata Ghosal, "8051 Microcontroller: Internals, Instructions, Programming and Interfacing", Pearson Education India, 2010.

23U405 DATA STRUCTURES AND ALGORITHMS

2 2 0 4

ALGORITHM ANALYSIS: Algorithms- properties — Recursive Algorithms- Analysis of Algorithms- Best case, Average case, Worst case – Asymptotic Notations. Data types – Abstract data types – Types of Data structures (6+6)

LINEAR DATA STRUCTURES: Arrays-operations – Memory Representation- Row Major and Column Major – Multi Dimensional Arrays – Sparse Matrix, Dense Matrix. Stack: Array implementation – operations-Applications – Checking of well-formedness of Parenthesis- Infix to Postfix Conversions. (6+6)

QUEUES AND LISTS: QUEUES: Queue Operations- Circular Queue - Array Implementation of Queue, LINKED LIST: Types-

Singly Linked List – Circularly Linked List – Doubly Linked List–List operations-linked stack-linked queue (6+6)

NONLINEAR DATA STRUCTURES: Trees-Terminologies - Binary trees – Representations – Operations – Traversals- Inorder, Preorder and Postorder- Binary Search Trees – Insertion and deletion. Graph: Terminologies -Breadth First Search algorithm-Depth First Search Algorithm. (6+6)

SORTING AND SEARCHING: Bubble Sort – Insertion Sort – Radix Sort- Quick sort- Algorithms and Time Complexity. Linear Search – Binary Search – Hashing: Hash functions – Separate Chaining – Open Addressing – Linear Probing. (6+6)

Total L: 30 + T: 30 = 60

TEXT BOOKS:

1. Thomas H Cormen, Charles E Leiserson, Ronald L Rivest, Clifford Stein, "Introduction to Algorithms", The MIT Press , 2022.
2. Mark Allen Weiss, "Data Structures and Algorithm Analysis in C++", Pearson Education , 2012.

REFERENCES:

1. Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran. "Fundamentals of Computer Algorithms" Second Universities Press, 2011.
2. SahniSartaj. "Data Structures, Algorithms and Applications in C++",Silicon Press , 2009.
3. Aaron M Tanenbaum, Moshe J Augenstein, YedidyahLangsam. "Data structures using C and C++", PHI Learning , 2009.
4. G A V VijayalakshmiPai. "DataStructures and Algorithms Concepts, Techniques and Allocations", New Delhi: McGraw Hill Education(India) Private Limited , 2015.

23U410 MICROPROCESSORS AND MICROCONTROLLERS LABORATORY

0 0 4 2

LIST OF EXPERIMENTS:

1. Familiarization of 8085 microprocessor
2. Arithmetic operations using 8085 microprocessor
3. Code conversion using 8085 microprocessor
4. Sorting and searching using 8085
5. Programmable Peripheral Interfacing
6. Arithmetic operations using 8051 microcontroller
7. Logical and Bit manipulation using 8051 microcontroller
8. Generation of delay using on-chip timer/counter and serial port
9. ADC and DAC interfacing
10. Stepper and DC motor control interface to 8051.

Total P:60

REFERENCES:

1. Mukhopadhyay A. K., "Microprocessor-Based Laboratory Experiments and Projects", Third Edition, I K International Publishing House Pvt. Ltd,2010.
2. Department of Instrumentation and Control Systems Engineering, "Laboratory Manual",2023.

23U411 MEASUREMENT AND CONTROL LABORATORY

0 0 4 2

LIST OF EXPERIMENTS:

1. Measurement of resistance using Wheatstone bridge and Kelvin's Doublebridge
2. Measurement of capacitance and inductance using Schering Bridge and AndersonBridge
3. Extension of instrumentranges
4. Calibration of analog instruments
5. Calibration of digital instruments
6. Time response of first order system
7. Time response of second order system
8. Study of frequency response analysis of a system
9. Empirical modeling of real time systems
- 10.Design of compensation networks

Total P:60

REFERENCES:

1. Golding E W, WiddisFC, "Electrical Measurements and Measuring Instruments", Reem Publications, New Delhi,2011.
2. K Ogata, "Modern Control Engineering", Pearson, Fifth Edition,2018
3. Departmentof Instrumentation and Control Systems Engineering, "Laboratory manual",2023.

23Q413 PROBLEM SOLVING

0 0 2 1

1. Algorithmic thinking, branching and repetition problems
2. Logical reasoning - data arrangements and relations
3. Solving problems based on coding and decoding, series, analogy, odd man out and visual reasoning
4. Problems based on ages, logical connectives, syllogisms, data interpretation and data sufficiency
5. Solving problems on clocks calendars, direction sense and cubes
6. Problems based on number system, percentages, simple and compound interest
7. Resume update

Total P: 30

REFERENCE:

1. R S Aggarwal, "Quantitative Aptitude for Competitive Examination", S Chand Publishing, New Delhi, 2017

23O414 INDIAN CONSTITUTION

2000

INTRODUCTION: Evolution of Indian Constitution; significance of constitution; Composition; Preamble and its Philosophy. (4)

RIGHTS, DUTIES AND DIRECTIVE PRINCIPLES: Fundamental Rights- Writs and Duties, Directive Principles of State Policy (5)

UNION GOVERNMENT: Union Government, President and Vice President, Houses of the Parliament and their functions; Types of Bills, Stages of passing of Bill into an Act, Veto Power, Constitution Amendment Procedure, Various Amendments made and their significance for India. (6)

STATE GOVERNMENT AND FEDERALISM: Composition of State Legislature; Powers, Functions and Position of Governor, Function of Chief Ministers, Council of Ministers; The Indian Federal System, Administrative Relationship between Union and States (8)

JUDICIARY: Supreme Court, High Court; District Court and Lower Courts - Functions and Powers – Judges – Qualifications and Powers - Judicial Review. (7)

Total L:30

TEXT BOOKS:

1. Subash C. Kashyap, "Our Political System", National Book Trust, 2011.
2. Praveenkumar Mellali, E., "Constitution of India, Professional Ethics and Human Rights", Sage Publications India Pvt. Ltd., 2015.

REFERENCES:

1. Brijji Kishore Sharma, "Introduction to the Constitution of India", Prentice Hall of India, 2010.
2. Basu D.D., "Introduction to the Constitution of India", Prentice Hall of India, 2016.
3. Jain. M. C., "The Constitution of India", Law House, New Delhi, 2001.
4. Shukla. V. N., "Constitution of India", Eastern Book Company Ltd., New Delhi, 2011.

23TC02தமிழரும் தொழில்நுட்பமும் / TAMILS AND TECHNOLOGY

1001

அலகு – I நெசவு மற்றும் பாணைத் தொழில்நுட்பம்:

3

சங்க காலத்தில் நெசவுத் தொழில் – பாணைத் தொழில்நுட்பம் – கருப்பு சிவப்பு பாண்டங்கள் – பாண்டங்களில் கீறல் குறியீடுகள்.

அலகு – II வடிவமைப்பு மற்றும் கட்டிடத் தொழில்நுட்பம்:

3

சங்க காலத்தில் வடிவமைப்பு மற்றும் கட்டுமானங்கள், சங்க காலத்தில் வீட்டுப் பொருட்களில் வடிவமைப்பு – சங்க காலத்தில் கட்டுமான பொருட்களும் நடுகல்லும் – சிலப்பதிகாரத்தில் மேடை அமைப்பு பற்றிய விவரங்கள் – மாமல்லபுரச் சிற்பங்களும், கோவில்களும் – சோழர் காலத்துப் பெருங்கோயில்கள் மற்றும் பிற வழிபாட்டுத் தலங்கள் – நாயக்கர் காலக் கோயில்கள் – மாதிரி கட்டமைப்புகள் பற்றி அறிதல், மதுரை மீனாட்சி அம்மன் ஆலயம் மற்றும் திருமலை நாயக்கர் மஹால் – செட்டிநாட்டு வீடுகள் – பிரிட்டிஷ் காலத்தில் சென்னையில் இந்தோ-சாரோசெனிக் கட்டிடக் கலை.

அலகு – III உற்பத்தித் தொழில் நுட்பம்:

3

கப்பல் கட்டும் கலை – உலோகவியல் – இரும்புத் தொழிற்சாலை – இரும்பை உருக்குதல், எஃகு – வரலாற்றுச் சான்றுகளாக செம்பு மற்றும் தங்க நாணங்கள் – நாணயங்கள் அச்சடித்தல்- மணி உருவாக்கும் தொழிற்சாலைகள் – கல்மணிகள், கண்ணாடி மணிகள் – சுடுமண் மணிகள் – சங்கு மணிகள் – எலும்புத் துண்டுகள் – தொல்லியல் சான்றுகள் – சிலப்பதிகாரத்தில் மணிகளின் வகைகள்.

அலகு – IV வேளாண்மை மற்றும் நீர்ப்பாசனத் தொழில் நுட்பம்:

3

அணை, ஏரி, குளங்கள், மதகு – சோழர்காலக் குழுவித் தூம்பின் முக்கியத்துவம் – கால்நடை பராமரிப்பு – கால்நடைகளுக்காக வடிவமைக்கப்பட்ட கிணறுகள் – வேளாண்மை மற்றும் வேளாண்மைச் சார்ந்த செயல்பாடுகள் – கடல்சார் அறிவு – மீன்வளம் – முத்து மற்றும் முத்துக்குளித்தல் – பெருங்கடல் குறித்த பண்டைய அறிவு – அறிவுசார் சமூகம்.

அலகு V – அறிவியல் தமிழ் மற்றும் கணித்தமிழ்:

3

அறிவியல் தமிழின் வளர்ச்சி – கணித்தமிழ் வளர்ச்சி – தமிழ் நூல்களை மின்பதிப்பு செய்தல் – தமிழ் மென்பொருட்கள் உருவாக்கம் – தமிழ் இணையக் கல்விக்கழகம் – தமிழ் மின் நூலகம் – இணையத்தில் தமிழ் அகராதிகள் – சொற்குவைத் திட்டம்.

TOTAL : 15 PERIODS

TEXT-CUM-REFERENCE BOOKS

1. தமிழக வரலாறு – மக்களும் பண்பாடும் – கே.கே.பிள்ளை (வெளியீடு: தமிழ்நாடு பாடநூல் மற்றும் கல்வியியல் பணிகள் கழகம்).
2. கணினித் தமிழ் – முனைவர் இல.சுந்தரம் (விகடன் பிரசுரம்).
3. கீழடி – வைகை நதிக்கரையில் சங்ககால நகர நாகரிகம் (தொல்லியல் துறை வெளியீடு)
4. பொருறை – ஆற்றங்கரை நாகரிகம். (தொல்லியல் துறை வெளியீடு)
5. Social Life of Tamils (Dr.K.K.Pillay) A joint publication of TNTB & ESC and RMRL – (in print)
6. Social Life of the Tamils – The Classical Period (Dr.S.Singaravelu) (Published by: International Institute of Tamil Studies.)
7. Historical Heritage of the Tamils (Dr.S.V.Subatamanian, Dr.K.D.Thirunavukkarasu) (Published by : International Institute of Tamil Studies).
8. The Contributions of the Tamils to Indian Culture (Dr.M.Valarmathi) (Published by: International Institute of Tamil Studies).
9. Keeladi – ‘Sangam City Civilization on the banks of river Vaigai’ (Jointly Published by : Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
10. Studies in the History of India with Special Reference to Tamil Nadu (Dr.K.K.Pillay) (Published by : The Author)
11. Porunai Civilization (Jointly Published by : Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
12. Journey of Civilization Indus to Vaigal (R.Balakrishnan) (Published by : RMRL) – Reference Book.

SEMESTER - V

23U501 INDUSTRIAL INSTRUMENTATION

3 0 0 3

PRESSURE MEASUREMENT: Terminologies – Units - Manometers-Different Types-Bourdon tube-Bellows- Diaphragms – Electrical Methods: Elastic elements with LVDT and strain gauges- Capacitive type pressure gauge - Piezo resistive pressure sensors-Low pressure measurement: McLeod gauge-Thermal conductivity gauge-Ionization gauge - Dead weight tester (8)

TEMPERATURE MEASUREMENT: Units - Filled-in systems - Bimetallic thermometer – Electrical methods; RTD and compensation techniques – Thermistors and its linearization - Thermocouples: laws, burnout detection, cold junction compensation – Thermopile- IC Temperature sensors – Radiation methods: Pyrometers and its types, Fibre optic temperature measurement (9)

FLOW MEASUREMENT: Units -Variable head type: Orifice plate -Venturi tube- Flow nozzle- Pitot tube– Variable area meter: Rotameter - Positive displacement flow meters: Reciprocating piston, oval gear and helix type flow meters-Turbine flow meter- Mass flow meters: Angular momentum mass flow meter, Coriolis mass flow meter, Thermal mass flow meter- Electromagnetic flow meter-Ultrasonic flow meters-Laser Doppler anemometer system-Vortex shedding flow meters - Target flow meter. (13)

LEVEL MEASUREMENT: Sight glass -Float type level indication- Level switches – Air purge system - Differential pressure method – Electrical methods of level measurement using resistance, capacitance, nuclear radiation and ultrasonic sensors. (7)

MEASUREMENT OF VISCOSITY, HUMIDITY AND MOISTURE: Viscosity measurement: Say bolt viscometer, Rotameter type viscometer – Humidity terms – dry and wet bulb psychrometers-hotwire electrode type hygrometer-dew cell – electrolysis type hygrometer – moisture measurement – different types of moisture measurement – Electronic recorders – Introduction to pneumatic instruments. (8)

Total L:45

TEXT BOOKS:

1. Walt Boyes, 'Instrumentation Reference Book', Elsevier Science, 2010.
2. Donald P Eckman, 'Industrial Instrumentation', CBS Publisher, 2004.

REFERENCES:

1. Liptak B.G, "Process Measurement and Analysis", Fourth Edition, Chilton Book Company, 2003.
2. William C Dunn, 'Fundamentals of Industrial Instrumentation and Process Control, Second Edition', McGraw-Hill Education, 2018.
3. B. E. Noltingk, 'Instrument Technology: Mechanical measurements', Fourth Edition, Butterworths, 2016
4. John P Bentley, "Principles of Measurement Systems", Fourth Edition, Pearson Education, 2020.

23U502 DIGITAL SIGNAL PROCESSING

3 0 0 3

DISCRETE TIME SIGNALS AND SYSTEMS: Motivation — Sampling and Quantization of signal-Discrete time (DT) sequences —Different representations of DT sequence-Operations on DT sequence - LTI system - properties —Linear and circular convolution-correlation. (9)

TRANSFORMS: Discrete Fourier Transform – Properties – Fast Fourier Transform, Decimation in Time FFT algorithm- Decimation in Frequency FFT algorithm- Frequency response of LTI system. (9)

LTI SYSTEMS AS FREQUENCY SELECTIVE FILTERS: Filter Characteristics-Digital resonators – Notch filters – Comb filters – All pass filters – Moving Average filters. (9)

FIR FILTERS: Characteristics, symmetry, linear phase and types – Design of FIR filter using windowing technique Frequency sampling technique- introduction to optimal FIR filter design- Realization of FIR filter. (9)

IIR FILTERS AND FINITE WORD LENGTH EFFECT: Design of analog prototype filter–Types of IIR filter–Frequency transformation–Impulse Invariant technique–Bilinear transformation technique – Realization of IIR filter- Effect of coefficient quantization-Effect of round-off noise in digital filters-Limit cycle due to round-off and truncation. (9)

Total L: 45

TEXT BOOKS:

1. John G. Proakis, Dimitris G. Manolakis, "Digital Signal Processing, Principles, Algorithms and Applications", Fifth edition Pearson, 2021.
2. Lonnie C. Ludeman, "Fundamentals of Digital Signal Processing", Wiley, 2009.

REFERENCES:

1. Alan V. Oppenheim, Ronald W. Schaffer, "Discrete Time Signal Processing", Pearson, 2014.
2. Sanjit K. Mitra, "Digital Signal Processing: A Computer based Approach", McGraw Hill Education, 2014.
3. Sophocles Orfanidis, "Introduction to Signal Processing", Prentice Hall, 2009.
4. Boaz Porat, "A Course in Digital Signal Processing", Wiley, 2012.

23U503 DIGITAL CONTROL SYSTEMS

3 1 0 4

INTRODUCTION TO DIGITAL CONTROL: Industrial and Embedded Control – Digital Computer as a Compensator Device - Advantages of Digital Computers –Implementation Problems in Digital Control – Digital Implementation of Analog Compensators – Sampling rate selection – Limits of Digital Implementation of Analog Compensators - Modeling the Digital Computer, Sampler, Zero-order Hold, Z-Transform, Inverse Z-Transform (9+3)

TRANSFER FUNCTION MODELS AND DYNAMIC RESPONSE: Model of a Plant – Model of Closed Loop Sampled Data System – Dynamic Response – S-plane to Z-plane mapping - Transfer Functions – Block Diagram Reduction - Pulse Transfer function of the Feedback System – Stability on the Z-plane – Bilinear Transformation – Steady State Errors – Design using Root Locus Plots – Design using Frequency Response Plots - Implementation of Digital Compensator (9+3)

STATE SPACE REPRESENTATIONS OF DISCRETE TIME SYSTEMS: Solution of The Time - Invariant Discrete- Time State Equation, State Transition Matrix, Z-Transform Approach to The Solution of State Equation, Discretization of Continuous- Time State- Space Equations , Controllability and Observability of Discrete- Time Systems, Conditions, Principle of Duality. (9+3)

DESIGN VIA POLE PLACEMENT: Necessary and Sufficient Condition for Pole Placement, Ackerman's Formula, State Observers – Necessary and Sufficient Condition for State Observation. Full Order State Observer, Error Dynamics of the Full Order State Observer, Design of Prediction Observer. (9+3)

DESIGN OF MINIMUM-ORDER OBSERVER: State Feedback Control System with Minimum- Order Observer. Diophantine Equation, Polynomial Equation approach to Control System Design. Design of Model Matching Control Systems. (9+3)

Total L: 45 +T: 15 =60

TEXT BOOKS:

1. Norman S. Nise, "Control Systems Engineering", Sixth Edition, John Wiley and Sons, 2011.
2. B. C. Kuo, "Digital Control Systems", Second Edition, Oxford Univ Press, Inc., 1992.

REFERENCES:

1. F. Franklin, J.D. Powell, and M.L. Workman, "Digital control of Dynamic Systems", Addison – Wesley Longman, Inc., Menlo Park, CA, 1998.
2. Gopal, "Digital Control and State Variable Methods", Tata McGraw Hill, India, 1997.
3. C. H. Houpis and G.B. Lamont, "Digital Control Systems", McGraw Hill, 1985.
4. John S. Baey, "Fundamentals of Linear State Space Systems", First edition, McGraw – Hill, 1999.

23U504 EMBEDDED SYSTEM DESIGN

3 0 0 3

EMBEDDED SYSTEMS OVERVIEW: Embedded system Design process – Design Challenge: Optimizing Design Metrics– Processor Technology–ICT Technology–Design Technology–Trade-offs. (9)

ARCHITECTURE AND PROGRAMMING: Hardware architecture – Software architecture – Application Software- Process of Generating executable image – Development/Testing Tools (9)

INTERFACING AND NETWORKS: Parallel I/O Interface –Serial Communications Interface (SCI) - Device Driver Arbitration, Multilevel Bus Architecture–Serial protocols: I2C, CAN, FireWire, USB–Wireless protocols: IrDA, Bluetooth, IEEE 802.11 - Internet Enabled systems -Sensor Networks (9)

RTOS FOR EMBEDDED SYSTEMS: Interrupt driven systems-Need for Real-time Operating System-RTOS Concepts Tasks, Context switching, Interrupt latency, Memory management, Scheduling, Task synchronization- Introduction to RTOS APIs- Power optimization strategies for processes-Basic design using RTOS-Response time calculation-Comparison of commercial

RTOSs.

(9)

EMBEDDED BASED CONTROL SYSTEM: Introduction to digital control systems – Open-Loop and Closed-Loop control system – General control systems and PID controllers – Software coding of a PID controller - PID Tuning - Practical issues related to computer – control implementations.

(9)

Total L:45

TEXTBOOKS:

1. Frank Vahid, Tony Givargis, "Embedded System Design: A unified hardware software introduction", John Wiley & Sons, 2018.
2. Wayne Wolf, "Computers as Components - Principles of Embedded Computer System Design", Fourth Edition, Morgan Kaufmann Publisher, 2017.

REFERENCES:

1. K.V.K.K.Prasad, "Embedded Real-Time Systems: Concepts, Design and Programming", Dreamtech Press, 2014.
2. Jonathan W. Valvano, "Embedded Systems–Real Time Interfacing", Fifth Edition, Cengage Learning, 2017.
3. Raj Kamal, "Embedded Systems: SOC, IOT, AI and Real Time Systems", Fifth Edition, Tata McGraw-Hill Education, 2020.
4. K V Shibu, "Introduction to Embedded Systems", Second Edition, McGraw Hill Education India Private Limited, 2020.

23U510 SIGNAL PROCESSING AND DIGITAL CONTROL LABORATORY

0 0 4 2

LIST OF EXPERIMENTS:

SIGNAL PROCESSING BASED EXPERIMENTS:

1. Generation of signals and operation on signals
2. Sampling and Quantization, discretization of continuous time systems
3. Transform domain analysis of signals and systems
4. Design of FIR filter
5. Design of IIR filter

(30)

DIGITAL CONTROL EXPERIMENTS:

6. Implementation of position and velocity form of PID controller.
7. Synthesis and implementation of digital PID controller.
8. Design of PID controller based on graphical analysis method.
9. Design of observer.
10. Design and implementation of pole placement controller.

(30)

Total P: 60

REFERENCES:

1. Department of Instrumentation and Control Systems Engineering, "Laboratory Manual", 2023.
2. B.C Kuo, Digital Control Systems, Second Edition, Oxford Univ Press, Inc., 1995.

23U511 EMBEDDED SYSTEMS LABORATORY

0 0 4 2

LIST OF EXPERIMENTS:

1. General Purpose Input- Output Programming.
2. Basic I/O Interfacing: Matrix Keypad, 7 segment LED, and LCD Display
3. Polling and interrupt service mechanism using Timer/ Counter Peripheral.
4. Interfacing of stepper motor using Pulse Width Modulation.
5. Interfacing of Analog Sensor using ADC peripheral.
6. Data Transmission between two different devices using USART, SPI & I2C.
7. Data Transmission between two different devices using Zigbee protocol.
8. Microcontroller based controller design for real time system.
9. Smart Monitoring and Control using LoRa.
10. Interfacing of Image Sensors with Single Board Computers.

Total P: 60

REFERENCES:

1. NXP, "LPC2148 & LPC1768 Datasheets".
2. ATMEL, "ATMEL Datasheet".
3. Dragino, "LoRa Product Datasheets".
4. NVIDIA, "Jetson Product Datasheets".
5. Department of Instrumentation and Control Systems Engineering, "Laboratory Manual", 2023.

23Q513 APTITUDE SKILLS

0 0 2 1

1. Reading comprehension
2. Sentence correction, Sentence completion and Para-jumbles
3. Vocabulary, Articles, Prepositions and Interrogatives
4. Critical reasoning
5. Ratio and Proportion, Profit and loss, Partnerships and averages
6. Permutation, Combination and Probability
7. Time, Speed and Distance
8. Resume progress check

Total P: 30

REFERENCES:

1. Norman Lewis, "Word Power Made Easy", Goyal Publisher, New Delhi, 2011.
2. P C Wren and H Martin, "High School English Grammar and Composition", S Chand Publishing, New Delhi, 2017.
3. R S Aggarwal, "Quantitative Aptitude for Competitive Examination", S Chand Publishing, New Delhi, 2017.

SEMESTER - VI

23U601 INSTRUMENTATION SYSTEM DESIGN

4 0 0 4

DESIGN OF DATA ACQUISITION SYSTEM: Need for data acquisition - Factors affecting performance of sensors- Signal conditioning - Resolution and Sampling rate considerations for ADC- Sampling techniques - Selection of ADC (12)

LINEARIZATION AND COMPENSATION TECHNIQUES: Analog linearization- Linear interpolation- Piece-wise linearization- Microcontroller-based linearization. Compensation methods: Offset compensation- Error and Drift Compensation- Lead wire compensation- Cold junction compensation. (12)

DESIGN OF FLOWMETERS: Orifice sizing –Liquid, gas and steam services – Factors affecting orifice plate sizing- Rotameter design- Pitot tube sizing- Design of venturimeter- Selection of flowmeters. (12)

DESIGN OF TRANSMITTERS: Flow transmitter- Level transmitter- RTD and thermocouple based temperature transmitter - pH transmitter- 2-wire transmitter current loops-4-wire transmitter current loops- Troubleshooting current loops- Smart transmitters. (12)

DESIGN OF ACTUATORS: Energy dissipation in a turbulent fluid stream- Importance of control valve sizing- Control valve sizing- Liquid, gas and steam services- Selection criteria for control valve- Variable Frequency Drive- Thyristor power controller. (12)

Total L:60

TEXT BOOKS:

1. Manabendra Bhuyan, "Intelligent Instrumentation: Principles and Applications", First Edition, CRC Press, 2017.
2. Subhas Chandra Mukhopadhyay, "Intelligent Sensing, Instrumentation and Measurements", Springer, 2013.

REFERENCES:

1. John P Bentley, "Principles of Measurement Systems", Fourth Edition, Pearson Education, 2020.
2. Gregory K McMillan and Douglas M Considine, "Process/ Industrial Instruments and Controls Handbook", Fifth Edition, Tata McGraw Hill, 2009.
3. Norman A Anderson, "Instrumentation for Process Measurement and Control", Third Edition, CRC Press, 2005.
4. Curtis D. Johnson, "Process Control Instrumentation Technology", Eighth Edition, Pearson Education India, 2015.

23U602 INDUSTRIAL PROCESS CONTROL

3 1 0 4

PROCESS DYNAMICS : Piping and Instrumentation Diagram : Instrument terms and symbols - Feedback control concepts - Classification of variables - Servo and regulator Control - Process Characteristics: Degrees of freedom, Self-regulation, Process lag, Dead time, Inverse response - Continuous and batch process - Mathematical model of first order level, pressure, hydraulic and thermal processes - Higher order Process : Interacting and non-interacting systems - Linearization of nonlinear models - Approximation of higher order transfer functions - Development of empirical models from process data: Fitting first and second order models using step tests and PRBS. (9+3)

BASIC CONTROL ACTIONS AND CONTROLLERS : Characteristics of on-off, proportional, single-speed floating, integral and derivative control modes - P, I, P+D and P+I+D control modes - Integral windup and prevention - Auto / Manual transfer: Bumpless transfer- Electronic controllers to realize various control actions - Response of controllers for test inputs - Effects of P, I and D actions on closed loop response - Selection of controller modes for control of level, pressure, flow and temperature process. (9+3)

FINAL CONTROL ELEMENT: Current to pressure converter - Pneumatic and electric actuators - Valve positioner - Pneumatic control valve: parts, classification and mathematical model - Characteristics of control valves: Inherent and Installed characteristics - Commercial valve bodies - Cavitation and flashing - Control valve types: flow control valve, pressure control valve and directional control valve - Motorized Control valve. (9+3)

CONTROLLER DESIGN: Model based controller design - Direct synthesis method- Internal Model Controller (IMC) - Evaluation criteria: IAE, ISE, ITAE - 1/4 decay ratio criterion - Determination of optimum settings for mathematically described processes using time response and frequency response methods - Online controller tuning: Process reaction curve method, Ziegler Nichols method, Damped oscillation method – Relay autotuning. (9+3)

MULTI LOOP & MULTIVARIABLE CONTROL : Feed forward control - Ratio control - Cascade control - Time delay compensation: Smith Predictor Control- Inferential control - Split range control - Selective Control - Introduction to Adaptive Control - Multi Input Multi Output Processes: Process interactions and control loop interactions - Pairing of controlled and manipulated variables – Design of Decoupler- Model Predictive Controller (MPC) - Case studies: Binary distillation column control, boiler drum level control and chemical reactor control. (9+3)

Total L:45 + T: 15 = 60

TEXT BOOKS:

1. George Stephanopoulos, "Chemical Process Control: An Introduction to theory and Practice", First Edition, Pearson Education India, 2008.
2. Dale. E. Seborg, Thomas F. Edgar, Duncan A. Mellichamp and Francis J. Doyle, "Process Dynamics and Control", Fourth Edition, John Wiley & Sons, 2016.

REFERENCES:

1. Donald P. Eckman, "Automatic Process Control", First Edition, Wiley India Pvt Ltd, 2021.
2. Donald R. Coughanowrand Steven E. LeBlanc, "Process Systems Analysis and Control", Third Edition, McGraw Hill Education, 2017.
3. Bela G. Liptak, "Instrument Engineers Handbook: Process Measurement and Analysis", Fifth Edition, CRC Press, 2016.
4. Curtis D. Johnson, "Process Control Instrumentation Technology", Eighth Edition, Pearson Education India, 2015.

23U603 ARTIFICIAL INTELLIGENCE FOR MEASUREMENT AND CONTROL**3 0 0 3**

FOUNDATIONS OF ARTIFICIAL INTELLIGENCE: Concept and importance of Artificial Intelligence, Machine Learning and Data Science - Human intelligence vs. Machine intelligence - AI Basics: Divide and Conquer, Greedy, Branch and Bound, Gradient Descent - Problem solving methods in AI - Knowledge representation. (8)

EXPERT SYSTEMS: Introduction - Expert system architecture- Functions of various parts - Mechanism and role of inference engine - Types of Expert system - Tuning of expert systems - Expert PID Control. (8)

DATA ANALYSIS TECHNIQUES: Regression analysis: Simple linear regression, Multiple linear regression, K-nearest neighbors regression – Multicollinearity in regression analysis - Classification techniques: Clustering, Association rules analysis - Logistic regression in classification. (10)

MACHINE LEARNING: Supervised learning - Unsupervised learning - Reinforcement learning. Artificial Neural Network: Architectures- Multi Layer Perceptron - Backpropagation learning algorithm - Activation functions - Types of Loss function - Optimization algorithm: Gradient descent, Stochastic gradient descent - Batch normalization and Dropout- Neural network control - Adaptive control based on neural approximation. (11)

SENSORS WITH ARTIFICIAL INTELLIGENCE: Multidimensional intelligent sensors - ANN-based intelligent sensors: Soft sensing, Fault detection - Fuzzy logic-based intelligent sensors - AI for prognostic instrumentation. (8)

Total L: 45**TEXT BOOKS:**

1. Lavika Goel, "Artificial Intelligence: Concepts and Applications", Wiley Publishers, 2021.
2. Ethem Alpaydin, "Introduction to Machine Learning", MIT Press, Fourth Edition, 2020.

REFERENCES:

1. Manabendra Bhuyan, "Intelligent Instrumentation: Principles and Applications", CRC Press, 2017.
2. Jinkun Liu, "Intelligent Control Design and MATLAB Simulation", Springer, 2018.
3. Thomas A. Runkler, "Data Analytics: Models and Algorithms for Intelligent Data Analysis", Springer Verlag, Second Edition, 2016.
4. John D. Kelleher, "Deep Learning", MIT Press, 2019.

23U610 INSTRUMENTATION SYSTEM DESIGN LABORATORY**0 0 4 2****LIST OF EXPERIMENTS:**

1. Study of zero elevation and zero suppression in differential pressure transmitter
2. Design of alarm circuit
3. Linearization of thermistor and cold junction compensation of thermocouple
4. Design of temperature transmitter
5. Calculation of discharge coefficient of orifice plate, venturi and pitot tube
6. Implementation of thermistor Linearization using graphical programming
7. Creation of GUI for temperature data acquisition and alarm interface circuit
8. Instrument control
9. Reusing existing code with graphical programming
10. Implementation of multithreading

Total P: 60**REFERENCES:**

1. Jovitha Jerome, "Virtual Instrumentation Using LabVIEW", PHI, 2010.
2. Department of Instrumentation and Control Systems Engineering, "Laboratory Manual", 2023.

23U611 PROCESS CONTROL LABORATORY**0 0 4 2****LIST OF EXPERIMENTS:**

1. First principle modeling and validation of interacting and non-interacting systems
2. Black-box modeling of real-time process

3. Experimental study and analysis of ON/OFF, Proportional controller response on Flow/Level/Pressure/Temperature control loops
4. Experimental study and analysis of PI, PD, PID controller response on Flow/Level/Pressure/Temperature control loops
5. Experimental study on Inherent and Installed characteristics of control valves
6. Experimental study on characteristics of control valve with and without valve positioner
7. Online controller tuning using open loop and closed loop methods
8. Model-based controller tuning for mathematically described processes
9. Design and implementation of Feed forward plus feedback controller, Cascade control and Inferential control
10. Design and Implementation of PID controller for MIMO process

Total P: 60

REFERENCES:

1. Stephanopoulos G, "Chemical Process Control : An Introduction to theory and Practice", First Edition, Pearson Education India, 2008.
2. Department of Instrumentation and Control Systems Engineering, "Laboratory Manual", 2023.

23U620 INNOVATION PRACTICES

0 0 2 1

Students will

- Learn about current practices in product / process development Identify key issues, priorities, and tasks
- Perform literature survey, including Patents
- Perform feasibility analysis
- Develop prototypes and test or re-use recent innovations systematically and submit a report

Total P:30

23Q613 ENHANCING PROBLEM SOLVING ABILITY WITH CODE

0 0 2 1

ENHANCING PROBLEM SOLVING ABILITY WITH CODE:

1. Compilation, Namespace, Header file, Data types, Variables, Declaration, Scope of variables
2. Input / Output, Type Conversion, Operators
3. For, While, Do-while, break, continue, .
4. Decision Making
5. Problem solving Pattern Programming
6. Arrays
7. Call by value & Call by reference, with and without arguments
8. Functions, Recursion & Strings
9. Structures & Union
10. Command Line Argument
11. Structure using Pointers
12. Handling Stress
13. Handling Peer pressure
14. Resume progress check

Total P: 30

REFERENCES:

1. Gayle Laakmann McDowell, "Cracking the Coding Interview: 150 Programming Questions and Solutions", 5th Edition, S Chand Publishing, New Delhi, 2015.
2. John Mongan, Noah Kindler and Eric Giguère, "Programming Interviews Exposed: Secrets to Landing Your Next Job", 5th Edition, Wrox, New Delhi, 2018.

SEMESTER – VII

23U701 LOGIC AND DISTRIBUTED CONTROL SYSTEMS

3 0 0 3

PROGRAMMABLE LOGIC CONTROLLER (PLC) : Introduction - Architecture of PLC -PLC Operation - Input/output modules - Sink and source I/O cards - Programming methods - Basic programming instructions: Bit-level logic Instructions, Instruction addressing, Branch instructions and Internal relay instructions - Ladder diagram for Logic gates - Concept of Latching and Unlatching - Timer instruction-Counter instruction. (9)

ADVANCED FUNCTIONS IN PLC: Program control instructions: Master Control Reset, Jump and Subroutines - Data manipulation instructions: Data transfer, Data compare - Math instructions: Addition, Subtraction, Multiplication and Division- Sequencer instructions- PID functions - Networking of PLC - PLC installation, maintenance and troubleshooting. (9)

SUPERVISORY CONTROL AND DATA ACQUISITION SYSTEM (SCADA): SCADA Architecture: Hardware and Software- Types of SCADA system- Functions of SCADA - Elements of SCADA: Remote terminal unit, Master terminal unit and HMI- Application areas of SCADA. (9)

DISTRIBUTED CONTROL SYSTEM (DCS) AND COMMUNICATION FACILITIES: Evolution of DCS - Architecture of DCS- Hardware units of DCS- Programming techniques- Operator interfaces - Engineering interfaces- Remote operation - Redundancy concept- Interoperability and Interchangeability- Integration of PLC with DCS. Communication facilities: HART protocol, Wireless HART, Foundation Fieldbus and Profibus- Case study of any one DCS. (9)

RECENT TRENDS IN AUTOMATION: Introduction to Networked Control System - Plant wide control - Cloud based Automation - OPC Vs OPC UA- Safety PLC- Safety Instrumented Function - Safety Integrity Level - Functional safety standards:

Total L:45**TEXT BOOKS:**

1. Frank D. Petruzella, "Programmable Logic Controllers", McGraw-Hill Education, Fifth Edition, New York, 2017.
2. Chanchal Dey and Sunit Kumar Sen, "Industrial Automation Technologies", CRC Press, 2020.

REFERENCES:

1. Stuart A. Boyer, "SCADA: Supervisory Control and Data Acquisition Systems", Fourth Edition, ISA Press, 2016.
2. John W Webb and Ronald A Reis, "Programmable Logic Controllers - Principles and Applications", Fifth Edition, Pearson Education, 2015.
3. Michael P Lukas, "Distributed Control Systems", Van Nostrand Reinhold Co., New York, First Edition, 1986.
4. Glisente Landrini, Basilio Abbamonte and Tino Vandecastelle, "Safety Instrumented Systems", Fifth Edition, GM International, 2017.

23U702 INDUSTRIAL INTERNET OF THINGS**3 0 0 3**

INTRODUCTION TO INDUSTRIAL IOT (IIOT) SYSTEMS: Industrial Revolutions- Role of Internet of Things (IoT) & Industrial Internet of Things (IIoT) in Industry- Support System for Industry 4.0- Smart Factories- Enabling Technologies of IoT- Industrial IoT Applications. (9)

IMPLEMENTATION SYSTEMS FOR IIOT: Sensors and Actuators for Industrial Processes- Sensor networks- Data Acquisitions on IoT Platform- Microcontrollers and Embedded PC roles in IIoT- Wireless Sensor nodes with Bluetooth- Wi-Fi- and LoRa Protocols and IoT Hub systems- Building Wireless connection into existing Modbus, HART & Profibus networks. (9)

IIOT PROTOCOLS: MQTT: Publish & subscribe - Architecture details- communication formats; MQTT-SN: Architecture & Topology - Difference between MQTT & MQTT-SN- Choosing a MQTT broker; CoAP: Architecture details- communication formats - usage example. (9)

IIOT DATA MONITORING & CONTROL: IoT Gate way- IoT Edge Systems and its Programming- Cloud computing- Real Time Dashboard for Data Monitoring- Data Analytics and Predictive Maintenance with IIoT technology. (9)

IOT SECURITY & CASE STUDIES OF IIOT SYSTEMS: IoT Security: IoT Security Challenges - Causes of IoT Security Breaches - Industrial Cyber Security IEC 62443 Standard- IoT Security Risks; Case Studies of IIoT Systems: Smart Energy meter - Monitoring environmental conditions in an apparel factory space - Predictive monitoring of CNC machine operation. (9)

Total L:45**TEXT BOOKS:**

1. Sudan Jha, Usman Tariq, Gyanendra Prasad Joshi, Vijender Kumar Solanki, "Industrial Internet of Things: Technologies, Design, and Applications", CRC Press, 2022.
2. Perry Lea, "IoT and Edge Computing for Architects: Implementing edge and IoT systems from sensors to clouds with communication systems, analytics, and security", Second Edition, Packt Publishing, 2020.

REFERENCES:

1. Anand Sharma, Sunil Kumar Jangir, Manish Kumar, Dilip Kumar Choubey, Tarun Shrivastava, S. Balamurugan, "Industrial Internet of Things: Technologies and Research Directions", First Edition, CRC Press, 2022.
2. Jean-Marie Flaus, "Cybersecurity of Industrial Systems", John Wiley & Sons, 2019.
3. Sravani Bhattacharjee, "Practical Industrial Internet of Things Security: A practitioner's guide to securing connected Industries", Packt Publishing Ltd, 2018.
4. Ronald L and Krutz, Industrial Automation and Control System Security Principles, ISA, 2016.

23U710 INDUSTRIAL AUTOMATION LABORATORY**0 0 4 2****LIST OF EXPERIMENTS:**

1. Implementation of basic logic operations and interlocks using PLC
2. Implementation of timer operations using PLC
3. Implementation of counter operations using PLC
4. Study of Analog and Digital I/O interfacing in PLC
5. Implementation of actuator control using PLC
6. Implementation of sequential control using PLC sequencer function
7. Implementation of PLC programming for practical applications
8. Study of interfacing smart field devices with DCS
9. Development of control logic using DCS
10. On-line monitoring and control of a process using DCS

Total P:60**REFERENCES:**

1. Frank D. Petruzella, "Programmable Logic Controllers", Tata McGraw Hill, Fifth Edition, New York, 2017.
2. Department of Instrumentation and Control Systems Engineering, "Laboratory Manual", 2023.

23U711 INDUSTRIAL INTERNET OF THINGS LABORATORY

0 0 4 2

LIST OF EXPERIMENTS:

1. Study of IoT System and its components.
2. Wireless Sensor Networks using Zigbee.
3. Wireless Sensor Networks using LoRa.
4. Wireless Sensor Networks using WiFi.
5. Communication between IoT devices through cloud using HTTP Protocol.
6. Communication between IoT devices through cloud using MQTT Protocol.
7. Closed loop control of a real - time system using cloud computing
8. Closed loop control of a real - time system using Fog Computing.
9. Closed loop control of a real - time system using Edge Computing.
10. Industrial IoT using PLC.

Total P:60

REFERENCES:

1. ESP32-WROOM-32 Datasheet Version 2.5
2. STM 32 (Rev 18) Datasheet July 2020
3. CC3200 Launchpad datasheet revised March 2020
4. Department of Instrumentation and Control Systems Engineering, "Laboratory Manual", 2023.

23U720 PROJECT WORK I

0 0 4 2

- Identification of a real life problem in thrust areas.
- Developing a mathematical model for solving the identified problem.
- Finalization of system requirements and specification.
- Proposing different solutions for the problem based on literature survey.
- Future trends in providing alternate solutions.
- Consolidated report preparation.

Total P:60

SEMESTER – VIII

23U820 PROJECT WORK II

0 0 8 4

The Project involves the following:

- I) Preparing a project—brief proposal including
 - Problem Identification
 - A statement of system / process specifications proposed to be developed (Block Diagram / Concept tree)
 - List of possible solutions including alternatives and constraints
 - Cost benefit analysis
 - Time line of activities
- II) A report highlighting the design finalization [based on functional requirements & standards (if any)]
- III) A presentation including the following:
 - Implementation Phase (Hardware / Software / both)
 - Testing & Validation of the developed system
 - Learning in the Project
- IV) Consolidated report preparation

Total P:120

PROFESSIONAL ELECTIVES (Regular, PE I to PE VI)

23U001 BIOMEDICAL INSTRUMENTATION

3 0 0 3

BIOELECTRIC POTENTIALS AND ELECTRODES: Cells and their structures - Transport of ions through cell membrane - Electrode-electrolyte interface, electrode- skin Interface - Origin of bio potential and its propagation - action and resting potential - Bioelectric potential - Bio potential electrodes – Types of electrodes: surface, needle and microelectrodes. (12)

CARDIO SYSTEM AND MEASUREMENTS: Heart and cardiovascular system – ECG – 12 lead system - principle of vector cardiography - blood pressure and its measurement – Measurements of heart sounds – PCG - Defibrillator: principle and types -pacemakers: principle and types. (9)

NERVOUS SYSTEM AND MEASUREMENTS: Central and peripheral nervous system - parts of human brain- Action potential of brain - brain wave – EEG- 10-20 electrode system – Nerve stimulator (8)

RESPIRATORY SYSTEM AND MEASUREMENTS: Physiology of respiratory system - Principle of pneumograph - Spirometer - ventilators: principle and types. (8)

MEDICAL IMAGING: X-ray imaging - Computed Tomography (CT) - Ultrasonic Scanner- Magnetic Resonance Imaging - Positron Emission Tomography (PET). (8)

Total L: 45

TEXT BOOKS:

1. Leslie Cromwell, Fred J Weibell, Erich A. Pfeiffer, "Biomedical Instrumentation and Measurements", Pearson Education, India, 2nd Edition, 2018.
2. Khandpur R S, "Handbook of Biomedical Instrumentation", McGraw Hill Education (I) P Ltd, Chennai, 5th Edition, 2018.

REFERENCES:

1. John G Webster, "Medical Instrumentation: Application and Design", John Wiley and Sons, 5th Edition, 2020.
2. Joseph J Carr, John M Brown, "Introduction to Biomedical Equipment Technology", Pearson Education, India, 4th Edition, 2006.
3. L.A.Geddes, L.E. Baker, "Principles of Applied Biomedical Instrumentation", John Wiley and Sons, 3rd Edition, 2009.
4. Richard Aston, "Principles of Biomedical Instrumentation and Measurement", Merril Publishing Company, New York, 1st Edition, 1990.

23U002 VIRTUAL INSTRUMENTATION

3 0 0 3

INTRODUCTION: Graphical System Design (GSD) model, Virtual Instrumentation (VI) model, virtual instruments and traditional instruments, hardware and software in virtual instrumentation, virtual instrumentation for test, control and design, graphical programming and textual programming. (9)

VI PROGRAMMING TECHNIQUES: Software environment, modular programming, repetition and loops, arrays, clusters, structures, plotting data, strings and file I/O. (9)

DATA ACQUISITION: DAQ system, Signal conditioning, DAQ hardware configuration- analog I/O, digital I/O, counter/timer, DAQ software architecture, grounding technique. (9)

APPLICATION MODULES AND TOOLKITS: Machine vision, motion control, data logging and supervisory control, control design and simulation tools – control design tool, system identification and simulation interface tool. (9)

INTERFACE STANDARDS: Serial and Parallel communication, RS232, RS422, RS485, GPIB, Firewire, Communication protocol overview - Industrial Ethernet, CAN. (9)

Total L:45

TEXT BOOKS:

1. Jovitha Jerome, "Virtual Instrumentation using LabVIEW", PHI Learning Pvt Ltd, New Delhi, 2010.
2. Steve Mackay, Edwin Wright, John Park, and Deon Reynders, "Industrial Data Networks", Elsevier, Boston, 2004.

REFERENCES:

1. Gary Johnson and Richard Jennings, "LabVIEW Graphical Programming", McGraw Hill Education, New Delhi, 4th Edition, 2013.
2. Sanjay Gupta and Joseph John, "Virtual Instrumentation using LabVIEW: Principle and Practices of Graphical Programming", Tata McGraw-Hill Education, New Delhi, 2nd Edition, 2010.
3. William Buchanan, "Computer Buses Design and Application", CRC Press, Boca Raton, 2000.
4. Clyde F Coombs, "Electronic Instruments Handbook", McGraw Education, 3rd Edition, 1999.

23U003 FIBER OPTICS AND LASER INSTRUMENTATION

3 0 0 3

OPTICAL FIBERS AND THEIR PROPERTIES: Characteristics of optical radiation, luminescence - Fiber materials and their characteristics - Principles of light propagation through a fiber - Types of optical fibers - Properties - Transmission characteristics - Absorption losses - Scattering losses - Dispersion - Fibers splicing, connector and couplers – Optocouplers – Optrodes (9)

OPTICAL SOURCES AND DETECTORS : LED — LED power and efficiency - Structures, planar, dome, ELED, SLED, super luminescent LEDs, characteristics and applications - General characteristics of photodetectors - Photodiode - Junction photodiodes - Heterojunction diode and PIN diode - APD - Special detectors –Schottky barrier diode - Photo- transistor and photo-thyristor -Solar cells (9)

INDUSTRIAL APPLICATION OF OPTICAL FIBERS: Fiber optic sensors - Fiber optic instrumentation system - Application in instrumentation: interferometric method of measurement of length - Moiré fringes - Measurement of pressure, temperature, current, voltage, liquid level and strain - Fiber optic gyroscope - Cavity dumping –Polarization maintaining fibers (9)

LASER FUNDAMENTALS: Characteristics of lasers – Three level and four level lasers – Properties of laser - Laser modes-Resonator configuration —Q switching and mode locking — Cavity dumping – Types of lasers : gas lasers, solid lasers, liquid lasers, semiconductor lasers (9)

INDUSTRIAL APPLICATIONS OF LASER : Laser for measurement of distance, length, velocity, acceleration, current, voltage and atmospheric effect - Material processing: laser heating, welding, melting, scribing, splicing and trimming of materials, removal and vaporization - Hologram: Principle of holography — Methods and holographic components - Holographic interferometry and applications, holography for nondestructive testing - Medical applications of lasers: laser and tissue

interaction — Laser instruments for brain surgery, plastic surgery and oncology

(9)

Total L:45

TEXT BOOKS:

1. John M Senior, "Optical Fiber Communication: Principles and Practice", Prentice Hall, 2010.
2. Khare R P, "Fibre Optics and Optoelectronics", Oxford Press, 1st Edition, 2004.

REFERENCES:

1. Wilson, Hawkes J F B, "Introduction to Opto Electronics", Prentice Hall of India, 2001.
2. Nambiar K R, "Laser: Principles, Types and Applications", New Age International, 2010.
3. S.C. Gupta, "Opto Electronic Devices and Systems", Prentice Hall of India, 2010.
4. Donald A, Sterling Jr, "Technicians Guide to Fiber Optics", Vikas Publishing House, 2009.

23U004 POWER PLANT INSTRUMENTATION

3 0 0 3

OVERVIEW OF POWER GENERATION: Methods of power generation: Hydro, Thermal, Nuclear, Solar and Wind power, Ocean Energy System, Geothermal Energy, Biomass Energy, Fuel cells. Building Blocks of Thermal power plant – Subcritical and Supercritical boilers – Operating Pressure and Temperature ranges – Overview of Instrumentation System in Thermal power plant (11)

MEASUREMENTS IN POWER PLANTS: Measurement of feed water flow, Fuel flow, Airflow and Steam flow with correction factor - Steam pressure and temperature measurement-Turbine speed and vibration measurement. (8)

ANALYZERS IN POWER PLANTS: Analysis of impurities in feed water and steam: Dissolved oxygen analyzer – Chromatography- pH meter - Fuel analyser – Flue gas oxygen analyser - Pollution monitoring instruments – SO_x and NO_x measurements. (8)

CONTROL LOOPS IN BOILER: Boiler drum level control – Deaerator level control - Main steam and reheat steam temperature control – Super heater steam temperature control - Steam bus pressure control - Combustion control – Air/fuel ratio control – Furnace draft control — Distributed control system in power plants – Interlocks in boiler operation. (10)

NUCLEAR POWER PLANT INSTRUMENTATION: Different types of Nuclear power plant, Nuclear reactor control loops, Reactor dynamics, Control and Safety instrumentation, Reliability aspects. (8)

Total L: 45

TEXT BOOKS:

1. David Lindsley, "Power Plant Control and Instrumentation", Institution of Electrical Engineers, 2000.
2. Sam G Dukelow, "The Control of Boilers", Instrument Society of America, 2nd Edition, 1991.

REFERENCES:

1. Jain R K, "Mechanical and Industrial Measurements", Khanna Publishers, New Delhi, 2009.
2. Bela G Liptak, "Process Measurement and Analysis", CRC press, Vol. 1, 2003.
3. Swapan Basu, Ajay Kumar Debnath, "Power Plant Instrumentation and Control Handbook", Academic Press, 2nd edition, 2019.
4. Nag P K, "Power Plant Engineering", Tata McGraw-Hill Education, 3rd edition, 2007.

23U005 INDUSTRIAL CHEMICAL PROCESSES

3 0 0 3

UNIT OPERATIONS: Unit operations - Transport of liquids, solids and gases adjusting particle size of bulk solids – Mixing processes – Separation processes – Chemicals from petroleum products. (9)

COMBUSTION PROCESSES: Combustion processes – Heat exchangers – Energy balance - Material balance – Evaporators – Crystallization. (9)

OTHER OPERATIONS: Drying – Distillation – Refrigeration process – Chemical reactions. (9)

CASE STUDY – I: Operations in the manufacture of paper and pulp – Operations in steel industry. (9)

CASE STUDY – II: Operations in thermal power plant – Operations in pharmaceutical industry and leather industry. (9)

Total L:45

TEXT BOOKS:

1. Balchan J G and Mumme K I, "Process Control Structures and Applications", Van Nostrand Reinhold Company, New York, 1998.
2. Waddams A L, "Chemicals from petroleum", John Wiley & Sons, 3rd Edition, 1973.

REFERENCES:

1. Austin G T, "Shreve's Chemical Process Industries", McGraw Hill Education, 5th Edition, 2017.
2. Liptak B G, "Instrument Engineers Handbook: Process Measurement and Analysis", CRC Press, 4th Edition, 2003.
3. Luyben W C, "Process Modelling, Simulation and Control for Chemical Engineers", McGraw Hill Education, New York, 3rd Edition, 1967.
4. Norris Shreve R, "Chemical Process Industries McGraw Hill Education, Moscow, 1967.

23U006 THERMODYNAMICS AND FLUID MECHANICS

3 0 0 3

BASIC CONCEPTS OF THERMODYNAMICS: System, property, state and equilibrium, process and cycle, work, heat and other forms of energy. Zeroth law and application, first law - statement, applications to closed and open systems, statements - Heat engine, refrigerator and heat pump, reversibility and irreversibility, Carnot cycle and Carnot theorem, second law, entropy - Thermo Dynamic Cycles: Air standard cycles - Otto cycle, Diesel cycle, Dual cycle, Brayton cycle. Rankine cycle (12)

BASIC CONCEPT OF FLUID MECHANICS : Introduction – Classification – Types of fluids – Properties – Laws of pressure – Atmospheric, gauge, absolute pressure, pressure measurement – Manometers – Mechanical gauges - Flow of fluids: Introduction – Types of fluid flow – Velocity – Rate equation of continuity – Energy of a liquid in motion – Head of a liquid (9)

FLOW THROUGH CIRCULAR PIPES: Pipes in series and parallel. Reynolds number, Darcy-Weisbach equation, minor losses-Sudden expansion, sudden contraction and losses in pipe fittings (9)

HYDRAULIC TURBINES AND PUMPS: Impulse type, Pelton wheel, reaction type, Francis, Kaplan and Propeller – Principle of operation, performance of turbines, draft tube. Hydraulic pumps: Classification - Reciprocating and centrifugal pumps - Performance studies, fluid coupling and torque converter (9)

COMPRESSORS: Classification, reciprocating and centrifugal compressors, applications, characteristics, surging and stalling (6)

Total L: 45

TEXT BOOKS:

1. Nag P K , "Engineering Thermodynamics", Tata McGraw Hill, New Delhi, 4th Edition, 2008.
2. Frank M White , "Fluid Mechanics", Tata McGraw Hill, New Delhi, Reprint, 2011.

REFERENCES:

1. Kothandaraman C P and Rudramoorthy R , "Basic Fluid Mechanics", New age International Publishers, Chennai, Reprint, 1999.
2. Kumar D S , "Fluid Mechanics and Fluid Power Engineering", Kataria S K and Sons, New Delhi, 9th Edition, 2018.
3. Holman J P , "Thermodynamics", McGraw Hill Book Company, New York, 4th Edition, 1988.
4. Yunus A Cengel , Michael A Boles , "Thermodynamics: An Engineering Approach", Tata McGraw Hill, Reprint, 2006.

23U007 INSTRUMENTATION AND CONTROL IN PETROCHEMICAL INDUSTRIES

3 0 0 3

PETROLEUM PROCESSING: Petroleum exploration - Recovery techniques - Refining of crude oil - Constituents of Crude Oil - Refining Capacity in India - Consumption of Petroleum products in India. (9)

CONTROL OF DISTILLATION COLUMN: P and I diagram of petroleum refinery - Atmospheric Distillation of Crude oil - Vacuum Distillation process - Control of Distillation Column - Temperature Control -Feed Control - Reflux Control - Reboiler Control. (9)

CONTROL OF CHEMICAL REACTORS AND DRYERS: Temperature and Pressure Control of chemical reactors - Control of Dryers - Batch Dryers - Continuous Dryers. (9)

CONTROL OF HEAT EXCHANGERS AND EVAPORATORS: Variables and Degrees of freedom - Liquid to Liquid Heat Exchangers - Steam Heaters - Condensers - Reboilers and Vaporizers-Types of Evaporators. (9)

CONTROL OF PUMPS: Centrifugal pump: On-Off Level control - Pressure control - Flow control - Rotary pumps: On-Off Level control. Reciprocating Pumps: On-Off control. Effluent and Water Treatment Control: Chemical Oxidation - Chemical Reduction - Neutralization - Precipitation - Biological control. (9)

Total L:45

TEXT BOOKS:

1. Liptak B G , "Instrument Engineer's Handbook, Vol. 2: Process Control and Optimization", CRC Press, 4th Edition, 2018.
2. Waddams A L , "Chemicals from petroleum", John Wiley & Sons, 3rd Edition, 1973.

REFERENCES:

1. Balchan J G and Mumme K I , "Process Control Structures and Applications", Van Nostrand Reinhold Company, New York, 1998.
2. Austin G T , "Shreve's Chemical Process Industries", McGraw Hill Education, 5th Edition, 2017.
3. Andrew W G, Williams H B , "Applied Instrumentation in the Process Industries", Houghton Mifflincom, Houston, 3rd Edition, 2002.
4. Wiseman P , "Petrochemicals", Ellis Horwood, Chichester, 1986.

23U008 POWER ELECTRONICS AND DRIVES

3 0 0 3

REVIEW OF POWER ELECTRONIC DEVICES: Power diodes and transistors. Thyristors, GTO, IGBT, other devices in thyristor family. Thyristor protection circuits – Thyristor triggering circuits – series and parallel operation of thyristors, commutation techniques. (9)

CONVERTERS: Principle of phase-controlled rectifier operation. Single phase thyristor converters with R, RL loads and freewheeling diode. Three phase thyristor converters with R and RL loads. Dual converters. Introduction to cycloconverters and AC voltage regulators. (9)

CHOPPER AND INVERTER: Choppers - Chopper classification - step up and step down choppers. Single phase Voltage Source Inverter – Harmonics- Three phase bridge inverters - voltage control in single phase inverters - Current source inverters - Series and Parallel inverters. (9)

DC AND AC DRIVES: Basic characteristics of DC motor - Control of DC motor - Regenerative and dynamic braking Closed loop control scheme - Speed torque characteristic of induction motor - stator voltage control – Sensor less vector control –Flux vector control- Static rotor resistance control - Slip power recovery scheme - Self control of synchronous motor. (9)

APPLICATIONS: Switched mode power supplies, Uninterrupted power supplies, High Voltage DC transmission – control of HVDC converters, Solid state relays, Static circuit breakers. Introduction to FACTS controllers – Introduction to electric vehicle chargers and systems for electric vehicles. (9)

Total L: 45

TEXT BOOKS:

1. Rashid M H , "Power Electronics – Circuits, Devices and Applications," Pearson Education India, New Delhi, 4th Edition, 2017.
2. Dubey G.K , "Fundamentals of Electric Drives", Narosa Publishing House, New Delhi, 2nd Edition, 2011.

REFERENCES:

1. Bimal K Bose , "Modern Power Electronics and AC Drives", Pearson Education India, New Delhi, 2017.
2. Dubey G.K., Doradla S R, Joshi A, Sinha RMK , "Thyristorised Power Controllers", New Age International New Delhi, 2009.
3. Vedam Subramaniam , "Electrical Drives Concepts and Applications", Tata McGraw-Hill, New Delhi, 2011.
4. Bimbhra P.S , "Power Electronics", Khanna Publishers, New Delhi, 2015.

23U009 ROBOTICS AND AUTOMATION

3 0 0 3

INTRODUCTION TO ROBOTIC SYSTEMS: Structure of a Robot, Classification of Robots: Cartesian, Cylindrical, Spherical,Articulated, SCARA - Accuracy, Resolution and Repeatability of Robots, Degrees of Freedom of Serial and Parallel Manipulators, Robot Application in Manufacturing: Material Transfers - Machine Loading and Unloading - Processing Operations – Assembly and Inspection. (9)

TRANSFORMATIONS AND KINEMATICS: Homogeneous Coordinates, Coordinate Reference Frames, Homogeneous Transformations for the Manipulator, D-H Representation, Forward and Inverse Problem of Manipulator Kinematics (9)

DRIVES AND SENSORS: Hydraulic and Pneumatic Systems, Mechanical Power Drive, Rotary to Linear Motion Conversion Mechanisms, electric drive systems: DC Motor, Servo Motor and Stepper Motor, Internal and External State Sensors, Touch and Tactile Sensors, Force and Torque Sensors, Proximity and Range Sensors, Vision Systems, Robot End Effectors, Gripper Force Analysis. (9)

DYNAMICS AND CONTROL: Differential Motion of Manipulators, Trajectory Planning, Manipulator Dynamics, Jacobian in terms of D-H Matrices, Manipulator Control, Controller Architecture, Robot Programming. (9)

COMPUTER AIDED MANUFACTURING APPROACHES: Robot Interface, Networking and Bus Standards, Flexible Manufacturing Systems (FMS), Computer Integrated Manufacturing (CIM) - Role of Robots in FMS and CIM – Case Studies. (9)

Total L: 45

TEXT BOOKS:

1. Saeed B. Niku , "An Introduction to Robotics: Analysis, Systems and Applications", Wiley India, New Delhi, 3rd Edition, 2020.
2. Nagrath I J, Mittal R K , "Robotics and Control", Tata McGraw Hill Education (I) P Ltd, Chennai, 2016.

REFERENCES:

1. Mikell P. Groover, Mitchell Weiss, Roger N. Nagel and Nicholas G. Odrey , "Industrial Robotics", McGraw-Hill Education (I) P Ltd, Chennai, 2nd Edition, 2019.
2. Richard D Klafter, Michael Negin , "Robotics Engineering: An Integrated Approach", Prentice Hall of India, New Delhi 2009.
3. S K Saha , "Introduction to Robotics", McGraw-Hill Education (I) P Ltd, Chennai, 2nd Edition, 2017.
4. Mikell P. Groover , "Automation, Production Systems and Computer Integrated Manufacturing", Pearson Education, Chennai, 4th Edition, 2020.

23U010 FUNDAMENTALS OF PNEUMATICS AND HYDRAULICS

3 0 0 3

INTRODUCTION TOPNEUMATIC CONTROL: Characteristics of compressed air - Elements of pneumatic control system - Pneumatic Actuators: Linear and rotary types - End position cushioning, sealing (6)

PNEUMATIC VALVES: Symbolic representation as per ISO 1219 and ISO 5599 - Direct and indirect actuation pneumatic cylinders - Use of memory valve - Flow control valves and speed control of cylinders supply air throttling and exhaust air

throttling - Use of quick exhaust valve - Signal Processing Elements - Use of Logic gates – OR and AND gates - Pressure dependent controls: types - Time dependent controls : principle, construction – Introduction to hydro pneumatics (12)

INTRODUCTION TO HYDRAULIC POWER : Elements of hydraulic system - Source of hydraulic power: Pumping theory - Classification of pumps - Gear pumps, vane pumps and piston pumps - Performance of pumps - Selection of pumps (9)

HYDRAULIC ACTUATORS AND VALVES: Linear hydraulic actuators - Hydraulic motors: gear motors, vane motors, piston motors - Control valves: Directional control valves, Constructional features and working principle – Pressure control valves: Direct and pilot operated types - Flow control valves (9)

HYDRAULIC CIRCUIT DESIGN AND ANALYSIS: Control of single and double acting hydraulic cylinders - Speed control of hydraulic actuators - Regenerative circuit - Pump unloading circuit- cylinder synchronizing circuits (9)

Total L: 45

TEXT BOOKS:

1. Ian C Turner, "Engineering Applications of Pneumatics and Hydraulics", CRC Press, 2nd Edition, 2020.
2. Andrew Parr , "Hydraulics and Pneumatics: A technician's and engineer's guide", Butterworth Heinemann, 3rd Edition, 2011.

REFERENCES:

1. Anthony Esposito , "Fluid Power with applications", , Pearson education, 7th Edition, 2013.
2. Majumdar, S.R , "Pneumatic Systems – Principles and Maintenance", Tata McGraw Hill, 1st Edition, 2017.
3. Srinivasan. R , "Hydraulic and Pneumatic Control", Tata McGraw - Hill Education, 2nd Edition, 2012.
4. Ilango Silvaraman, "Introduction to Hydraulics and Pneumatics", Prentice Hall India Pvt. Limited, 1st Edition, 2017.

23U011 VLSI DESIGN

3 0 0 3

VLSI DESIGN METHODOLOGY: VLSI design process - Layout styles: Full-custom - Semi-custom approaches. Electrical Properties of MOS and CMOS Circuits: MOS Transistor - Threshold voltage - Basic DC equations - Second order effects – Small signal AC characteristics. nMOS and CMOS inverters - Inverter delay – Power consumption in CMOS gates: Static dissipation - Dynamic Dissipation. Pass transistor - Transmission gate. (10)

VLSI FABRICATION TECHNIQUES: CMOS processes - n well - p well - Twin tub - Silicon on insulator. Design rules – Mead Conway design rules for the silicon gate nMOS - CMOS process - CMOS. Sheet resistance - Resistance estimation - Capacitance estimation - Driving large capacitive loads. Layer representations - Stick diagrams - nMOS design style - CMOS design style - Simple layout examples. (8)

LOGIC DESIGN: Switch logic- Pass transistor and transmission gate. Other forms of CMOS logic: Dynamic CMOS logic – Clocked CMOS logic - Precharged domino CMOS logic - Combinational logic design examples. Clocked sequential circuits - Two phase clocking - Charge storage - Dynamic register element - nMOS and CMOS Dynamic shift register - JK flip flop. (8)

SUBSYSTEM DESIGN PROCESS: General arrangement of a 4-bit arithmetic processor - Design of a 4-bit shifter - Design of an ALU subsystem - Implementation of ALU functions with an adder - Carry look ahead adder - Multipliers: Serial parallel multipliers - Pipelined multiplier array. (9)

VHDL : Introduction-identifiers-data objects- data types - operators- structural modeling - dataflow modeling- behavioral modeling- hardware modeling examples: encoder - clock divider - pulse shifter - adder - multiplexer - demultiplexer – decoder - parity generator and checker. (10)

Total L: 45

TEXT BOOKS:

1. Douglas A Pucknell, and Kamran, Eshraghian , "Basic VLSI design", Prentice Hall of India, New Delhi,3rd Edition, 2015.
2. Bhasker J, "VHDL Primer", Pearson Education, Noida, 3rd Edition, 2018.

REFERENCES:

1. Jan M Rabaey, A Chandrakasan and Nikolic B, "Digital Integrated Circuits: A Design Perspective", Pearson Education, Noida, 3rd Edition 2017.
2. James D. Plummer, Michael D. Deal and Peter B. Griffin, "Silicon VLSI Technology:Fundamentals Practice and Modeling", Pearson Education, 2017.
3. Neil H E Weste, David Money Harris, "CMOS VLSI Design: A Circuits and Systems Perspective", Pearson Education, New Delhi, 4th Edition, 2017.
4. Wayne Wolf, "Modern VLSI Design: System on Chip Design", Pearson Education, New Delhi, 2015.

23U012 DIGITAL IMAGE PROCESSING

3 0 0 3

DIGITAL IMAGE FUNDAMENTALS: Pixel-Relationship between pixels - Gray level - resolution - image sampling and quantization - - Image sensors - elements of image processing system. (9)

IMAGE TRANSFORM: Need for image transform - Fourier transform - Discrete Cosine Transform - KL transform - Singular Value Decomposition -Introduction to Wavelet transform. (9)

IMAGE ENHANCEMENT: Gray level transformation - histogram equalization - spatial domain filtering: smoothing, sharpening filters - frequency domain filters. (9)

IMAGE DENOISING: Types of noise in digital image - Methods to minimize impulse noise: Median filter and its variants - Spatial domain and frequency domain approach to minimize speckle noise - methods to minimize periodic noise. (9)

IMAGE SEGMENTATION AND COMPRESSION: Point, line and edge detection - Different edge detection operators - thresholding approach - region based segmentation - Watershed algorithm - lossless and lossy compression - transform based image compression- image compression standard. (9)

Total L: 45

TEXT BOOKS:

1. Rafael C Gonzalez and Richard E Woods, "Digital Image Processing", Pearson, New York, 4th Edition, 2018.
2. Anil K Jain, "Fundamentals of Digital Image Processing", Pearson Education Ltd, 4th Edition, 2019.

REFERENCES:

1. Milan Sonka, Vaclav Hlavac, Roger Boyle, "Image Processing, Analysis, and Machine Vision", Springer New York, 1st Edition, 2013.
2. Alan C Bovik, "Handbook of Image and Video Processing", Elsevier, 3rd Edition, 2010.
3. John W. Woods, "Multidimensional Signal, Image, and Video Processing and Coding", Academic Press, 2nd Edition 2011.
4. Kenneth R Castleman, "Digital Image Processing", Pearson, 2nd Edition, 2011.

23U013 COMPUTER ARCHITECTURE

3 0 0 3

BASIC COMPUTER ORGANISATION: Classification of Architectures: CISC and RISC- Instructions - Timing and Control - Instruction Cycle - Fetch and Decode - Execution - Processor clock- Measuring computer system performance (9)

CENTRAL PROCESSOR ORGANISATION: Introduction to CPU design- Hard wired control- Micro-programmed control- General register organization - Stack organization - Instruction formats - Addressing modes - Data transfer and manipulation - Program control - Address sequencer - Data path structure. (9)

ARITHMETIC PROCESSING: Number system and representation- IEEE 754 representation- Addition, Subtraction, Multiplication and division Algorithms- Role of Flag register- Fixed -point and Floating point Arithmetic operations - Design of Arithmetic units (9)

MEMORY AND INPUT/OUTPUT ORGANISATION: Basic concepts - Memory Hierarchy - Main memory - Auxiliary memory - Associative memory - Cache and Virtual memory concepts - Performance considerations- Input - Output interface - Modes of transfer : Asynchronous Data transfer - Direct memory access. (9)

INTRODUCTION TO PARALLEL PROCESSING: Parallelism in uniprocessor systems- Pipeline execution- Instruction level parallelization- Parallel Computer architecture classifications - Data level parallelization - Graphical Processing Unit - Typical applications. (9)

Total L: 45

TEXT BOOKS:

1. David A. Patterson, John L. Hennessy, "Computer Organization and Design: The Hardware/Software Interface", Elsevier, Boston, 4th Edition, 2009.
2. Morris Mano. M , "Computer System Architecture", Prentice Hall of India, New Delhi , 3rd Edition, 2008.

REFERENCES:

1. John P Hayes, "Computer Architecture and Organization", Tata McGraw Hill, New Delhi, 8th Edition, 2002.
2. William Stallings, "Computer Organization and Architecture - Designing for Performance", Prentice Hall of India, New Delhi, 11th Edition, 2019.
3. Behrouz Parhami , "Computer Architecture from up to Super Computer", Oxford press, Kolkata, Reprinted, 2014.
4. Carl Hamacher, Zvonko Vranesic, Safwat zaky , " Computer Organization and Embedded Systems ", Tata McGraw-Hill, New York, 6th Edition, 2012

23U014 OPERATING SYSTEMS

3 0 0 3

INTRODUCTION: Operating system objectives and functionalities - Types of Operating System - Structure of Operating system - Computer architecture support to operating systems: - Instruction execution, Interrupts, Memory hierarchy, - Cache memory, Direct Memory Access - Multiprocessor and Multicore organization - Time sharing - system programs and calls (8)

MEMORY MANAGEMENT: Single contiguous allocation - Partitioned allocation - Paging - Virtual memory concepts - Swapping - Demand paging - Page replacement algorithms - Segmentation - Segmentation with paging. (8)

PROCESS MANAGEMENT : Introduction to processes - Scheduling objectives - Scheduling Criteria - Types of scheduling algorithms - Performance comparison - Inter- process communications - Synchronization - Semaphores - Types of Semaphores - Deadlock: - Principles of Deadlock - Deadlock Prevention - Deadlock Avoidance - Deadlock Detection and Recovery (12)

DEVICE INPUT OUTPUT AND FILE MANAGEMENT : Principles of I/O hardware and software - Device controllers - Device drivers - Interrupt driven device management - Interaction between operating system, - drivers and devices, File Systems - Files-Directories - File system implementation - Allocation methods - Security - Protection mechanisms. (8)

REAL-TIME OPERATING SYSTEMS : Characteristics of real-time operating systems - classification of real-time systems, - architectures of real-time systems, - micro-kernels, Memory management schemes - scheduling in RTOS - rate monotonic scheduling, - Inter-task communication, Shared data problem, - priority inversion - Selection of RTOS - Design and implementation of a multitasking application using RTOS (9)

Total L: 45

TEXT BOOKS:

1. Dhamdhere D M , "Operating Systems: A Concept-based Approach", Tata McGraw Hill, New Delhi, 2nd Edition, 2011.
2. William Stallings , "Operating Systems: Internals and Design Principles", Pearson, Noida, 2nd edition, 2018.

REFERENCES:

1. Silberschatz A, Galvin P and Gagne G , "Operating Systems Concepts", 6th edition, John Wiley and Sons, New York, 2010.
2. Andrew S Tanenbaum , "Modern Operating System", Pearson, Noida, 3rd edition, 2018.
3. Jane W S Liu , "Real Time Systems", Pearson, Noida, 3rd edition, 2016.
4. Deitel H M , "Operating Systems", Prentice Hall of India, New Delhi, 3rd edition ,2009

23U015 SAFETY INSTRUMENTED SYSTEMS

3 0 0 3

INSTRUMENTATION STANDARDS: Significance of codes and standards – Overview of various types of codes and standards - Introduction of various instrumentation standards – review, interpretation and significance of specific standards - Examples: Usage of standards on specific applications. (9)

INTRODUCTION TO SAFETY INSTRUMENTATION: Hazards and Risk – Process Hazards Analysis (PHA) – Safety Life Cycle - Allocation of safety functions to protective layers - SIS design and engineering. (9)

PROTECTION LAYERS: Process plant design - Process control system - Alarm systems - Physical protection. Mitigation layers:- Containment system - Scrubbers and flares - Fire and gas systems - Evacuation procedure. (9)

SAFETY INTEGRITY LEVEL: SIL determination methods - ALARP - Risk matrix - Risk graph - LOPA - Examples for design of SIL. (9)

SELECTION OF TECHNOLOGY: Relay systems - Solid-state systems - Microprocessor based systems - PLC based systems - Safety PLCs - Safety system complexity - Communication with other systems. (9)

Total L: 45

TEXT BOOKS:

1. Paul Gruhn, Harry Cheddie , "Safety Instrumented Systems: Design, Analysis, and Justification", ISA, 2nd Edition, 2006.
2. Glisente Landrini, Basilio Abbamonte, Tino Vandecapelle , "Safety Instrumented System", GM International, 5th Edition, 2017.

REFERENCES:

1. W.M.Goble, Harry Cheddie , "Safety Instrumented Systems Verification: Practical Probabilistic Calculations", ISA, 2005.
2. Lawrence D. Goettsche , "Maintenance of Instruments and Systems", ISA, 2nd Edition, 2005.
3. Swapan Basu , "Safety Instrumentation Systems", Academic Press, 2017.
4. Bela G. Liptak , "Instrument Engineers' Handbook", CRC Press, 4th Edition, 2003.

23U016 PRODUCT DESIGN AND DEVELOPMENT

3 0 0 3

INTRODUCTION: Product Development- Successful Product Development-Duration and Cost- Challenges. Product Planning: Planning Process- Identifying Customer Needs. (10)

PRODUCT SPECIFICATIONS: Establishing Target Specification- Setting the Final specification. Concept Generation: The Activity of Concept Generation. Concept Selection: Concept Screening-Concept Scoring- Concept Testing. (8)

PRODUCT ARCHITECTURE: Implications-Establishment- Delayed Differentiation - Platform Planning - Related System Level Design Issues. Industrial Design: Assessing the Need-The Impact-The Industrial Design Process-Management and Assessment. Design for Manufacturing. Prototyping – Basics – Principles - Prototyping Technologies - Planning. Robust Design. (10)

PATENTS AND INTELLECTUAL PROPERTY: Formulate a Strategy and Plan- Study Prior Inventions- Outline Claims- Write the Description of the Invention- Refine Claims- Pursue Application- Reflect on the Results and the Process. (7)

PRODUCT DEVELOPMENT ECONOMICS: Elements of Economic Analysis - Build a Base - Case Financial Model- Perform Sensitivity Analysis- Use Sensitivity Analysis to Understand- Consider the influence of the Qualitative Factors on Project Success- Carrying out Qualitative Analysis. Managing Projects: Understanding and Representing Tasks-Baseline Project Planning-Project Execution-Postmortem-Project Evaluation - Certification. (10)

Total L:45

TEXT BOOKS:

1. Karl. T. Ulrich and Steven D Eppinger, "Product Design and Development", Irwin McGraw Hill, 7th Edition, 2020.
2. Kevin Otto, Kristin Wood, "Product Design", Pearson Education, 2nd Edition, 2013.

REFERENCES:

1. Ali Jamnia, "Introduction to Product Design and Development for Engineers" Taylor and Francis, 1st Edition, 2018.
2. A C Chitale and R C Gupta, "Product Design and Manufacturing", Prentice Hall of India, New Delhi, 6th Edition, 2014.
3. Timjones, "New Product Development", Butterworth Heinmann, 1st Edition, 1997.
4. GeofferyBoothroyd, Peter Dewhurst and Winston Knight, "Product Design for Manufacture and Assembly", Taylor and Francis, 3rd Edition, 2010.

23U017 ADVANCED DIGITAL SIGNAL PROCESSING**3 0 0 3**

MULTIRATE SIGNAL PROCESSING: Review of discrete time system - Decimation- Interpolation –time-domain characterization-frequency-domain characterization- Multirate identities – Polyphase representations. (9)

FILTERBANK: Two channel filter bank – Perfect Reconstruction Criterion – Polyphase representation – M-channel filter bank – Modulated filter bank. Applications of filterbank in subband coding of speech and image signal. (9)

TIME-FREQUENCY ANALYSIS: Fourier Transform – Limitation of Fourier Transform – Short Time Fourier Transform (STFT) – Properties of STFT – Limitations of STFT. (9)

TIME – SCALE ANALYSIS: Introduction to Wavelets – Continuous Wavelet Transform – Properties – Discrete Wavelet Transform – Relation between Wavelet and Filter bank – Applications of Wavelets. (9)

ADAPTIVE FILTER : Need for Adaptive filter - Wiener filter - limitations - Gradient Descent algorithm - LMS algorithm - variants of LMS algorithm - introduction to RLS algorithm - Applications of adaptive filter (9)

Total L: 45**TEXT BOOKS:**

1. VaidyanathanP.P, "Multirate Systems and Filter Banks", Pearson Education, 3rd Edition ,2023.
2. VikramM.Gadre, Aditya S. Abhyankar, "Multiresolution and Multirate Signal Processing", McGraw Hill, 1st Edition ,2017.

REFERENCES:

1. Behrouz Farhang-Boroujeny , "Adaptive Filters: Theory and Applications", Wiley, 2nd Edition, 2013.
2. Sidney Burrus, Ramesh A. Gopinath, Hailong Guo , "Introduction to Wavelets and Wavelet Transforms: A Primer", Prentice Hall, 2nd Edition, 2005.
3. Martin Vetterli and JelenaKovacevic, "Wavelets and Subband Coding", Create Space Independent Publishing Platform, 3rd Edition, 2018.
4. Simon Haykin, "Adaptive Filter Theory", Pearson, 5th Edition 2014

23U018 ANALYTICAL INSTRUMENTATION**3 0 0 3**

INTRODUCTION TO SPECTROSCOPY: UV-Visible Spectroscopy - Electromagnetic radiation and its interaction with matter - Absorption spectroscopy - Absorption laws - Types of electronic transitions - Radiation sources - Monochromators - Filters - Prisms - Diffraction gratings - Detectors - Choice of solvents for UV -Visible spectrometers - single-beam and double-beam instruments. (9)

INFRA-RED SPECTROSCOPY: Basic principles - Hooks law - Calculation of wave numbers - Degrees of freedom - Types of vibrations - IR sources - Cells - Detectors - Sample preparation. Analysis using Attenuated Total Reflectance (ATR) - Atomic absorption spectrometry (AAS) - Wavelength choice - Sources - Cells - Detectors. Flame emission spectrometry. Atomic fluorescence spectrometry. (9)

NMR AND X-RAY SPECTROSCOPY: Nuclear magnetic Resonance (NMR) spectroscopy - Basic principles -The NMR phenomenon - Magnetic moments - Number of signals - Chemical shift - Continuous wave NMR spectrometer - Pulsed Fourier Transform NMR spectrometer - NMR applications. X-ray absorption methods - X-ray fluorescence methods - X-ray diffraction. Radioactive measurement - Units of radioactivity - Application of radio nuclides in analysis - Radioactivity detectors. (9)

CHROMATOGRAPHY: Basic principles of chromatography - Types of Chromatography - Gas chromatography - Column details - Detectors for chromatography - Thermal conductivity detector - Flame ionization detector - Flame photometric detector - Electron capture detector - Effect of temperature programming - High pressure liquid chromatography (HPLC). Mass spectroscopy - Instrumentation - Base peak - Molecular ion peak - Metastable peak - Isotopic peaks - Applications. (9)

SAMPLING AND ELECTROCHEMICAL TECHNIQUES: Sample collection for gas, liquid and solid analysis. Ion selective electrodes - Glass and reference electrodes - pH measurement - pH meter and its calibration. Electrical conductivity measurement - Measuring circuit. Flue gas analysis for pollution control - Measurement of Oxygen, Carbon monoxide, Carbon dioxide, NOX and SOX, dust and smoke measurement, Continuous Emission Monitoring system. (9)

Total L:45**TEXT BOOKS:**

1. Willard H H, Merit L.L, Dean J A and Seattle F L, "Instrumental Methods of Analysis", CBS Publishing and Distribution, New Delhi, 7th Edition, 1996.
2. Ewing G W, "Instrumental Methods of Analysis", McGraw-Hill, New York, 5th Edition, 2013.

REFERENCES:

1. Robert D Braun, "Introduction to Instrumental Analysis", BSP Books P Ltd, Hyderabad, 2nd Edition, 2019.

2. Skoog D A, James Holler F and Stanley R Crouch, "Principles of Instrumental Analysis", Cengage India P Ltd, New Delhi, 7th Edition, 2018.
3. Liptak B G, "Instrument Engineers Handbook: Process Measurement and Analysis", CRC Press, Boca Raton, 5th Edition, 2016.
4. Khandpur R S, "Handbook of Analytical Instruments", McGraw-Hill Education (I) P Ltd, Chennai, 5th Edition, 2018.

23U019 SMART SENSORS AND ACTUATORS

3 0 0 3

SENSOR SYSTEMS IN ENGINEERING: Role of sensors and sensor systems – Innovative sensor Technologies – Application scenarios - Instrumentation Process – Instrumentation Steps – Application examples. Smart sensor basics - General sensing system – Classical sensor Model – Smart sensor model – Monolithic integrated smart sensor - Hybrid integrated smart sensor (9)

SIGNAL CONDITIONING FOR SMART SENSORS: Instrumentation Amplifier – Step mode operational amplifier – Rail to rail Op-Amp – Switched Capacitor Amplifier – 4 to 20 mA Signal Transmitter – Inherent power supply rejection – Separate Versus Integrated Signal Conditioning – Digital Conversion. (9)

INTEGRATED SMART SENSORS: Monolithic sensor interface – MCU's for sensor interface – DSP for sensor interface – Techniques and system considerations: Linearization – PWM Control – Auto-zero and Auto range – Diagnostics. Software tools and support – Sensor integration – Alternative views of smart sensing. (9)

MICRO MACHINED ACTUATORS: Micro valves – Micro motors – Micro Pumps – Micro dynamometer – Micro steam engines – Actuators in other semiconductor materials – Various Micromachined structures: Cooling channels – Micro optical actuator – Micro grippers. (8)

COMMUNICATION AND STANDARDS FOR SMART SENSORS: Automotive Protocols: SAEJ 1850 – CAN protocol – Industrial networks. Industrial usage of CAN – Protocols in Silicon: MI-Bus. IEEE 1451 family of standards – Extending the system to network. (10)

Total L:45

TEXT BOOKS:

1. Randy Frank, "Understanding Smart Sensors", Artech House, 3rd Edition, 2013.
2. Ananthasuresh G K, Vinoy K J, Gopalakrishnan S, Bhat K N, Aatre V K, "Micro and Smart Systems", Wiley Publishers, 2011.

REFERENCES:

1. Subhas Chandra Mukhopadhyay, "Smart Sensors, measurement and Instrumentation", Springer Heidelberg, New York, 2013.
2. Clarence W de Silva, "Sensors and Actuators: Control Systems Instrumentation", CRC Press, New York, 2nd Edition, 2014.
3. Tai Ran Hsu, "MEMS and Microsystems: Design and Manufacture", Tata Mcgraw Hill Publishing Co Ltd, 2017.
4. Gerord C M Meijer, "SmartSensor Systems", John Wiley and Sons, 2008.

23U020 ELECTRIC VEHICLE SYSTEM MODELING AND CONTROL

3 0 0 3

ARCHITECTURE AND MODELING: Classification of electric vehicles- benefits and challenges- general architecture- power flow- Basic components: Prime mover, Motors with converters, Energy storage systems, Transmission systems- Modeling of IC engine, motor, transmission system and vehicle body. (9)

ENERGY STORAGE SYSTEM MODELING AND CONTROL: Methods of determining the State of Charge (SoC)- Current and voltage based SoC Determination Methods- Combination of SoCs Estimated Through Different Approaches- Estimation of battery power availability- Cell balancing - Battery system efficiency- Utilization of solar power. (9)

ENERGY MANAGEMENT STRATEGIES: Rule-based Energy Management Strategy- Fuzzy-logic-based Energy Management Strategy- Cost-function-based Optimal Energy Management Strategy- Optimal Energy Management Strategy Incorporated with Cycle Pattern Recognition- Driving Cycle/Style Pattern Recognition Algorithm. (9)

ELECTRIC VEHICLE CONTROL LOOPS: Engine torque fluctuation dumping control- High-voltage bus spike control- Thermal control- Traction torque control- Anti-rollback control- Active suspension control- - Noise and vibration control (9)

DESIGN CONSIDERATIONS: Design requirements of motor drives: DC drives, Induction motor drives, Permanent magnet drives, Switched Reluctance motor drives- Aerodynamic considerations- Consideration of vehicle mass- Alternate energy resources (9)

Total: L:45

TEXTBOOKS

1. Liu Wei, "Hybrid electric vehicle system modeling and control", Wiley & Sons, Second Edition, 2017.
2. Emanuele Crisostomi, Robert Shorten, Sonja Stüdli and Fabian Wirth, "Electric and Plug-in Hybrid Vehicle Networks, Optimization and Control" Taylor & Francis, 2017.

REFERENCES

1. K.T. Chau, "Electric Vehicle Machines and Drives: Design, Analysis and Application", Wiley-IEEE press, 2015.
2. James Larminie and John Lowry, "Electric Vehicle Technology Explained", Wiley-Blackwell, 2003.
3. C.C. Chan and K.T.Chau, "Modern Electric Vehicle Technology", Oxford University Press, 2001.
4. Kwang Hee Nam, "AC Motor Control and Electrical Vehicle Applications", CRC Press, 2018.

23U021 COMPUTER NETWORKS

3 0 0 3

INTRODUCTION TO DATA COMMUNICATION: Characteristics and Components of Data Communication, Protocols and Standards, Network topology, Transmission Impairment, Network performance, Transmission media-guided media and unguided media. (9)

COMPUTER COMMUNICATION ARCHITECTURE: Network Models, Layered tasks, OSI model-functions of layers in OSI model, Switching-Circuit switched networks, Datagram networks and Virtual circuit networks. (9)

DATA LINK LAYER & MEDIUM ACCESS: Introduction to LANs- Link Layer Addressing, Datalink layer protocols- Medium access control techniques - CSMA, CSMA/CD, Token passing, Connecting Devices. (9)

NETWORK LAYERS: Introduction to TCP/IP Protocol suite, Logical Addressing- IPv4- Classful and Classless Addressing, IPv6 addresses, IPv4 and IPv6 Header format, Transition from IPv4 to IPv6. (9)

TRANSPORT AND APPLICATION LAYER: Client- Server Architecture, Connectionless protocols- UDP, Connection oriented protocols- TCP, Application Layer protocols – WWW and HTTP, Network Security. (9)

Total L: 45

TEXTBOOKS:

1. Behrouz A. Forouzan, Sophia Chung Fegan , "Data Communications and Networking", 4th Edition, Tata McGrawHill, 2017.
2. Douglas E. Comer, "Internetworking with TCP/IP Principles, Protocols and Architecture Volume-I", 6th Edition, Pearson, 2013.

REFERENCES:

1. Larry L. Peterson, Bruce S. Davie, "Computer Networks: A Systems Approach", Fourth Edition, Elsevier Publishers, 2007.
2. Stallings W , "Data and Computer Communications", Tenth Edition, Prentice Hall, New Delhi, 2017.
3. Andrew S. Tanenbaum, David J. Wetherall , "Computer Networks", Fifth Edition, Pearson, 2013.
4. Buchanan W , "Computer Buses", Arnold, 2000.

23U022 DIGITAL TWIN

3 0 0 3

FUNDAMENTALS OF DIGITAL TWIN: Overview of Digital Twins Historical context and evolution – Definition and Concept of Digital Twin - Digital Twins maturity models - Significance in digital transformation and modern industries – Benefits and Challenges. (9)

KEY ENABLING TECHNOLOGIES FOR DIGITAL TWIN: Definition and Fundamentals: IoT and Sensor Networks – Data Analytics and Machine Learning – Cloud Computing and Edge Computing – Extended Reality - Case studies. (9)

MODELING AND SIMULATION IN DIGITAL TWIN: Different types of Digital Representation of Physical Systems – Approaches for different Simulation Techniques – Real-Time Monitoring and Control applications. (9)

DATA INTEGRATION AND INTEROPERABILITY: Challenges in Data Integration – Interoperability Standards and Protocols – Integration with Enterprise Systems. (9)

SECURITY AND ETHICAL CONSIDERATIONS: Basics of Cybersecurity, Necessity of Cybersecurity in Digital Twins – Privacy and Data Governance – Risk Management and Resilience. (9)

Total: L:45

TEXTBOOKS

1. Ranjan Ganguli, Sondipon Adhikari, Souvik Chakraborty, Mrittika Ganguli, "Digital Twin: A Dynamic System and Computing Perspective", CRC Press, 2023.
2. Diego Galar, Uday Kumar, and Raul Poler "Digital Twin: Manufacturing Excellence through Virtual Factory Replication", Springer, 2020.

REFERENCES

1. Perry Lea, "Internet of Things for Architects", Packt Publishing, 2018.
2. Andrea D'Ambrogio, Andreas Tolk, Daniele Gianni, "Modeling and Simulation-Based Systems Engineering Handbook", CRC Press, 2018.
3. Anthony Giordano, "Data Integration Blueprint and Modeling", IBM Press, 2011.
4. Liming Wang, Stefan Poslad, and Boon-Chong Seet, "Cybersecurity and Privacy in Cyber-Physical Systems: Foundations, Principles and Applications" CRC Press, 2017.

23U023 INSTRUMENTATION FOR FLOWSHEETS

3 0 0 3

FLOWSHEETING: Introduction, Symbols, Flow sheet presentation with examples, Manual flow sheet calculation, Constraints and their applications in flow sheet calculations, Types of flow sheets, Synthesis of steady state flow sheet, Flow sheeting software. (9)

PHYSICAL PROPERTY CALCULATIONS: The data cycle- computerized physical property systems- physical property calculations- Degrees of freedom in a flow sheet: degrees of freedom- independent stream variables-degrees of freedom for a unit-degrees of freedom for a flowsheet. (9)

FLWSHEETING BY EQUATION SOLVING METHODS: Selection, decision and convergence of variables in a flow sheet with simple and complex examples. (9)

FLWSHEET APPLICATIONS: P & I D development, typical stages of P & I D, Applications of P & I D in design stage- Construction stage - Commissioning stage - Operating stage - Revamping stage - Applications of P & I D in HAZOPS and Risk analysis. (9)

INSTRUMENTATION AND FLWSHEET: Flow sheet with recycle- The P& I Diagram- Basic Symbols and Layout- General Instruments and Controller symbols- Type of Instruments- Valves - Selection of Valves- Pumps, Compressor-Selection of Pumps and Compressors. (9)

Total L: 45

TEXT BOOKS

1. A.N. Westerberg, H.P.Hutchison, R.L. Motard, P. Winter, "Process Flowsheeting", Cambridge University Press, 2011.
2. Moe Toghræi, "Piping and Instrumentation Diagram Development", John Wiley & Sons Inc., 2019.

REFERENCES

1. GawinTowler, Ray Sinnott, "Chemical Engineering Design", Elsevier, 2012.
2. Max. S. Peters and K.D.Timmerhaus, "Plant Design and Economics for Chemical Engineers", McGraw Hill, Inc., New York, 2017.
3. Paul Benedek, "Steady State Flowsheeting of Chemical Plants", Elsevier Scientific Publishing Company,1980.
4. Avinashkumar.V.Karre, "Piping and Instrumentation Diagram- A Stepwise Approach", Walter de Gruyter, 2023.

23U024

INDUSTRIAL CYBERSECURITY

3 0 0 3

INTRODUCTION: Industrial security environment - Industrial Control System (ICS) – Purdue model -Operation Technology Vs Information Technology - Cyberattacks: Threat sources and steps to successful cyberattacks - Issues in ICS security - Information security - Approaches of information security – Applications – Overview of ISA/IEC 62443 Standards- Introduction to Cyber Resilience Act (CRA). (9)

RISK ANALYSIS: Risk identification - classification and assessment, Addressing risk: Cybersecurity Management System (CSMS), organizational security, physical and environmental security, network segmentation, access control, risk management and implementation. (9)

ACCESSING THE CYBERSECURITY OF ICS: Identifying the scope of the ICS - generation of cybersecurity information - identification of vulnerabilities - risk assessment - evaluation of realistic threat scenarios - Gap assessment-capturing Ethernet traffic - documentation of assessment result. (9)

CYBERSECURITY DESIGN AND IMPLEMENTATION: Cybersecurity lifecycle - conceptual design process - detailed design process - firewall design remote access design - intrusion detection design - Applications of AI in Cybersecurity. (9)

TESTING AND MAINTENANCE: Developing test plans - cybersecurity factory acceptance testing- site acceptance testing - network and application diagnostics and troubleshooting - cybersecurity audit procedure - ICS incident response- ICS attack Case studies. (9)

Total L: 45

TEXT BOOKS:

1. Charles J. Brooks, Philip A. Craig Jr., "Practical Industrial Cybersecurity", Wiley, 2022.
2. Bipin Lokegaonkar and Anand Shinde, "Industrial Cybersecurity: A Practical Approach to Operational Technology Protection", Notion Press, 2024.

REFERENCES:

1. Ronald L and Krutz, "Industrial Automation and Control System Security Principles", ISA, 2017.
2. Steve Mustard, "Industrial Cybersecurity: Case Studies and Best Practices", ISA, 2022.
3. Jean-Marie Flaus, "Cybersecurity of Industrial Systems", Wiley, 2019.
4. Keith Stouffer, Michael Pease, Chee Yee Tang, Timothy Zimmerman, Victoria Pillitteri, Suzanne Lightman, Adam Hahn, Stephanie Saravia, Aslam Sherule, Michael Thompson, "Guide to Operational Technology (OT) Security", NIST, U.S Department of Commerce, 2023.

PROFESSIONAL ELECTIVE VERTICALS FOR HONORS

CONTROL AND INDUSTRIAL AUTOMATION

23U031 OPTIMAL AND ADAPTIVE CONTROL SYSTEMS

3 0 0 3

OPTIMAL CONTROL: Statement of optimal control problem - Problem formulation and types of optimal control – Selection of performance measure, cost function and norms – Hamilton Jacobi Equation - Pontryagin's minimum principle – State inequality constraints – Necessary conditions for optimality –optimal control problems by Transfer function approach and State variable approach. (10)

LINEAR QUADRATIC CONTROL PROBLEMS: Choice of weighting matrices- LQG control – Matrix Ricatti equation and solution methods of state regulator and discrete systems – Optimal control law - Optimal estimation. (8)

IDENTIFICATION METHODS: Conventional techniques of identification – Identifications of systems with dead time – Discrete systems – ARMA process – Discrete state model – Least squares techniques – Recursive Least Squares algorithms – Minimum variance method. (9)

INTRODUCTION TO ADAPTIVE CONTROL: Development of adaptive control problem-The role of Index performance (IP) in adaptive systems- Development of IP measurement process model. Pole placement design - Self tuning controller – Minimum variance controller. (8)

ADAPTIVE CONTROL OF DETERMINISTIC SYSTEMS : Gain scheduling controller – Model reference adaptive control – Adaptive predictive control – The MIT rule – Determination of adaptation gain - Minimum prediction error adaptive controls – Adaptive control of time varying systems. BIBO Stability – Model free adaptive control - Applications of adaptive control. (10)

Total L: 45

TEXT BOOKS:

1. Donald E. Kirk, "Optimal Control Theory: An Introduction", Dover Publications, New York, 2004.
2. Karl J Astrom, Bjorn Wittenmark , "Adaptive Control, Dover Publications, New York, 2nd Edition, 2013.

REFERENCES:

1. Desineni Subbaram Naidu, "Optimal Control Systems", CRC Press, 2009.
2. Arun K. Tangirala, "Principles of System Identification: Theory and Practice", CRC Press, 1st Edition, 2014.
3. Nagrath I J, Gopal M , "Control Systems Engineering", New Age International Pvt Ltd, 6th Edition, 2018.
4. LjungL, "System Identification: Theory for the User", Prentice Hall, Englewood Cliffs, 1999.

23U032 NON - LINEAR SYSTEMS THEORY

3 0 0 3

NONLINEAR SYSTEMS: State space representation of nonlinear systems - Autonomy - Basic characteristics of nonlinear systems - Bifurcation - Chaos - Limit cycles -Types of nonlinear elements (9)

MATHEMATICAL BACKGROUND: Manifolds - Tangent and cotangent Spaces - Vector fields and flows - Lie bracket and Lie derivatives - Distributions and co distributions - Frobenius theorem (9)

PHASE PLANE ANALYSIS: Concepts of phase plane analysis - Phase portraits - Construction of phase portrait - Isocline and delta methods - Singular points - Phase plane analysis of linear system and nonlinear system - Existence of limit cycles (9)

DESCRIBING FUNCTION: ANALYSIS: Describing function fundamentals - Computing describing functions for common nonlinearities in control systems - Describing functions analysis of nonlinear systems - Stability analysis (9)

STABILITY ANALYSIS: Linearization method - Stability analysis based on Lyapunov's direct method - Krasovski's method - Variable gradient method (9)

Total L: 45

TEXT BOOKS:

1. Jean-Jacques E. Slotine , "Applied Nonlinear Control", Prentice Hall Englewood Cliffs, New Jersey, 1991.
2. Khalil, H.K , "Nonlinear Systems", Pearson Education, India, 3rd edition, 2014.

REFERENCES:

1. Vidyasagar.M , "Nonlinear System Analysis", Prentice Hall Englewood Cliffs, New Jersey, 1978.
2. Strogatz, S. H , "Nonlinear Dynamics & Chaos, with Applications to Physics, Biology, Chemistry and Engineering", Westview Press, 2nd Edition, 2014.
3. Harry G. KwatnyGilmer L. Blankenship, "Nonlinear Control & Analytical Mechanics: a Computational Approach", Springer, 2017.
4. M.Gopal , "Digital Control and State Variable Methods", McGraw Hill Education, 4th Edition, 2017.

23U033 SYSTEM IDENTIFICATION

3 0 0 3

INTRODUCTION TO IDENTIFICATION AND MODELS : System Identification Procedure - Identifiability - Signal to Noise ratio - Over fitting - Models: Definition of a model, Classification of models - Models for discrete time linear time invariant Systems - Models for time varying systems and nonlinear systems (9)

GOODNESS OF ESTIMATORS: Types of estimation problems - Fisher information, Bias, Variance, Efficiency, Sufficiency, Cramer Rao's inequality - Asymptotic bias, Mean square error, Consistency (9)

ESTIMATION METHODS: Method of moments estimators - Least squares estimators - Nonlinear least squares - Maximum likelihood estimators - Bayesian estimators (9)

IDENTIFICATION OF NON-PARAMETRIC AND PARAMETRIC INPUT – OUTPUT MODELS : Identification of Non-Parametric Input Output Models:- Impulse response estimation - Step response estimation - Estimation of frequency response function - Identification of Parametric Input Output Models:- Prediction error minimization (PEM) methods - Properties of the PEM estimator (9)

STATISTICAL ELEMENTS OF MODEL BUILDING: Informative Data - Input design for identification – Data preprocessing - Model development (9)

Total L: 45

TEXT BOOKS:

1. Arun K Tangirala, "Principles of System Identification: Theory and Practice", CRC Press, Boca Raton, 1st Edition, 2014.
2. Lennart Ljung, "System Identification: Theory for the User", Prentice Hall, Englewood Cliffs, 2nd Edition, 1999.

REFERENCES:

1. Michel Verhaegen, Vincent Verdult, "Filtering and System Identification: A Least Squares Approach", Cambridge University Press, 2007.
2. Johan Schoukens, Rik Pintelon, Yves Rolain, "Mastering System Identification in 100 Exercises", Wiley-IEEE Press, 1st Edition, 2012.
3. Thomas Kailath, Ali H Sayed, Babak Hassibi, "Linear Estimation", Pearson, 1st Edition, 2000.
4. Jer Nan Juang, "Applied System Identification", Pearson, 1994.

23U034 ROBUST CONTROL

3 0 0 3

MATHEMATICAL PRELIMINARIES: - Norms: vector, matrix, signals and systems; Linear systems: controllability and stabilizability, observability and detectability, poles and zeros of transfer function matrix, LMI and Riccati equation - singular value decomposition(SVD) (4)

UNCERTAINTY AND ROBUSTNESS: Concepts of model uncertainty, including both parametric and dynamic uncertainty - Fundamental concept of robustness and the relationship between physical systems and mathematical models - Co prime Factorization and stabilizing controllers, SVD application to perturbation analysis. (5)

MODELLING OF UNCERTAIN SYSTEMS: Unstructured Uncertainties - Parametric Uncertainty - Linear fractional transformations and canonical forms - Structured Uncertainty - Robust stability and performance problems. (9)

ROBUST DESIGN SPECIFICATIONS: Small gain theorem and Robust Stabilization, Performance Consideration, Structured Singular Values. H – infinity design: Mixed Sensitivity H-infinity Optimization. H-infinity suboptimal solutions, Discrete time case. (9)

H-INFINITY LOOP SHAPING DESIGN: Robust Stabilization against normalised Coprime Factor Perturbations, Loop Shaping Design, Mixed Optimization Design Method with LSDP. μ - Analysis and Synthesis: Consideration of Robust performance, μ -synthesis- D-K Iteration method, μ -K Iteration method. (9)

LOWER ORDER CONTROLLERS: Absolute-error Approximation Methods, Reduction via Fractional Factors, Relative-error Approximation Methods, Frequency Weighted Approximation Methods. (9)

Total L: 45

TEXTBOOKS:

- 1 K. Zhou, J. C. Doyle and K. Glover, "Robust and Optimal Control", Prentice-Hall, Englewood Cliffs, NJ, 1995
- 2 Mackenroth U "Robust Control Systems, Theory and Case Studies", Springer India Pvt. Ltd, New Delhi, 2010.

REFERENCES:

1. Ian Postlethwaite, Sigurd Skogestad "Multivariable Feedback Control: Analysis and Design", Wiley, 2015.
2. Gu D W, Petkov P and Konstantinov M M, "Robust Control Design with MATLAB", Springer, 2013.
3. Kemin Zhou and John Doyle, "Essentials of Robust Control", Prentice-Hall Inc., 1998.
4. Zhou K, Doyle J C and Glover K, "Robust and Optimal Control", Prentice-Hall Inc., 1996.

23U035 SOLID STATE DRIVES

3 0 0 3

FUNDAMENTALS OF ELECTRICAL DRIVES AND CHARACTERISTICS: Introduction to Electrical drives, Fundamental torque equations, Speed torque conventions and multi-quadrant operation, Components of load torques, Nature and classification of load torques. (9)

DC MOTOR DRIVES: State space model of DC motor drive, Single-phase and Three-phase drives: Half converter, Semi converter, Full converter and Dual converter fed drives- Two quadrant and four quadrant chopper-controlled drives – Closed loop control of dc drives. (9)

INDUCTION MOTOR DRIVES: Performance characteristics, Stator Control: Stator voltage control, Rotor voltage control, Frequency control, Voltage and frequency control, Current control, Voltage, current and frequency control - Rotor resistance control: Conventional methods, Static rotor resistance control - Slip power recovery: Static Kramer drive, Static Scherbius drive. (9)

VECTOR CONTROL OF INDUCTION MOTOR DRIVES: Principle of vector control – Direct vector control - Flux vector estimation – Indirect vector control – Vector control of line-side PWM rectifier – Stator flux oriented vector control – Vector control of current fed inverter drive. (9)

SYNCHRONOUS AND SPECIAL DRIVES: Synchronous Motor Drives: Open loop volts/hertz control, Self control model Permanent magnet AC motor drives, Brushless dc motor drives, Sensorless control - Stepper motor and Switched reluctance motor drives. (9)

Total: L: 45

TEXTBOOKS:

1. Gopal K Dubey, "Fundamentals of Electric Drives", Narosa Publishing House, New Delhi, 2005.
2. Bimal K Bose, "Power Electronics and Variable Frequency Drives - Technology and Application", IEEE Press, New York, 1997.

REFERENCES

1. Ion Boldea and Nasar S A, "Electric Drives", CRC Press LLC, New York, 1999.
2. Krishnan R, "Electric Motor Drives: Modelling, Analysis and Control", Prentice Hall of India, New Delhi, 2002.
3. Muhammad H Rashid, "Power Electronics Handbook", Academic Press, 2001.
4. S.K. Pillai, 'A First Course on Electrical Drives', Wiley Eastern Limited, 1993.

23U036 FAULT DIAGNOSIS AND CONTROL

3 0 0 3

INTRODUCTION TO FAULT DIAGNOSIS: Scope - Types of faults and different tasks of Fault Diagnosis and Implementation - Different approaches to FDD: Model free and Model based approaches - Classification of Fault and Disturbances- Different issues involved in FDD- Typical applications. (9)

ANALYTICAL REDUNDANCY CONCEPTS: Mathematical representation of Faults and Disturbances: Additive and Multiplicative types – Residual Generation: Detection, Isolation, Computational and Stability properties – Design of Residual generator – Residual specification and implementation. (9)

DESIGN OF STRUCTURED RESIDUALS: Introduction- Residual structure of single fault Isolation: Structural and Canonical structures- Residual structure of multiple fault Isolation: Diagonal and Full Row canonical concepts – Introduction to parity equation implementation and alternative representation. (9)

DESIGN OF DIRECTIONAL RESIDUALS: Introduction – Directional Specifications: Directional specification with and without disturbances – Parity Equation Implementation – Linearly dependent column. (9)

FAULT TOLERANT CONTROL: Passive Fault-tolerant control- Active Fault-tolerant control - Hybrid Fault-tolerant control - Actuator and Sensor Fault tolerance Principles: Compensation for actuator - Fault-tolerant control design against major actuator failures - Case studies. (9)

Total L: 45

TEXTBOOKS:

1. Janos J. Gertler, "Fault Detection and Diagnosis in Engineering systems", Second Edition, CRC Press, 1998.
2. Mogens Blanke, Michel Kinnaert, Jan Lunze, Marcel Staroswiecki, "Diagnosis and Fault-Tolerant Control", Springer, 2016.

REFERENCES:

1. Steven X. Ding, "Model based Fault Diagnosis Techniques: Schemes, Algorithms, and Tools", Springer Publication, 2012.
2. Rolf Isermann, "Fault-Diagnosis Systems an Introduction from Fault Detection to Fault Tolerance", Springer Verlag, 2006.
3. Hassan Noura, Didier Theilliol, Jean-Christophe Ponsart, Abbas Chamseddine, "Fault-Tolerant Control Systems: Design and Practical Applications", Springer Publication, 2009.
4. Leo H. Chiang, Evan L. Russell, Richard D. Braatz, "Fault Detection and Diagnosis in Industrial Systems", Springer London, 2012

23U037 MODEL BASED CONTROL

3 0 0 3

INTRODUCTION TO MIMO CONTROL: Introduction to MIMO Systems-Multivariable control-Multiloop Control-Multivariable IMC, IMCPID-Case studies. (9)

MODEL PREDICTIVE CONTROL: Introduction to Model Predictive Control - Model Predictive Control Elements – Generalized Predictive Control Scheme – Multivariable Generalized Predictive Control Scheme – Multiple Model based Model Predictive Control Scheme Case Studies. (9)

STATE SPACE MODEL BASED PREDICTIVE CONTROL SCHEME: State Space Model Based Predictive Control Scheme - Review of Kalman Update based filters – State Observer Based Model Predictive Control Schemes – Case Studies. (9)

CONSTRAINED MODEL PREDICTIVE CONTROL: Constraints Handling: Amplitude Constraints and Rate Constraints – Constraints and Optimization – Constrained Model Predictive Control Scheme – Case Studies. (9)

ADAPTIVE CONTROL SCHEME: Introduction to Adaptive Control-Gain Scheduling-Self tuning regulators–MARS-Adaptive Model Predictive Control Scheme –Case Studies. (9)

Total L: 45

TEXTBOOKS:

1. Coleman Brosilow, Babu Joseph, "Techniques of Model-Based Control", Prentice Hall PTR Pub, 2002.
2. E. F. Camacho, C. Bordons, "Model Predictive Control", 2nd Edition, Springer-Verlag London Limited, 2007.

REFERENCES:

1. K.J. Astrom and B. J. Wittenmark, "Adaptive Control", Second Edition, Pearson Education Inc., 2013.
2. Paul Serban Agachi, Zoltan K. Nagy, Mircea Vasile Cristea, and Arpad Imre-Lucaci, "Model Based Control Case Studies in Process Engineering", WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim 2007.
3. Ridong Zhang, Anke Xue Furong Gao, "Model Predictive Control Approaches Based on the Extended State Space Model and Extended Non-minimal State Space Model", Springer Nature Singapore Pte Ltd. 2019.
4. J.A. Rossiter "Model-Based Predictive Control A Practical Approach", Taylor & Francis eLibrary, 2005.

SENSORS AND INSTRUMENTATION

23U041 STATE ESTIMATION

3 0 0 3

INTRODUCTION TO STATE ESTIMATION: Review of state observers for deterministic System- Least squares estimation- Recursive least squares- Wiener filter- Propagation of states and covariances. (8)

KALMAN FILTER: Discrete Kalman filter- Kalman filter generalization: Correlated process and measurement noise- Colored process and measurement noise- Constrained Kalman filter- Limitations of Kalman filter. (9)

NONLINEAR KALMAN-BASED FILTERS: Linearized Kalman filter – Extended Kalman filter –Second order Extended Kalman filter – Constrained Extended Kalman filter- Unscented Kalman filter - Unscented transformation - Simplex unscented transformation – Spherical unscented transformation - Constrained Unscented Kalman filter. (10)

PARTICLE FILTER: Introduction - SIS filtering algorithm - Degeneracy phenomenon - SIR Particle filter - Implementation issues: - Sample Impoverishment – Selection of proposal density: Particle filter with EKF as proposal - Unscented Particle filter. (10)

CASE STUDIES: Kalman filter- Extended Kalman filter- Unscented Kalman filter- Particle filter. (8)

Total: L:45

TEXTBOOKS:

1. Dan Simon, "Optimal State Estimation Kalman, H-infinity and Non-linear Approaches", John Wiley and Sons, 2006.
2. Branko Ristic, Sanjeev Arulampalam, Neil Gordon, "Beyond the Kalman Filter: Particle filters for Tracking Applications", Artech House Publishers, Boston, London, 2004.

REFERENCES:

1. Mohinder S. Grewal, Angus P. Andrews, "Kalman Filtering: Theory and Practice with MATLAB", John Wiley and Sons, 2015.
2. Chaw-Bing Chang, Keh-Ping Dunn, "Applied State Estimation and Association" MIT Press, 2016.
3. Bruce P. Gibbs, "Advanced Kalman Filtering, Least-Squares and Modeling: A Practical Handbook" John Wiley and Sons, 2011.
4. Arthur Gelb, "Applied Optimal Estimation", MIT Press, 1974.

23U042 MICRO ELECTRO MECHANICAL SYSTEMS

3 0 0 3

OVERVIEW OF MEMS AND MICROSYSTEMS: MEMS and Microsystem, Typical MEMS and Microsystems Products, Evolution of Microfabrication, Microsystems and Microelectronics, Multidisciplinary Nature of Microsystems, Miniaturization. Applications and Markets. (9)

WORKING PRINCIPLES OF MICROSYSTEMS: Introduction, Microsensors, Microactuation, MEMS with Microactuators, Microaccelerometers, Microfluidics. Engineering Science for Microsystems Design and Fabrication: Introduction, Molecular Theory of Matter and Inter-molecular Forces, Plasma Physics, Electrochemistry. (9)

ENGINEERING MECHANICS FOR MICROSYSTEMS DESIGN: Introduction, Static Bending of Thin Plates, Mechanical Vibration, Thermo mechanics, Fracture Mechanics, Thin Film Mechanics, Overview on Finite Element Stress Analysis. (9)

SCALING LAWS IN MINIATURIZATION: Introduction, Scaling in Geometry, Scaling in Rigid-Body Dynamics, Scaling in Electrostatic Forces, Scaling in Fluid Mechanics, Scaling in Heat Transfer. (9)

OVERVIEW OF MICROMANUFACTURING: Introduction, Bulk Micromanufacturing, Surface Micromachining, The LIGA Process, Summary on Micromanufacturing. (9)

Total: L:45

TEXTBOOKS:

1. Tai-Ran Hsu, "MEMS and Micro systems: Design, Manufacture and Nanoscale Engineering", Second Edition, Wiley.
2. Chang Liu, "Foundations of MEMS", Second edition, Pearson, 2011.

REFERENCES:

1. Hans H. Gatzert, Volker Saile, Jurg Leuthold, "Micro and Nano Fabrication: Tools and Processes", Springer, 2015.
2. Dilip Kumar Bhattacharya, Brajesh Kumar Kaushik, "Microelectromechanical Systems (MEMS)", Cengage Learning.
3. Marc Madou, Fundamentals of Microfabrication and Nanotechnology, Third Edition, CRC Press, 2011
4. Stephen D. Senturia, "Microsystem Design", Springer, 2004

23U043 MULTISENSOR DATA FUSION

3 0 0 3

MULTISENSOR DATA FUSION: Introduction –Sensors and sensor data - Use of multiple sensors - Fusion applications. The inference hierarchy: Output data - Data fusion model- Architectural concepts and issues - Benefits of data fusion. Mathematical tools used: Algorithms, co-ordinate transformations, rigid body motion. Dependability and Markov chains- Meta-heuristics. (9)

ESTIMATION: Kalman filtering - Practical aspects of Kalman filtering - Extended Kalman filter. Decision level identify fusion - Knowledge based approaches. Data information filter- Extended information filter. (8)

DECENTRALIZED ESTIMATION: Decentralized and scalable decentralized estimation - Sensor fusion and approximate agreement - Optimal sensor fusion using range trees recursively - Distributed dynamic sensor fusion - Limitations of fully connected decentralization. (9)

ALGORITHMS FOR DATA FUSION: Taxonomy of algorithms for multisensor data fusion - Data association- Identity declaration. High performance data structures: Tessellated, trees, graphs and function. Representing ranges and uncertainty in data structures - Designing optimal sensor systems with in dependability bounds - Implementing data fusion system. (10)

STATE ESTIMATION AND LOCALIZATION FOR SELF-DRIVING CARS: Use of Kalman filter variants- H^∞ filtering- GNSS/INS sensing for position and orientation estimation- Basics of LIDAR sensing - Fusion of sensor data for an autonomous vehicle state estimator. (9)

Total: L:45

TEXTBOOKS:

1. David L. Hall, Sonya A H McMullen, "Mathematical Techniques in Multisensor Data Fusion", Artech House, Boston, Second Edition, 2004.
2. R.R. Brooks, S.S. Iyengar, "Multisensor Fusion: Fundamentals and Applications with Software", Prentice Hall Inc., New Jersey, 1998.

REFERENCES:

1. Jitendra R Raol, "Data Fusion Mathematics: Theory and Practice", CRC Press, 2016.
2. Yaakov Bar-Shalom, Peter K Willett, and Xin Tian, "Tracking and Data Fusion: A Handbook of Algorithms", YBS Publishing, 2011.
3. Thor I. Fossen, Kristin Y. Pettersen and Henk Nijmeijer, "Sensing and Control for Autonomous Vehicles: Applications to Land, Water and Air Vehicles", Springer, 2017.
4. Dan Simon, "Optimal State Estimation Kalman, H-infinity and Non-linear Approaches", John Wiley and Sons, 2006.

23U044 ADVANCED SENSING TECHNIQUES

3 0 0 3

INTRODUCTION TO SENSORS IN AUTOMATION: Introduction - Different Phases of automation. Importance of sensor/smart sensor in automation. Features of Advanced sensing techniques - Sensor classifications according to the energy domains. Introduction of advanced sensing materials. Properties (physical, electrical, chemical, biological) of materials in different domains. (9)

DESIGN AND MODELING TECHNIQUES: Design and modeling-Design and modeling issue in advanced sensing technique. Introduction of different mathematical tools used in sensor design. Analytical design from given specification, conformal mapping- Optimization techniques used in sensor design. Numerical design -FEM, FDM. Concept of Feedback in sensing. Fabrication and packaging- issues. (9)

CHEMICAL SENSING ELEMENTS: Chemical sensor: characteristics, specific difficulties related to chemical sensor, Classification of Chemical sensing mechanism. Metal oxide chemical sensor, electrochemical sensors, conductive sensors, amperometric sensors, enhanced catalytic gas sensors. (9)

BIOLOGICAL SENSING ELEMENTS: Sensor Terminologies in human body, enzyme sensors, biochemical sensor, electrodes are biophysical sensing, polarizable and nonpolarizable electrodes – body surface recording electrodes. (9)

ADVANCED SENSOR TECHNOLOGY: Introduction: Lab on chip/sensor platform technology. Role of PCA, LDA, Neural network in designing sensor array. Temperature cycle mode of sensing -virtual sensor array. Case study: gas sensing platform, liquid sensing. (9)

TEXTBOOKS:

1. Ahmed Barhoum, Zeynep Altintas, "Advanced Sensor Technology: Biomedical, Environmental, and Construction Applications", Elsevier Science, 2022.
2. Florinel-Gabriel Bănică, Chemical Sensors and Biosensors: Fundamentals and Applications, John Wiley & Sons, Ltd., 2012.

REFERENCES:

1. Sabaree Solomon, "Sensors Handbook", McGraw Hill, 1998.
2. Eugenio Lannone, "Labs on Chip-Principles, Design and Technology", CRC Press, Taylor& Francis group, 2017.
3. Eun Sok Kim, "Fundamentals of Microelectromechanical Systems", McGraw Hill, 2021.
4. Martin Davis Adams, "Sensor Modelling, Design and Data Processing for Autonomous Navigation", World Scientific, 1999.

23U045 MEDICAL IMAGING SYSTEMS

3 0 0 3

X RAYS: Nature of X-rays- Block diagram of X-Ray Equipment: XRay Tube, the collimator, Bucky Grid, power supply- Digital Radiography X-ray Image Intensifier tubes – Fluoroscopy – Digital Fluoroscopy. Angiography, cine Angiography. Digital subtraction Angiography. Mammograph. (9)

COMPUTED TOMOGRAPHY: Principles of tomography, CT Generations, X- Ray sources- collimation- X- Ray detectors – Viewing systems – spiral CT scanning – Ultra fast CT scanners. Image reconstruction techniques – back projection and iterative method. (9)

MAGNETIC RESONANCE IMAGING: Fundamentals of magnetic resonance- Interaction of Nuclei with static magnetic field and Radio frequency wave- rotation and precession – Induction of magnetic resonance signals – bulk magnetization – Relaxation processes T1 and T2. Block Diagram approach of MRI system – system magnet (Permanent, Electromagnet and Superconductors), generations of gradient magnetic fields, Radio Frequency coils, shim coils, Electronic components. (9)

NUCLEAR IMAGING: Radioisotopes- alpha, beta, and gamma radiations- Radiation detectors –ionization chambers, proportional counter, GM counter and scintillation Detectors, Gamma camera – Principle of operation, collimator, photomultiplier tube, X-Y positioning circuit, pulse height analyzer. Principles of SPECT and PET. (9)

RADIATION THERAPY AND RADIATION SAFETY: Radiation therapy – linear accelerator, Telegamma Machine. Stereotactic radiosurgery (SRS) – Stereotactic radiotherapy (SRT) – Recent Techniques in radiation therapy – radiation measuring instruments Dosimeter – electronic dosimeter – Radiation protection in medicine. (9)

Total: L:45

TEXTBOOKS:

1. Andreas Maier, Stefan Steidl, Vincent Christlein, Joachim Horneegger, "Medical Imaging Systems: An Introductory Guide", Springer, 2018.
2. Khin Wee Lai, Dyah Ekashanti Octorina Dewi "Medical Imaging Technology", Springer Singapore, 2015.

REFERENCES:

1. Isaac Bankman, I. N. Bankman, Handbook Of Medical Imaging: Processing and Analysis (Biomedical Engineering), Academic Press, 2000.
2. Khandpur R.S, "Handbook of Biomedical Instrumentation", Tata McGraw – Hill, New Delhi, 2003.
3. Jacob Beutel, M. Sonka, "Handbook of Medical Imaging", Volume 2. Medical Image Processing and Analysis, SPIE Press, 2000.
4. J. L. Prince and J. M. Links, "Medical Imaging Signals and Systems", Pearson Prentice Hall, 2006.

23U046 SENSOR SYSTEMS DESIGN

3 0 0 3

SIGNAL CONDITIONING: Design of signal conditioning circuits for resistive, capacitive, thermal transducers for improving linearity, sensitivity and other required specifications and performance through hardware and software methods - Linearization, A/D conversion, temperature compensation. (9)

SENSOR INTERFACING: Noise analysis of interface circuits - Current, frequency, period or pulse-width modulation conversion - Review of transmitters - EMI and EMC design consideration for sensor interfacing circuit design. (9)

DATA ACQUISITION SYSTEM: Introduction - Issues related to interfacing of static and dynamic sensors - Design of data acquisition for a given measurement application through theory and practical approach - Introduction to Sensor buses and sensor network protocols. (9)

DIGITAL AND INTELLIGENT SENSORS: Position encoders - Resonant sensors - Variable oscillators - Smart sensors and digital sensor system design: Technologies and design methodology, IEEE 1451 standard and frequency sensors. (9)

DIRECT SENSOR-MICROCONTROLLER INTERFACING: Interfacing for resistive and capacitive transducers: design and practical implementation - Universal frequency to digital converter - Universal sensors and transducer interface - features and performance, future trends in sensor circuit design. (9)

Total: L:45

TEXTBOOKS:

1. Ramon Pallas Areny, John G. Webster, "Sensors and Signal Conditioning", 2nd Edition, John Wiley and Sons, 2000.
2. Kirianaki N.V., Yurish S.Y., Shpak N.O., Deynega V.P., "Data Acquisition and Signal Processing for Smart Sensors", John Wiley & Sons, Chichester, UK, 2002.

REFERENCES

1. Ferran Reverter, Ramon Pallas Areny, "Direct Sensor-to Microcontroller Interface Circuits: Design and Characterization", Marcombo S.A., 2005.
2. S.Y. Yurish and M.T. Gomes, "Smart Sensors and MEMS", Springer Verlag, 2005.
3. A. Custodio, R. Bragos, R. Pallas-Areny, "A Novel Sensor-Bridge-to-Microcontroller Interface", in Proceedings of IEEE Instrumentation and Measurement Technology Conference, Budapest, Hungary, 21-23 May, 2001.
4. H.R. Taylor, "Data Acquisition for Sensor Systems", Springer, 2010.

SIGNAL ANALYSIS

23U051 WAVELETS AND THEIR APPLICATIONS

3 0 0 3

TIME-FREQUENCY REPRESENTATION: Function space-basis function – Review of Fourier transform and its properties – Shortcoming of Fourier Transform-Short-Time Fourier Transform (STFT) – Heisenberg uncertainty principle-property of STFT- Limitations of STFT. (9)

CONTINUOUS WAVELET TRANSFORM (CWT): Admissibility condition – Forward and inverse CWT expression- Properties of CWT- Time and frequency resolution of CWT - Redundancy of CWT –Applications of CWT. (9)

DISCRETE WAVELET TRANSFORM: Relationship between wavelet and filter bank – Two channel filter bank-Tree structure filter bank – Multi- Resolution Analysis (MRA). (9)

VARIANTS OF WAVELETS: Wavelet Packet – Multiwavelet – Ridgelet – Curvelet – Sychro squeezing wavelet – Dual-tree complex wavelet. (9)

APPLICATIONS: Signal Denoising – Signal compression - Power line signal analysis – Subband coding of speech signal – Biomedical signal analysis using wavelets. (9)

Total: L: 45

TEXT BOOKS:

1. Stephane Mallat, "A Wavelet Tour of Signal Processing", Academic Press, 2000.
2. Raghuveer M. Rao and Ajit S. Bopardikar, "Wavelet Transforms", Prentice Hall, 1998.

REFERENCES:

1. Martin Vetterli, Jelena Kovacevic, "Wavelets and Subband Coding", Prentice Hall, 1995.
2. Jaideva C. Goswami, Andrew K. Chan, "Fundamentals of Wavelets: Theory, Algorithms, and Applications", Wiley-Interscience, 2008.
3. Gilbert Strang, Truong Nguyen, "Wavelets and Filter Banks", Wellesley-Cambridge Press, 1996.
4. Eugenio Hernandez, Guido Weiss, "A First Course on Wavelets", CRC Press, 2000.

23U052 PATTERN RECOGNITION AND CLASSIFICATION

3 0 0 3

INTRODUCTION: Introduction to Machine Perception- Pattern and features- Pattern Recognition Systems: Sensing- Segmentation and Grouping- Feature Extraction- Classification- Post Processing- The Design Cycle- Training and learning in pattern recognition systems: Supervised Learning – Unsupervised Learning- Reinforcement Learning- Pattern recognition approaches- Applications of Pattern Recognition. (9)

DIMENSIONALITY REDUCTION: Singular Value Decomposition- Principal Component Analysis- Linear Discriminated Analysis: Gradient descent procedures- Fisher discriminant analysis- Multiple discriminant analysis. (9)

STATISTICAL PATTERN RECOGNITION: Gaussian model – Supervised learning – Parametric estimation – Maximum likelihood estimation – Bayesian parameter estimation – Perceptron algorithm – LMSE algorithm – Problems with Bayes approach – Pattern classification by distance functions – Maximum distance pattern classifier- Decision Boundaries. (9)

CLUSTER ANALYSIS: Unsupervised learning – Clustering for unsupervised learning and classification – K-means clustering algorithm – Hierarchical clustering procedures – Graph theoretic approach to pattern clustering – Validity of clustering solutions- Criterion functions for clustering- Techniques for clustering. (10)

FEATURES EXTRACTION AND RECENT ADVANCES: Entropy minimization – Karhunen –Loeve transformation – Neural network structures for pattern recognition – Unsupervised learning – Self organizing networks – Fuzzy pattern classifiers – Genetic algorithm. (8)

Total: L: 45

TEXT BOOKS

1. Richard O Duda, Peter E Hart and David G Stork, "Pattern Classification" John Wiley, Second Edition, 2001.
2. Earl Gose, Richard Johnson baugh and Steve Jost, "Pattern Recognition and Image Analysis", Pearson, 2015.

REFERENCES

1. Kevin P. Murphy, "Probabilistic Machine Learning: An Introduction", The MIT Press, 2022.
2. S.Theodoridis and K.Koutroumbas, "Pattern Recognition", Academic Press, Fourth Edition, 2009.
3. Christopher M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2006.
4. Robert Schalkoff, "Pattern Recognition: Statistical, Structural and Neural Approaches", John Wiley & Sons, 1992.

23U053 DEEP LEARNING FOR SIGNAL ANALYSIS

3 0 0 3

INTRODUCTION TO DEEP LEARNING FOR SIGNAL ANALYSIS: Introduction to signals- time series- Elements of stochastic processes and signal processing- Introduction to Deep learning computation- Examples of practical Deep learning for Signal Analysis data and scenarios. (9)

DEEP LEARNING FUNDAMENTALS: Regression and classification problems- Optimization algorithms for deep learning- Propagation- computational graphs- automatic differentiation- Multilayer perceptron and backpropagation- Layers, blocks, and activations- Training networks. (9)

DEEP LEARNING FOR SIGNAL ANALYSIS OF IMAGES AND STRUCTURED SIGNALS: Convolutional neural networks (CNNs) - Deep learning architectures based on blocks - Visual Geometry Group (VGG) - VGG Very Deep Convolutional Networks (VGGNet) - Residual and dense networks- Advanced CNN architectures- Applicative examples. (9)

DEEP LEARNING FOR SIGNAL ANALYSIS OF SEQUENTIAL AND TIME-VARYING SIGNALS: Recurrent layers- Encoder-decoder architectures- Attention mechanism- Transformer- Applicative examples. (9)

FAST AND EFFICIENT DEEP LEARNING FOR SIGNAL ANALYSIS: Low-complexity approaches and sparsification techniques- Model compression and acceleration- Quantized and efficient architectures- Applicative examples using sensor signals, images, smart grids, biomedical signals, audio, speech and music, edge-AI, and IoT data. (9)

Total: L: 45

TEXT BOOKS

1. Aston Zhang and Zachary C. Lipton and Mu Li and Alexander J. Smola, "Dive into Deep Learning", 2020.
2. SergiosTheodoridis, "Machine Learning: A Bayesian and Optimization Perspective", Second edition, Elsevier, 2020.

REFERENCES

1. Kevin P. Murphy, "Probabilistic Machine Learning: An Introduction", The MIT Press, 2022.
2. Sebastian Raschka and Vahid Mirjalili, "Python Machine Learning", Packt Publishing, Third edition, 2019.
3. Ian Goodfellow, Yoshua Bengio and Aaron Courville, "Deep Learning", The MIT Press, 2016.
4. Christopher M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2006.

23U054 BIOMEDICAL SIGNAL PROCESSING

3 0 0 3

BIOMEDICAL SIGNALS: Action potential and its generation, origin and characteristics of Electrocardiogram (ECG), Electroencephalogram (EEG), Electromyogram (EMG), Phonocardiogram (PCG)- objective of biomedical signal analysis – principle of computer aided diagnostics. (9)

FILTERING FOR REMOVAL OF ARTIFACTS: Random noise, structured noise, motion artefact, powerline interference, Maternal interference in fetal ECG – Time domain and frequency domain filters- Adaptive and Optimal filter - Non-linear filter. (9)

CARDIOLOGICAL SIGNAL PROCESSING: Introduction to electrocardiography, acquisition, lead system, ECG features and their estimation. Pre-processing, QRS Detection Methods, Rhythm analysis, Arrhythmia detection Algorithms, automated ECG analysis. ECG pattern recognition, Heart rate variability analysis, clinical applications. (9)

NEUROLOGICAL SIGNAL PROCESSING: Introduction to brain potential and EEG Signals, its origin, characteristics, frequency division, and evoked potentials. Pattern recognition of alpha, beta, theta and delta waves in EEG waves, sleep stages, epilepsy detection. (9)

MODELLING OF BIOMEDICAL SYSTEMS: Design of notch filter using pole-zero approach - Parametric System modelling – Autoregressive model – Autocovariance method - Levinson-Durbin algorithm- Computation of model parameters – Model order selection. (9)

Total: L: 45

TEXT BOOKS

1. Rangaraj M. Rangayyan, "Biomedical Signal Analysis", Wiley, 2016.
2. Eugene N. Buruce, "Biomedical Signal Processing and Signal Modelling", Wiley, 2007.

REFERENCES

1. Willis J. Tompkins, "Biomedical Digital Signal Processing", Prentice Hall, 2004.
2. D.C. Reddy, "Biomedical Signal Processing: Principles and Techniques", Tata McGraw Hill, 2005.
3. Leif Sornmo, Pablo Laguna, "Bioelectrical signal Processing in Cardiac and Neurological Applications", Academic Press, 2005.
4. Iyad Obeid, Ivan Selesnick and Joseph Picone, "Biomedical Signal Processing: Innovation and Applications", Springer Nature Switzerland, 2022.

23U055 MEDICAL IMAGE PROCESSING

3 0 0 3

MEDICAL IMAGE MODALITIES: Principle and characteristics of X-ray, Computed Tomography (CT), Magnetic Resonance Imaging (MRI), Positron Emission Tomography (PET), ultrasound. (9)

IMAGE RECONSTRUCTION FROM PROJECTION: Radon transform – Fourier-slice theorem – Filtered back projection – Fan-beam filtered back projection –Challenges: image reconstruction with incomplete and noisy data. (9)

SEGMENTATION: Threshold based segmentation- region based approach – clustering based approach – Mean shift algorithm- semantic segmentation – Atlas-based segmentation – Graph cut - Deep learning based image segmentation – challenges in image segmentation. (9)

IMAGE REGISTRATION: Rigid body transformation, principal axes registration - Landmark based approach – warping and non-linear image registration – Mutual information based registration - Deep learning based image registration. (9)

SHAPE ANALYSIS AND CLASSIFICATION: Shape description – Chain code - Topological descriptor - Fourier descriptor – Hough transform- Active Shape models – Active Appearance models – machine learning based image classification – deep learning based image classification. (9)

Total: L: 45

TEXT BOOKS:

1. Rangaraj M. Rangayyan, "Biomedical Image Analysis", CRC Press, 2010.
2. Issac Bankman, "Handbook of Medical Imaging", Academic Press, 2004.

REFERENCES

1. Atam P. Dhawan, "Medical Image Analysis", Wiley-Blackwell, 2003.
2. S. Kevin Zhou, Hayit Greenspan and Dinggang Shen, "Deep learning for Medical Image Analysis", Academic Press, 2017.
3. Geoff Dougherty, "Digital Image Processing for Medical Applications", Cambridge University Press, 2009.
4. Joseph V. Hajnal, Derek L.G. Hill, "Medical Image Registration", CRC Press, 2019.

23U056 MACHINE VISION

3 0 0 3

DIGITAL IMAGE FORMATION: Fundamentals of image formation- Transformation: Orthogonal, Euclidean, Affine and Projective- Image sensors –Lenses - Sources of lighting – Line scan, area scan camera, thermal camera - camera calibration. (7)

FEATURE EXTRACTION: Edge based features- Scale Invariant Feature Transform (SIFT) – Speeded Up Robust Feature (SURF) – Histogram of Oriented Gradient (HOG) – Scale-space analysis – Image pyramid. (10)

PATTERN ANALYSIS: Clustering: K-Means, K-Medoids, Mixture of Gaussians. Classification: Supervised, Unsupervised, Bayes and Artificial Neural Network (ANN) models. Dimensionality reduction: Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA) and Independent Component Analysis (ICA). (12)

MOTION ANALYSIS: Background subtraction and modelling, optical flow, spatio-temporal analysis, Motion parameter estimation- Inference of human activity from image sequence. (8)

OBJECT DETECTION: Introduction to convolution neural network – R-CNN – YOLO, SSD for object detection and classification. (8)

Total: 45

TEXT BOOKS:

1. Richard Szeliski, "Computer Vision: Algorithms and Applications", Springer Nature Switzerland AG, 2022.
2. David A. Forsyth, Jean Ponce, "Computer Vision: A Modern Approach", Pearson Education, 2015.

REFERENCES:

1. Carsten Steger, Markus Ulrich, and Christian Wiedemann, "Machine Vision: Algorithms and Applications", Wiley, 2017.
2. Richard Duda, Peter Hart and David Stork, "Pattern Classification", Wiley, 2000.
3. Simon J.D. Prince, "Computer Vision", Cambridge University Press, 2012.
4. Ian Goodfellow, YoshuaBengio and Aaron Courville, "Deep Learning", The MIT Press, 2016.

LANGUAGE ELECTIVES

23G001 COMMUNICATION SKILLS FOR ENGINEERS

0 0 4 2

COMMUNICATION CONCEPTS: Process of Communication – Inter and Intrapersonal Communication – Essentials for effectiveness (9)

ORAL COMMUNICATION: Oral presentations with visual aids and Group discussions. (16)

FOCUS ON SOFT SKILLS: Etiquette – Work Place etiquette – Telephone etiquette- Body Language – Critical Reasoning and Conflict Management based on Case Studies – Group Communication- Meetings -Interview Techniques (14)

TECHNICAL WRITING: Technical Writing Principles - Style and Mechanics - Technical Definitions – Physical, Functional and Process Descriptions — Technical Report Writing – Preparing Instructions – Interpretation of Technical Data (14)

BUSINESS CORRESPONDENCE: Writing Emails, Preparing Resumes. (7)

Total: 60

TEXT BOOKS:

1. Course materials prepared by the Faculty, Department of English.

REFERENCES:

1. Jeff Butterfield, "Soft Skills for Everyone", Cengage Learning, New Delhi, 2020.
2. Sabina Pillai and Agna Fernandez, "Soft skills and Employability Skills", Cambridge University Press, New Delhi, 2019.
3. Prashant Sharma, "Soft Skills Personality Development for Life Success", BPB Publications, New Delhi, 2021.
4. Shoba K N and Praveen Sam D, "Technical English", Cambridge University Press, New York, 2020

23G002 BASIC GERMAN

0 0 4 2

Guten Tag! - Learning: To greet, learn numbers till 20, practice telephone numbers & e mail address, learn alphabet, speak about countries & languages ; Vocabulary: related to the topic; Grammar: W – Questions, Verbs & Personal pronouns I. (10)

Freunde, Kollegen und ich - Learning: To speak about hobbies, jobs, learn numbers from 20; Vocabulary: related to the topic; Grammar: Articles, Verbs & Personal pronouns II, sein & haben verbs, ja/nein Frage, singular/plural. (10)

In der Stadt – Learning: To know places, buildings, question, know transport systems, understand international words; Vocabulary: related to the topic; Grammar: Definite & indefinite articles, Negotiation, Imperative with Sie. (12)

Guten Appetit! – Learning: To speak about food, shop, converse; Vocabulary: related to the topic; Grammar: Sentence position, Accusative, Accusative with verbs. (13)

Tag für Tag and Zeit mit Freunden – Learning: To learn time related expressions, speak about family, ask excuse, fix appointments on phone, birthdays, understand & write invitations, converse in the restaurant ; Vocabulary: related to the topic; Grammar: Preposition – am, im, um, von...bis, Possessive articles, Modalverbs. (15)

Total : (60)

TEXT BOOKS:

1. Dengler, Stefanie et al., *Netzwerk A1.1*, Klett-Langenscheidt GmbH, München, 2013.

REFERENCES:

1. Dengler, Stefanie et al., *Netzwerk A1*, Klett-Langenscheidt GmbH, München, 2013.
2. Sandra Evans, Angela Pude, Franz Specht-Menschen A1–Hueber Verlag ,2012.
3. Hermann Funk, Christina Kuhn, Silke Demme, Studio d A1 , Goyal Publishers & Distributors Pvt. Ltd ,2009.
4. Rosa-Maria Dallapiazza, Eduard von Jan, Til Schönherr, *Tangram Aktuell 1 (Deutsch als Fremdsprache)* , Max Hueber Verlag ,2004.

23G003 BASIC FRENCH

0 0 4 2

Unité 1: inviter et répondre à une invitation, Pronoms sujets, L'article définis, l'article indéfinis, Conjugation : présent, adjectifs possessifs, interrogation, décrire les personnes. La vie de quatre parisiens de profession différentes (15)

Unité 2: Exprimer l'ordre et l'obligation demander et commander, l'adjectif possessifs, l'articles partitif, l'article démonstratif, négation ne....pas, l'article contracté, verbe pronominaux, prépositions. (15)

Unité 3: Raconter et reporter-donner son avis, Futur simple, pronom complètement d'objet direct, passé composé, plusieurs région de France, imparfait, pronom y/en, imparfait (15)

Unité 4: Demander l'autorisation-passé récent, futur proche, La vie administrative et régionale, Pluriel des noms, moyens de transport (15)

Total : 60

TEXT BOOKS:

1. Christine Andant étal, À propos (livre de l'élève), Langer, New Delhi, 2012.

REFERENCES:

1. Dondo Modern French Course ---Mathurin Dondo
2. Modern French Grammar---Margaret Lang and Isabelle Perez.

23G004 BASIC JAPANESE**0 0 4 2****Course contents :**

Orientation Session, Geographic & Socio, economic perspective to Japan, Japanese people and culture and Basic greetings and responses (3)

Basic script, Method of writing hiragana and katakana, and Combination sounds and simple words (3)

Topic marker "wa", Desu / dewa arimasen cupolas, Interrogative particle "ka", Grammar particles "mo", "no", " " Introducing some one: "Kochira wa ~" and Self introductions: Hajimemashite" (3)

Demonstratives "Kore", "Sore", "Are", Demonstrative "Kono", "Sono", "Ano", Possessive noun particle "no" and Japanese apartments: Greeting your neighbour (3)

Place markers "Koko", "Soko", "Asoko", Direction markers "Kochira", "Sochira", "Achira" and Japanese department stores: Asking for and buying something (3)

Asking for and telling the time, Particle "ni (at)" for time, kara (from) ~ made (until), Particle "to (and)", Time periods: Days of the week, months, time of day, Verbs (Present / future and past tense) and Telephone enquiry: Asking for a phone no. And business hours (3)

Destination particle "e", Particles "de (mode of transportation)" and "to (with)" and Japanese train station: Asking for Fare and track no. / types of trains (3)

Direct object particle "o", Particle "de (place of action)", Verbs ("~masen ka", "~mashou") and "Ohanami" Cherry blossom viewing (3)

Particle "de (by means of)", Particle "ni (to)", ,Aemasu (give) and Moraimasu (receive) and Visiting a Japanese house (3)

Adjectives ("i" and "na" type), Adjectives (Positive and negative useage), Particle "ga (however, but)", "Dore which?" and Leaving a room, thanking someone for hospitality (3)

Likes and dislikes, Potential verbs (wakarimasu and dekimasu), "Kara (~ because)", Adverbs and Asking someone out over the phone (3)

Verbs denoting presence: "Imasu" and "arimasu", Particle "ni (in)", "Dare (who?)", Adverbs ("Chikaku ni ~"), Particle "dare mo (negative ~ no one)", Dare ka (anyone), dare ga (who), Nani ka (anything), nani ga (what) - ~ya (and) ~ nado (etc.) and Asking for directions (3)

Counters and Counting suffixes (3)

Introduction to Adjectives (na and ii type), Different usages of adjectives, Comparison, Likes and dislikes and Going to a trip (3)

Need and desire (ga hoshii), Wanting to ... (Tabeti desu), Going for a certain purpose (mi -ni ikimasu) and Choosing from a menu (3)

Verb groups, I, II and III and Exercises to group verbs (3)

Please do (te kudasai), Present continuous tenses (te imasu), Shall I? (~ mashou ka) and Describing a natural phenomenon (It is raining) (3)

To grant permission (~te mo ii desu), Asking for permission (~ te mo ii desu ka) and Should not do (~ te wa ikemasen) (3)

Describing a continuing state and Describing a habitual action (2)

Roleplays in Japanese (2)

A demonstration on usage of chopsticks and Japanese tea party (2)

Total: 60**TEXT BOOK**

1. Minna no nohongo – Romaji ban (first 10 lessons of this book)

REFERENCE

- 1 Minna no Nihongo I Honsatsu Roma – ji ban (Main Textbook Romanized Version). International publisher – 3A Corporation, Tokyo, Indian distributor – Goyal Publishers & Distributors, New Delhi.

ONE CREDIT COURSES

23UF01 ADVANCED INDUSTRIAL AUTOMATION SYSTEMS

1 0 0 1

PROGRAMMABLE AUTOMATION CONTROLLERS : Multidisciplined Controllers-System Model-Backplane Communications - Different Form Factors - Networking Options - Operating System - Task Program and Routine as per IEC 61131-3 - Controller Data Features - Programming Software - Anatomy and Memory Structure - Operating System Priorities - Concept of Connections for Industrial Automation Systems - Understanding Controller Communication to the external World (4)

HANDS ON PRACTICE: Logix Controller programming Software with examples on Process Control and PID Tuning (3)

INDUSTRIAL NETWORKING : Evolution of Industrial Networking - Need for Industrial Networking - Network functions - Open Vs Proprietary Industrial Networking - Netlinx - Seamless Industrial Networking - CIP and Netlinx - Netlinx networks - Implementation and application paradigm of Device net, Control net and Ethernet/IP networks - Applications (2)

DEMONSTRATION: Hands on Practice on Device net/Controlnet/Ethernet/IP Networks. (3)

HMI SYSTEMS AND VISUALIZATION SOLUTIONS: Integrated Architecture and Factory Talk Manufacturing Information platform - One Architecture for Plant wide Control and Information - Role of Industrial Visualization - Visualization fit within an Automation Environment - View technology Cornerstones - View Technologies - Machine level and PC Based platform (3)

Total L:15

REFERENCES:

1. John W. Webb and Ronald A. Reis, "Programmable Logic Controllers: Principles and Applications", 5th Edition, Prentice Hall India publication, 2002.
2. Instruction Manual, "Control and Configuration software", Rockwell Automation, Bangalore,
3. Frank D Petronella, "Programmable Logic Controllers", 5th Edition, Mc Graw Hill publication, 2010.
4. John Park, Steve Mackay, "Practical Industrial Networking for Engineers and Technicians", IDC Technologies, 2011.

23UF02 SYSTEM DESIGN AND IMPLEMENTATION

1 0 0 1

INTRODUCTION TO CONTROL SYSTEM DESIGN AND SIMULATION : Developing a plant model, transfer function model, State space model - Overview of Control Design Toolkit-Control Design VIs, Model representation, Model forms - Constructing Transfer function models, state space models and zero pole gain models (3)

MODEL CONVERSION AND INTERCONNECTION : Converting Model Forms- Continuous to Discrete model conversion - Connecting Models in series, Connecting models in parallel - Placing models in closed loop configuration. (3)

TIME RESPONSE ANALYSIS : Calculating the time domain specification - Case study: Spring mass damper system, RLC circuit - Analyzing the time response of a system for an applied step and impulse inputs, Analyzing initial response of a given system (3)

FREQUENCY RESPONSE ANALYSIS : Frequency domain specifications, Bode frequency analysis - Gain margin, Phase margin - Nichols frequency analysis, obtaining frequency response data (3)

STABILITY ANALYSIS AND CLASSICAL CONTROLLER DESIGN : Determining stability : Root locus method, Pole zero map - PID Controller design and prototyping - Implementation of auto tuning PID controller. Case study - Speed control of a DC motor, Level and Flow control (3)

Total L:15

REFERENCES:

1. Dorf, Bishop, "Modern Control Systems", 11th Edition, Prentice Hall, 2017.
2. Ogata K, "Modern Control Engineering", Prentice-Hall of India Pvt Ltd., New Delhi, 1997.
3. Norman S Nise, "Control System Engineering", 4th Edition, John Wiley & sons, Inc, 2004.
4. Nagrath IJ, Gopal M, "Control System Engineering", 3rd Edition, Wiley and Sons, 2004.

23UF03 CALIBRATION TECHNIQUES

1 0 0 1

MEASUREMENT UNCERTAINTY: Background — random and systematic errors - type A and type B uncertainty - sensitivity coefficients — uncertainty evaluation (3)

CALIBRATION: Introduction - meaning — objectives - necessity of calibration - basic calibration process — various components of a calibration system (3)

STANDARDS AND STANDARDIZATION : Working standards, check standards and international standards - levels of standard accuracies, accuracy ratio between levels of calibration pyramid - Requirements of traceability - metrology standardization documents (3)

CALIBRATION TECHNIQUES : Introduction — Calibration Curve Method, Standard Additions Method – Internal Standard Method, Comparative technique—choosing calibration method—determining calibration intervals (3)

CALIBRATION SETUPS : Electrical calibration - Temperature calibration - Pressure and Flow calibration—demonstrations (3)

Total L:15

REFERENCES:

1. Stephanie Bell , "A Beginner's Guide to Uncertainty of Measurement", National Physical Laboratory, UK, 1999.
2. Allan. S. Moris , "Measurement and Calibration for Quality Assurance", Prentice Hall, 1991.
3. Mike Cable , "Calibration: A Technicians Guide", ISA Publisher, 2005.
4. International Organization for Standardization , "International Vocabulary of basic and general terms in Metrology", 2008.

23UF04 MOTION CONTROL SYSTEMS

1 0 0 1

INTRODUCTION : Motion Control System, Micro and nano level motion systems. (2)

MOTION SENSING TECHNOLOGIES : Optical and magnetic encoders, magnetic and optical readers, resolvers, MEMS based accelerometers / gyros, multi-axis force/moment sensors, piezoresistive sensors, PSD sensors, tactile sensors, nanometrology. (3)

ACTUATOR TECHNOLOGIES : PMDC Motor, BLDC Motor, Induction Motor, Variable Reluctance motors, voice coil motors, piezoelectric actuators, magnetic actuators, shape memory alloys, linear actuation, electro-pneumatic and electro-hydraulic actuators. (2)

POWER TRANSMISSION SYSTEMS AND LOADS : Planetary gears, cycloid gears, harmonic drives, series elastic drives, cable drives, ball screws, and CVT. Loads and motor sizing. (2)

CONTROL OF ACTUATORS : Force/Torque, speed, position control loops, importance of stiffness, damping and inertia in various applications, auto-tuning, motion profiling and trajectory generation. (2)

ADVANCED MOTION CONTROL STRATEGIES : Multi-axis motion systems, master-slave bilateral operations (2)

SAFETY IN MOTION SYSTEMS : Application of brakes and clutches and international regulations. (2)

Total L:15

REFERENCES:

1. Urs Kafader , "Selection of High Precision Microdrives", Maxon Academic Series, 2010.
2. Kok Kiong Tan, Tong Heng Lee , "Precision Motion Control: Design and Implementation", Springer, 2010.
3. Asif Sabanovic, Kouhei Ohnishi , "Motion Control Systems", Wiley-IEEE Press, 2011.

23UF05 ELECTRICAL METROLOGY

1 0 0 1

BASIC METROLOGY: Understanding metrology Vocabulary, Measurement significance and application - Common metrology terminologies - Calibration: Objective and benefits of calibration (3)

TRACEABILITY AND SI UNITS: Traceability and its importance, International Metrology Structure – Calibration Hierarchy - Importance of SI Units and types of SI Units (3)

STANDARDS USED IN ELECTRICAL CALIBRATION: Absolute, Primary, secondary, working and census standards – measurement setup - Interpretation of specifications of various calibration standards and Device Under Calibration (DUCs) (3)

CALIBRATION PROCEDURES FOR VARIOUS ELECTRICAL T & M INSTRUMENTS: Methods of Calibration: Oscilloscopes, clamp meter, Panel meters - Calibration procedure of Voltmeter, Multimeter, DMM - Case studies (3)

MEASUREMENT UNCERTAINTY: Overview on Uncertainty, reason for uncertainty and its importance - Uncertainty evaluation - best practice - Uncertainty in practice and Case studies (3)

Total L:15

REFERENCES:

1. International Organization for Standardization , "Guide to expression of Uncertainty in measurement", Switzerland, 1995.
2. International Organization for Standardization , "International Vocabulary of basic and general terms in Metrology", Switzerland, 1993.
3. International Organization for Standardization , "International Standard ISO 3534-1, statistics-Vocabulary and Symbols-part-I: Probability and General Statistical Terms", Switzerland, 1993.
4. National Accreditation Board for Testing and Calibration Laboratories, "Guidelines for Estimation and Expression of Uncertainty in Measurement—NABL-141", NABL, 2016.

23UF06 STANDARD PRACTICES FOR POWER PLANT INSTRUMENTATION

1 0 0 1

OVERVIEW OF POWER PLANT : Role of Instrumentation engineers in large power & process plants. – Evolution of Instrumentation (Gauges to DCS and beyond) (2)

STANDARDS AND CONTROL SCHEMES: Communication standards of Instrumentation - Standards of Instrumentation (National/International), Instrument numbering system, KKS coding, Typical control schemes in powerplants. (5)

SAFETY: Instrumentation in hazardous areas - Standards of Instrument enclosures. (3)

INTERNATIONAL STANDARDS: International certification for Instruments - Underwriters Laboratories/Canadian Standards Association/Verband der Elektrotechnik (UL/CSA/VDE), Instrument installation guidelines. (5)

Total L:15

REFERENCES:

1. Alan. S. Morris, "Measurement and Instrumentation Principles", Butterworth-Heinemann, 2011.
2. Bouwens A. J., "Digital Instrumentation", Tata McGraw-Hill Education, 2001.
3. Bela. G. Liptak, "Instrument Engineers' Handbook - Process Measurement and Analysis", Vol 1, CRC Press, 2003.
4. Bela. G. Liptak, "Instrument Engineers' Handbook - Process Control and Optimization", Vol 2, CRC Press, 2005.
5. Bela. G. Liptak, "Instrument Engineers' Handbook - Process Software and Digital Networks", Vol 3, CRC Press, 2011.
6. Sam G. Dukelow, "The Control of Boilers", 2nd Edition, Instrument Society of America, 1991.

23UF07 AUTOMOTIVE INSTRUMENTATION AND CONTROL

1 0 0 1

AUTOMOTIVE INSTRUMENTATION CLUSTER : Measurement and display techniques of Vehicle Speed, Engine Speed and Odometer, Fuel Level, Oil Pressure, Engine Temperature, Battery Status, Gear Shift Indication, Seat Belt Indication, Door Open indication. (3)

VEHICLE CONTROL FUNCTION : Cruise Control (Simple and Adaptive), Speed Limiter, Vehicle Stability control. (3)

ENGINE CONTROL FUNCTION : Air system Control, Fuel System Control, Ignition Control, Exhaust Control, Water temperature control (Electric fan and pump), Engine speed control and Engine speed limitation. (3)

BODY ELECTRONICS : Immobilizer for vehicle security, Power Window control, Wiper speed control (with and without rain sensing) Head lamp intensity control based on Ambient light sensing. (3)

AUTOMOTIVE COMMUNICATION PROTOCOLS : Controller Area Network (CAN), Local Interconnect Network (LIN), FlexRay, Media Oriented Systems Transport (MOST). (3)

Total L:15

REFERENCES:

1. Robert Bosch GmbH, "Automotive Handbook", 9th Edition, Bentley Publishers, 2014.
2. Konrad Reif, "Automotive Mechatronics: Automotive Networking, Driving Stability Systems, Electronics (Bosch Professional Automotive Information)", Springer Vieweg, 2015.

23UF08 AIRCRAFT INSTRUMENTATION

1 0 0 1

OVERVIEW : Introduction to aircraft instrumentation, Types of instruments, Location inside the cockpit. (2)

INTRODUCTION TO FLIGHT INSTRUMENTATION AND BAROMETRIC INSTRUMENTS : Pitot Static System, Air Speed Indicator (ASI), Altimeter, Rate of Climb Indicator (ROCI), Mach Meter. (3)

INTRODUCTION TO POWER PLANT INSTRUMENTATION AND ENGINE PARAMETER MONITORING INSTRUMENT : Auxiliary Engine Gas Temperature Indicator, Main Engine Rpm Gauge, Exhaust Gas Temperature Indicator (EGTI), Engine Oil Temperature Indicator, Engine Oil Pressure Indicator, Engine Vibration Measurer, Fuel Flow Meter, Fuel Quantity Gauge, Engine Regime Indicator, Hydraulic System Pressure/Temperature Indicator. (5)

INTRODUCTION TO NAVIGATION INSTRUMENTS AND GYRO INSTRUMENTS : Artificial Gyro Horizon, Combined Course Indicator (CCI), Flight Director Indicator (FDI), Null Indicator, Turn and Slip Indicator. (3)

INDEPENDENT INSTRUMENTS : Chronometer, Magnetic Compass, Bomb Bay Compartment Temperature Indicator, Cabin Temperature Indicator, Cabin Pressure Indicator. (2)

Total L:15

REFERENCES:

1. S Nagabhushan, "Aircraft Instrumentation and Systems", I. K. International Pvt Ltd, 2013.
2. EHJ Pallett, "Aircraft Instruments", Longman Scientific & Technical, 1988.
3. Max F Henderson, "Aircraft Instrument and Avionics", Jeppesen Sanderson, 1993.
4. Dale Crane, "A Pilot Guide to Aircraft and Their Systems", Aviation Supplies & Academics, Incorporated, 2002.

23UF09 AUTOMATIC FLIGHT CONTROL SYSTEM

1 0 0 1

INTRODUCTION : Aircraft Stability, Directional Divergence, Spiral Divergence, Dutch Roll and Controllability. (2)

NEED AND ROLE OF FLIGHT CONTROL SYSTEM : Elements of flight control system. Functional diagram of a basic closed loop flight control system. Purpose of Altitude hold and airspeed hold (4)

CLASSIFICATION OF FCS : Single axis, two axis and three axis. Functional components of three axis FCS. (2)

INTRODUCTION TO SENSORS/ERROR DETECTORS: Gyros (Vertical, Directional, Displacement, Rate), Altitude Sensor, Air speed Sensor, Compass/Heading Sensor, Accelerometers (INS), Synchronos. (3)

INTRODUCTION TO ACTUATORS: Electro Pneumatic Servo Actuators, Electro Hydraulic Servo Actuators, Electro Mechanical Servo Actuators, Position Control Servomechanism, Speed Control Servomechanism. (2)

SIGNAL PROCESSING: Functions of signal processing elements in a FCS, Signal Processing channel and its elements, Limiting Synchronizing, Gain and Adaptive Control, Feedback in a FCS. (2)

Total L:15

REFERENCES:

1. E H J Pallett, S Coyle, "Automatic Flight Control", 4th Edition, Oxford : Blackwell Science, 2005.
2. E.H.J. Pallett, "Aircraft Instruments", Himalaya Publishing House, New Delhi, 1993.
3. John P. Fielding, "Introduction to Aircraft Design", 4th Edition, Cambridge University Press, 1999.
4. L J Clancy, "Aerodynamics", Pitman Publishing, 1978.

HUMANITIES

23OFH01 CLOSED LOOP ECONOMY AND ENGINEERING SOLUTIONS

1 0 0 1

Introduction: Linear Economy – Limitations of Linear Economy – Emergence of Closed Loop Economy – Linear Vs Closed Loop Economy an Outlook – Significance of Closed Loop Economy. (2)

Elements of Closed Loop Economy: Concept of Loops – Innovation – Waste Reduction – Recovery and Recycling – Renewable Energy – Butterfly Diagram – Case Studies. (4)

Engineering and Closed Loop Solutions: Sustainable Product Design Methods – Zero Waste and Design Out Waste Method – Closed Loop Packaging Solutions – Value Optimization – Case Studies. (3)

Closed Loop Business Models: Shift from Take-Make-Dispose to Reduce-Reuse-Recycle Model – Reducing Negative Externalities – Extended Producer Responsibility (EPR) – Industrial Symbiosis – Case Studies of Successful Closed Loop Business Models by Industries. (4)

Circularity in India: Closed Loop Knowledge and Capacity Building in India – Role of Government – Regulatory Frameworks – Benefits of Closed Loop Systems for Indian Businesses. (2)

Total L: 15

REFERENCES

1. Yang, J. From Zero Waste to Material Closed Loop: The Way towards Circular Economy. Springer Nature, 2022.
2. Liu, L., & Ramakrishna, S. (Eds.). An Introduction to Circular Economy. Singapore: Springer Singapore, 2021.
3. Bhalla, S. G. Circular Economy (Re) Emerging Movement, Springer, 2020.
4. Mishra, A. Understanding Circular Economy, Bharti Publications, 2023.

MATHEMATICS

23OFM01 INTRODUCTION TO QUATERNION ALGEBRA

[15hrs]

QUATERNION-BACKGROUND: Quaternion-Hamilton discovery of quaternion - complex number representation vs quaternion representation. (3)

QUATERNION ALGEBRA: Quaternion's, addition-multiplication-Quaternion, Properties of quaternion - commutative, associative, additive inverse, multiplicative inverse, conjugate-quaternion, norm of quaternion, inverse-quaternion, rotation-quaternion, Reflection-Euler's Rotation, various rotation operators - Quater rotation using complex numbers, Rodriguez's formula for rotation, rotation matrices for rotation operations. Theorem - Rodriguez's Rotation - quaternion power - quaternion exponential - quaternion logarithm - quaternion differentiation - quaternion integration. (7)

QUATERNION APPLICATIONS: Dual quaternion - quaternion interpolation - Lerp (linear Interpolation) - Slerp (spherical linear interpolation) - quaternion in kinematics. Programming quaternion functions through computers. Application of quaternion's to computer graphics, robot kinematics and Quantum physics. (5)

REFERENCES:

1. J.B.Kuipers, "Quaternion's and Rotation Sequences", Princeton University Press 2020.
2. K.Shoemake, "Animating rotation with Quaternion curves", Computer Graphics, 19(3):245-254, 1985.
3. Phil Kim, "Rigid Body Dynamics for Beginners, Euler angles & Quaternions", Createspace Independent Publishing Platform 2013.
4. John Vince, "Quaternion's for Computer Graphics", Springer-Verlag 2021.
5. Ron Goldman, "Rethinking Quaternions, Theory and Computation", Morgan and Claypool, 2010.

PHYSICS**23OFP01 PHOTONICS****(1 0 0 1)**

Maxwell's equations and light propagation in anisotropic media. (5)

Electro optic effect and applications in optical modulation and switching. (5)

Nonlinear optical effects: Second harmonic generation, sum and difference frequency generation and parametric amplification and some applications (5)

Total L= 15 Hours**REFERENCES:**

1. A K Ghatak and K Thyagarajan, Optical Electronics, Cambridge India; 2017.
2. E Hecht and A R Ganesan, Optics, Pearson 2002
3. A Yariv Pochi Yeh, Photonics - Optical Electronics in modern communication, Oxford, 1997
4. M Born and E Wolf, Principles of Optics, Cambridge University press, UK 1999
5. R Boyd, Non Linear Optics, Academics Press, 2020

23OFP02 LASERS**(1 0 0 1)**

Einstein's A and B coefficients, population inversion and optical amplification. (5)

Rate equations for three level and four level lasers. (5)

Optical resonators, longitudinal and transverse modes. (5)

Ultrashort pulses using mode locking and Q-switching (5)

Total L=15 Hours**REFERENCES:**

1. E Hecht and A R Ganesan, Optics, Pearson 2002
2. A Yariv Pochi Yeh, Photonics - Optical Electronics in modern communication, Oxford, 1997
3. M Born and E Wolf, Principles of optics, Cambridge University press, UK 1999
4. R Boyd, Non linear optics, Academics Press, 2020

23OFP03 OPTICS**(1 0 0 1)**

Interference and interferometers, Fraunhofer and Fresnel diffraction. (5)

Role of diffraction in resolution of optical instruments such as microscope and telescope, Polarization and polarization components (5)

Spatial frequency filtering and applications in image processing, Holography and some applications (5)

Total L = 15 Hours**REFERENCES:**

1. V. Lakshminarayanan, Ajoy Ghatak and K Thyagarajan, Lagrangian Optics, Kluwer Academic Publishers, Boston, 2002.
2. E Hecht and A R Ganesan, Optics , Pearson 2002
3. M Born and E Wolf, Principles of optics, Cambridge University press, UK 1999
4. R Boyd, Non linear optics, Academics Press, 2020

23OFP04 FIBER OPTICS**(1 0 0 1)**

Ray treatment of optical guiding in optical fibers, Modes in optical fibers and their properties. (5)

Single mode and multimode fibers. Dispersion in optical fibers and its impact on transmission capacity. (5)

Nonlinear effects in optical fibers, Applications in communication and sensing. (5)

Total L = 15 Hours**REFERENCES:**

1. Ajoy Ghatak and K. Thyagarajan, Introduction To Fiber Optics, Cambridge University Press, 2017

2. K Thyagarajan and Ajoy Ghatak, Fiber Optic Essentials, Wiley-IEEE Press; 2007.
3. E Hecht and A R Ganesan, Optics , Pearson 2002
4. A Yariv Pochi Yeh, Photonics - Optical Electronics in modern communication, Oxford, 1997
5. R Boyd, Non linear optics, Academics Press, 2020

23OFP05 QUANTUM PHOTONICS

(1 0 0 1)

Quantization of electromagnetic waves and the concept of photons. (5)

Quantum states such as single mode states, multimode states, entangled states, squeezed states. (5)

Generation of non classical states of light; applications in quantum computing, quantum teleportation and quantum cryptography. (5)

Total = 15 Hours

REFERENCES:

1. C C Gerry and P L Knight, Introductory quantum optics, Cambridge University Press, 2005
2. K Thyagarajan and A Ghatak, Lasers : Fundamentals and applications, Springer, 2019
3. A Yariv, Quantum Electronics, John Wiley, USA, 1991
4. B E A Saleh and M C Teich, Fundamentals of Photonics, John Wiley & Sons. 2019

CHEMISTRY

23OFC01

BATTERY THERMAL MANAGEMENT SYSTEMS

1 0 0 1

INTRODUCTION TO BATTERY THERMAL MANAGEMENT - Basics of lithium-ion batteries - Thermal issues in batteries: heat generation, effects on performance, and degradation - Need for BTMS in EVs and stationary storage (3)

HEAT GENERATION AND THERMAL MODELING - Sources of heat in batteries (ohmic, reaction-based) -Governing equations and thermal modeling basics - Introduction to battery thermal simulation tools (COMSOL, ANSYS) (3)

BTMS TECHNOLOGIES - Air cooling (natural, forced) - Liquid cooling (direct, indirect) - Phase change materials (PCM) - Heat pipes and advanced materials - Hybrid cooling technologies - Comparative analysis of methods (3)

DESIGN AND INTEGRATION IN EVs - BTMS layout in electric vehicles - Integration with vehicle HVAC systems - Battery pack design considerations (cell spacing, airflow) - Packaging and space constraints (3)

SAFETY, STANDARDS, AND INNOVATIONS - Thermal runaway and propagation - Fire protection strategies - Regulatory standards (ISO 6469, SAE J2929, AIS 156) - Emerging technologies: immersion cooling, AI in BTMS, smart BTMS (3)

Total L: 15

REFERENCES:

1. Gregory L. Plett, Battery Management Systems, Artech house publisher, London, 2015.
2. Hafiz Muhammad Ali, Thermal Management for Batteries: From Basic Design to Advanced Simulation and Management Methods, Elsevier, 2024.
3. Ibrahim Dinçer, Halil S. Hamut, Nader Javani, Thermal Management of Electric Vehicle Battery System, John Wiley & Sons, 2017

23OFC02

HYDROGEN ENERGY FOR ELECTRICAL MOBILITY

1 0 0 1

INTRODUCTION TO HYDROGEN ENERGY - Properties of hydrogen - Hydrogen as an energy vector - Comparison with conventional fuels and batteries - Overview of hydrogen economy. (3)

HYDROGEN PRODUCTION TECHNOLOGIES - Electrolysis (alkaline, PEM, SOEC) - Steam Methane Reforming (SMR) - Biomass gasification - Green vs. Grey vs. Blue Hydrogen - Role of renewable energy in hydrogen production (3)

HYDROGEN STORAGE AND TRANSPORTATION - Compressed and liquefied hydrogen - Metal hydrides - Cryogenic storage - Hydrogen distribution and refueling infrastructure. (2)

HYDROGEN FUEL CELLS IN ELECTRIC MOBILITY - Principles of fuel cell operation (PEMFC, SOFC, etc.) - Fuel cell vs. battery electric vehicles - Design of hydrogen-powered electric vehicles - Case studies (e.g., Toyota Mirai, Hyundai Nexa). (3)

SAFETY, STANDARDS, AND POLICY FRAMEWORK - Hydrogen handling and safety concerns - Regulations, codes, and standards - National and international hydrogen roadmaps (India, EU, USA) - Public and private sector initiatives, Cost and scalability - Techno-economic barriers - Emerging technologies (e.g., solid hydrogen storage, micro fuel cells) - Role of hydrogen in decarbonizing heavy transport and aviation. (4)

Total L: 15

REFERENCES:

1. Bent Sørensen, Giuseppe Spazzafumo, 'Hydrogen and Fuel Cells: Emerging Technologies and Applications', Elsevier (2018)
2. Aldo Steinfeld, D. Yogi Goswami, "Handbook of Hydrogen Energy, CRS press (2019)

SELF DIRECTED LEARNING COURSES

23IC01 EXPERIENCING ENTREPRENEURSHIP

3 0 0 3

Entrepreneurship Development Process: Entrepreneurial Ecosystem - Entrepreneur, Enterprise and Environment, Entrepreneurial Motivation, Entrepreneurial Competencies. ED programme - Need and objectives, EDPs in India and its phases. Stages of ED Process - Discover, evaluate, developing a plan, resources, managing the company, harvesting (10)

Creativity, Idea Generation: Creativity: Right environment, Components of performance, tools and techniques, motivation. Design thinking: Process, heuristics approaches, behavioural change, empathy. Idea Generation: Process, methods, sources, entrepreneurial linkages. Systematic Inventive Thinking (SIT): Principles, tools Disruptive Innovation (8)

Product / Process / Technology / Service Development: Understanding Market Needs - Market Research Fundamentals, Identifying Customer Pain Points, Target Audience Analysis. Prototyping and Design - Rapid Prototyping, User-Centered Design Principles, Iterative Design Processes. Streamlining Operations - Value Stream Mapping, Lean and Agile Principles, Continuous Improvement Strategies. Service Development - Designing Customer-Centric Services, Service Innovation, Quality Assurance in Services. (10)

Business Plan and Feasibility Analysis: Market Analysis: Preparation of Business Plan/Model. Product/Service Feasibility Analysis – Methods and Tools. Industry and Competition Analysis: GTM Strategy – Viability - FAB Analysis - USP, Gap Analysis. Environment and Technical Analysis. Financial Feasibility Analysis: Concept of Costs, Profits and Losses, Manage Cash Flow, Analyze Financial Performance and Budgeting (8)

* **Capstone Project:** Students will prepare a business plan and feasibility analysis for a business of their choice (9)

Total L : 45hrs

References:

1. Rajeev Roy, Entrepreneurship. New Delhi: 3 rd Edition, Oxford University Press, 2020
2. Pareek, Udai Narain, and T. Venkateswara Rao. Developing motivation through experiencing: a trainers' guide for behavior simulation in motivation training. Oxford & IBH, 1991.
3. Rod Judkins, "The Art of Creative Thinking", Hodder & Stoughton, 2015
4. Peter Drucker, "Innovation and Entrepreneurship", Routledge, 2015
5. Osterwalder, A. and Pigneur, Y. Business model generation: a handbook for visionaries, game changers, and challengers (Vol. 1). John Wiley & Sons, 2010
6. Andy Polaine, Lavrans Løvlie, and Ben Reason, "Service Design: From Insight to Implementation" Rosenfeld Media, 2013.

23IC02 VENTURE DEVELOPMENT

3 0 0 3

Business Model Canvas: Customer Segments, Value Propositions, Channels and customer relationships, Revenue streams and key resources, Key activities and key partnerships, Cost structure (10)

Funding Opportunities: Government Funding and Incubators: Functions of an incubator. Venture capital and Angel Funding: Venture Capital Industry, Company Formation, Due diligence, Valuation and term sheets, Value addition and governance, Exits. (10)

Enterprise formation & legal compliances: Understanding Legal Structures / Compliance Requirements - Business Registration and Licensing, Intellectual Property Protection. Tax Compliance and Planning - Corporate Taxes and Business Deductions, External Audits and Compliance Checks. Exit Strategies and Legal Considerations - Succession Planning and Legal Requirements, Handling Legal Challenges during Business Exits. (15)

* **Capstone Project:** Students will demonstrate their ability to fully develop and communicate a venture concept (10)

Total L : 45hrs

REFERENCES:

1. Osterwalder, A. and Pigneur, Y. Business model generation: a handbook for visionaries, game changers, and challengers (Vol. 1). John Wiley & Sons, 2010
2. Rajeev Roy, Entrepreneurship, Oxford Higher Education, India, 2020
3. "A Note on the Venture Capital Industry," Harvard Business School (HBS # 9-295- 065, July 12, 2001)
4. Sumit Pahwa "Corporate Law Referencer 9/e", OakBridge Publishing Pvt. Ltd, 2023
5. Charantimath Poornima M, "Entrepreneurship Development and Small Business Enterprises", 3/e, Pearson Education, 2018

(* The activity will start during the 7th week of the respective courses)

EVOLUTION AND CONSTITUTIONAL FRAMEWORK : Historical Development – Kautilya's Arthashastra - Mughal administration - Legacy of British rule in politics and administration, revenue administration, district Administration, local self-government - Constitutional Framework- Role and Structure of Civil Services - Administrative Reforms and Commissions- Challenges of liberalisation, Privatization, Globalization - Good Governance: concept and application - New Public Management - E-Governance and Digital Administration (9)

STRATEGIC GOVERNANCE AND GLOBAL INTERFACE: PSU Management, Privatization and Disinvestment - Project Implementation Framework - Role of Regulatory Bodies and Public Accountability - Defense Policy Administration and Historical Evolution - Role of Civil Services in Strategic Security Governance - Crisis Leadership and Disaster Management Framework (9)

INDIA'S FOREIGN POLICY: Principles and Evolution - Role of Diplomacy and Bureaucracy in Global Affairs - Interface of Foreign and Domestic Policy - Administrative Preparedness in International Negotiations - Internal Security and Administrative Response. (9)

LEADERSHIP EXCELLENCE IN PUBLIC ADMINISTRATION: Strategic Vision and Institutional Continuity - Ethics and Crisis Leadership in Governance - Insights from Senior Leadership Roles in Public Service - Emotional Intelligence and Motivation for Administrators - Strategic Thinking and Change Management in Public Systems - Participatory Governance and Citizen Engagement - Case Studies on Indian Policy Decisions and Leadership. (9)

GOVERNANCE, ETHICS AND ACCOUNTABILITY: Values in Public Administration - Citizen Charter - Right to Information (RTI) - Anti-Corruption Mechanisms - Role of Civil Society and Media in Accountability - Code of Conduct and Code of Ethics for Civil Servants - Social Audit and Community Monitoring - Role of Parliamentary Committees (PAC, Estimates Committee) - E-Governance Tools for Transparency - Whistleblower Protection Framework - Administrative Reforms Commissions - Right to Public Services Acts (state-level initiatives) - Ethics Training and Capacity Building in Public Service (9)

Total L: 45

TEXT BOOKS:

1. Chandrasekhar, K. M. As Good as My Word: A Memoir (1st paperback ed., 312 pp.). HarperCollins, 2022.
2. Bidyut Chakrabarty Public Administration: The Indian Context. Orient BlackSwan, 2022
3. S.R. Maheshwari, Indian Administration – 7th Edition. Orient Blackswan, 2021

REFERENCES:

1. Alka Dhameja et al, Public Policy and Governance in India. Pearson Education, 2022
2. Partha Pratim Basu, Democratic Governance in India. Routledge, 2023
3. India.Ramesh K Arora, Indian Public Administration: Institutions and Issues. New Age International, 2023

INTRODUCTION TO INDIAN ECONOMY AND ECONOMIC PLANNING: Features of Indian Economy -- Sectors of the Economy and their contribution to growth -- Institutional Roles in Economic Policy Formulation RBI, Finance Commission, Economic Ministries, Regulatory Bodies, Research Institutions - History of Planning in India - Five-Year Plans - NITI Aayog - Local Economic Governance: Panchayati Raj Institutions and decentralization. (9)

PUBLIC FINANCE, FISCAL POLICY, INDIAN MONETARY SYSTEM AND MONETARY POLICY: Indian budgetary process and fiscal architecture - GST and the tax ecosystem: direct and indirect taxes - Fiscal deficit, revenue deficit, and FRBM Act - Public expenditure management - Role of RBI and Monetary Policy - Inflation, Interest Rates, and Credit Flow - Financial Inclusion and Digital Banking - Credit Access for MSMEs and rural enterprises. (9)

INTERNATIONAL ECONOMICS AND FOREIGN TRADE: International Trade Basic Concepts and Theories – Trade Policies and Instruments – Foreign Exchange Market - Global Trade Systems and India's Foreign Trade Policy - International Institutions: WTO, IMF, World Bank - Balance of Payments, Exchange Rates, and Trade Deficits - Global Investment Trends and FDI in India – Current Issues in International Economics (9)

ECONOMIC REFORMS AND SECTORAL POLICIES: Liberalization, Privatization, Globalization (LPG) - Sectoral Focus: Agriculture, Industry, Services – Challenges faced by the Sectors in India - Trade and Investment Policies - Start-up and Innovation Policy Ecosystem - Education and Skill Development Policy – Economics of Infrastructure Development and Policies in India - Inclusive and Sustainable Development - Policies on SDGs and Climate Finance – Viksit Bharath: The way Ahead (9)

SOCIAL JUSTICE AND RURAL DEVELOPMENT: Significance of social justice in economic development - Link between equity, poverty alleviation, and inclusive growth - Rural Poverty and Inequality - Rural development as a strategy for social justice - Key Government Programs and Policies: Agriculture Policy and Food Security - Social Inclusion and Marginalized Groups - Gender Equity and Women's Economic Empowerment - Land Reforms and Agrarian Justice - Education, Health, and Social Infrastructure for Inclusive Growth - Case Studies and Success Stories of Community-driven development models - Best practices from Indian states (9)

Total L: 45

TEXT BOOKS:

1. Debraj Ray, Development Economics – Updated Edition. Princeton University Press, 2022
2. C. Rangarajan, Essays on Macroeconomic Policy and Growth in India. Academic Foundation, 2022

3. Jean Drèze & Amartya Sen, *An Uncertain Glory: India and its Contradictions*. Penguin, 2021

REFERENCES:

1. Rathin Roy et al, *Fiscal Policy in India: Past, Present and Future*. Oxford University Press, 2021
2. Kaushik Basu, *Policy Maker's Journal: From New Delhi to Washington D.C.* Penguin, 2023
3. Arvind Subramanian & Josh Felman (2020). *India's Great Slowdown: What Happened? What's the Way Out?* Centre for Policy Research, 2020